Application of Integrals

Question1

Let the area of the region bounded by the curve $y = \max\{\sin x, \cos x\}$, lines $x = 0, x = \frac{3\pi}{2}$, and the x-axis be A . Then, $\mathrm{A} + \mathrm{A}^2$ is equal to ____。

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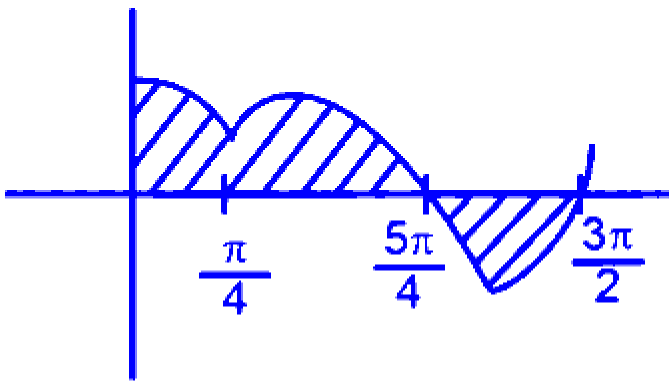
Answer: 12

Solution:

$$f(x) = \max(\sin x, \cos x)$$
$$= \begin{cases} \cos x, x \in \left(0, \frac{\pi}{4}\right) \\ \sin x, x \in \left(\frac{\pi}{4}, \frac{5\pi}{4}\right) \\ \cos x, x \in \left(\frac{5\pi}{4}, \frac{3\pi}{2}\right) \end{cases}$$

$\Rightarrow$



$$\Rightarrow \int_0^{\frac{3\pi}{4}} |f(x)| dx = \int_0^{\frac{\pi}{4}} (\cos x) dx + \int_{\frac{\pi}{4}}^{\frac{5\pi}{4}} (\sin x) dx + \int_{\frac{5\pi}{4}}^{\frac{3\pi}{2}} (\cos x) dx$$
$$= 3 \Rightarrow \mathrm{A} = 3$$
$$A + A^2 = 12$$

Question2

If the area of the region bounded by $16x^2 - 9y^2 = 144$ and $8x - 3y = 24$ is A, then $3(A + 6\log_e(3))$ is equal to ________.

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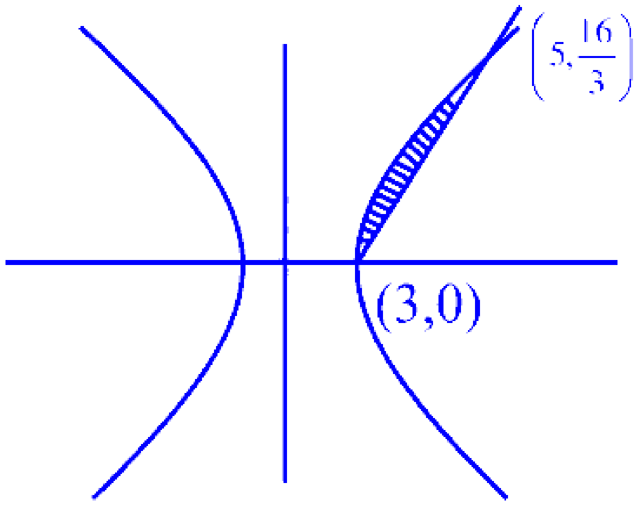
Answer: 24

Solution:



$16x^2 - 9y^2 = 144$

$16x^2 - (8x - 24)^2 = 144$

$16x^2 - 64(x-3)^2 = 144 \rightarrow x^2 - 4(x-3)^2 = 9$

$3x^2 - 24x + 45 \rightarrow x^2 - 8x + 15 = 0$

$x = 3, 5$

$\text{Area} = \int\limits_3^5 \sqrt{\frac{16x^2 - 144}{9}} - \frac{1}{2} \cdot 2 \cdot \frac{16}{3}$

$= \frac{4}{3} \int\limits_3^5 \sqrt{x^2 - 9} - \frac{16}{3}$

$= \frac{4}{3}\left(\frac{x}{2}\sqrt{x^2-9} - \frac{9}{2}\log_e\left(x+\sqrt{x^2-9}\right)\right)_3^5 - \frac{16}{3}$

$= \frac{4}{3}\left(\frac{5}{2}\cdot 4 - \frac{9}{2}\log_e 9 - \frac{3}{2}\cdot 0 + \frac{9}{2}\log_e 3\right) - \frac{16}{3}$

$\text{Area} = 8 - 6\log_e 3 = A$

$\therefore A + 6\ell n 3 = 8$

$\Rightarrow 3(A + 6\ell n 3) = 24$

Question3

Let $f : \mathbf{R} \to \mathbf{R}$ be a function such that $f(x) + 3f\left(\frac{\pi}{2} - x\right) = \sin x, x \in \mathbf{R}$. Let the maximum value of f on $\mathbf{R}$ be α. If the area of the region bounded by the curves $g(x) = x^2$ and $h(x) = \beta x^3, \beta > 0$, is α^2, then $30\beta^3$ is equal to _____ .

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Answer: 16

Solution:

$f(x) + 3f\left(\frac{\pi}{2} - x\right) = \sin x \qquad \ldots(1)$

Now replace x by $\frac{\pi}{2} - x$ in (1). This is a standard NCERT step to get a second equation in the same two terms.

$\Rightarrow f\left(\frac{\pi}{2} - x\right) + 3f(x) = \cos x \qquad \ldots(2)$

Now we solve the two linear equations (1) and (2) to find $f(x)$.

Multiply (1) by 3:

$3f(x) + 9f\left(\frac{\pi}{2} - x\right) = 3\sin x \qquad \ldots(3)$

Subtract (2) from (3):

$\big(3f(x) - 3f(x)\big) + (9-1)f\left(\frac{\pi}{2} - x\right) = 3\sin x - \cos x$

$8f\left(\frac{\pi}{2} - x\right) = 3\sin x - \cos x$

Now put $x \to \frac{\pi}{2} - x$ in this result to get $f(x)$:

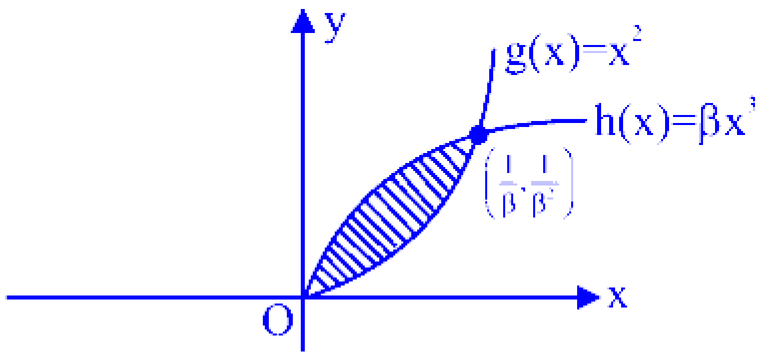
$8f(x) = 3\cos x - \sin x$

$f(x) = \frac{1}{8}(3\cos x - \sin x)$

To find the maximum value, write $3\cos x - \sin x$ in the form $R\cos(x+\phi)$, where $R = \sqrt{3^2+(-1)^2} = \sqrt{10}$. So the maximum value of $3\cos x - \sin x$ is $\sqrt{10}$.

$f_{\max} = \frac{\sqrt{10}}{8}$

$\mathrm{y = g(x) \& y = h(x)}$ intersect as shown in the figure



The curves are $g(x) = x^2$ and $h(x) = \beta x^3$ with $\beta > 0$. Their points of intersection come from $x^2 = \beta x^3$:

$x^2 = \beta x^3 \Rightarrow x^2(1-\beta x) = 0 \Rightarrow x = 0$ and $x = \frac{1}{\beta}$

For $0\beta x^3(since\beta x<1), sotheareabetweenthecurvesistheintegralof(x^2-\beta x^3)from0to \frac{1}{\beta}$.

$$\therefore \text{ Area bounded } = \Delta = \left|\int_0^{\frac{1}{\beta}} \left(\beta \mathrm{x}^3 - \mathrm{x}^2\right) \mathrm{dx}\right|$$

$$= \left|\left[\frac{\beta x^4}{4} - \frac{x^3}{3}\right]_0^{1/\beta}\right|$$

$$= \left|\left(\frac{\beta}{4}\cdot\frac{1}{\beta^4} - \frac{1}{3}\cdot\frac{1}{\beta^3}\right)\right|$$

$$= \left|\left(\frac{1}{4\beta^3} - \frac{1}{3\beta^3}\right)\right| = \frac{1}{12\beta^3} = \alpha^2 \text{ (given)}$$

$$\Rightarrow \frac{1}{12\beta^3} = \left(\frac{\sqrt{10}}{8}\right)^2 = \frac{10}{64} = \frac{5}{32}$$

$$\Rightarrow \beta^3 = \frac{1}{12}\cdot\frac{32}{5} = \frac{8}{15}$$

$$\Rightarrow 30\beta^3 = 30\cdot\frac{8}{15} = 16$$

Question4

The area of the region, inside the ellipse $x^2+4y^2=4$ and outside the region bounded by the curves $y=|x|-1$ and $y=1-|x|$, is :

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Options:

A.

$2\pi - 1$

B.

$3(\pi - 1)$

C.

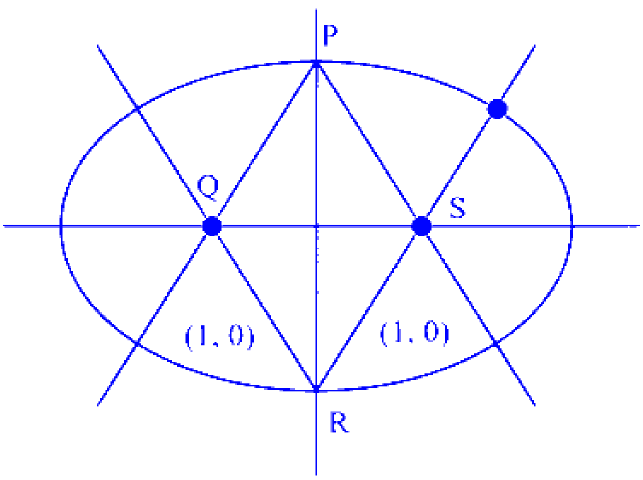
$2(\pi - 1)$

D.

$2\pi - \frac{1}{2}$

Answer: C

Solution:



$E : \dfrac{x^2}{4} + \dfrac{y^2}{1} = 1$

Area inside $E = 2\pi$

Area $PQRS = (\sqrt{2})^2$

$= 2$

$$\begin{aligned} \text{Required area } &= 2\pi - 2 \\ &= 2(\pi - 1) \end{aligned}$$

Question5

If the area of the region $\{(x, y) : 1 - 2x \leq y \leq 4 - x^2,\ x \geq 0,\ y \geq 0\}$ is $\dfrac{\alpha}{\beta}$, $\alpha, \beta \in \mathbb{N}, \gcd(\alpha, \beta) = 1$, then the value of $(\alpha + \beta)$ is:

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Options:

A.

73

B.

85

C.

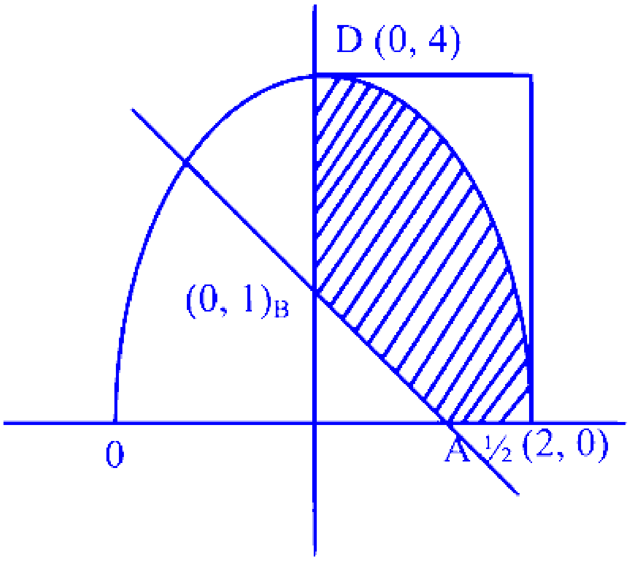
67

D.

91

Answer: A

Solution:



$\int_0^2 (4-x^2)dx - \frac{1}{2} \times 1 \times \frac{1}{2}$

Question6

Let the line $x=-1$ divide the area of the region $\left\{(x, y): 1+x^2 \leq y \leq 3-x\right\}$ in the ratio $m: n, \operatorname{gcd}(m, n)=1$. Then $m+n$ is equal to

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Options:

A.

27

B.

28

C.

25

D.

Answer: A

Solution:

$$m=\int_{-2}^{-1}(3-x)-\left(1+x^2\right)dx$$
$$m=2+\frac{3}{2}-\frac{7}{3}=\frac{7}{6} \quad M=\int_{-1}^{1}(3-x)-\left(1+x^2\right)dx=4-\frac{1}{2}(0)-\frac{1}{3}(1+1)=4-\frac{2}{3}=\frac{10}{3}$$
$$m:M=\frac{7}{6}:\frac{10}{3}=\frac{7}{20}$$
$$m+n=27$$

Question7

The area of the region $\mathrm{A}=\left\{(x,y):4x^2+y^2\leqslant 8 \text{ and } y^2\leqslant 4x\right\}$ is:

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Options:

A.

$\pi+\frac{2}{3}$

B.

$\frac{\pi}{2}+2$

C.

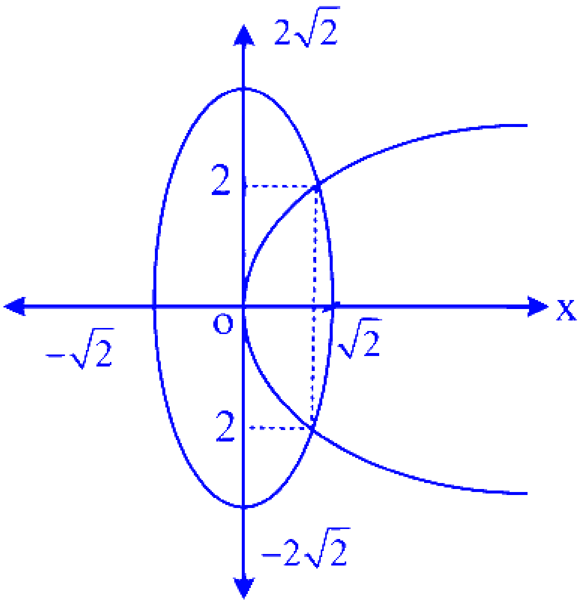
$\pi+4$

D.

$\frac{\pi}{2}+\frac{1}{3}$

Answer: A

Solution:



$A=2\int_0^1 2\sqrt{x}dx+2\int_1^{\sqrt{2}}\sqrt{8-4x^2}dx=\frac{8}{3}\left(X^{\frac{3}{2}}\right)_0^1+4\int_1^{\sqrt{2}}\sqrt{2-x^2}dx$

$\frac{8}{3}+4\times\frac{1}{2}\left[x\sqrt{2-x^2}+2\sin^{-1}\left(\frac{x}{\sqrt{2}}\right)\right]_1^{\sqrt{2}}$

$\frac{8}{3}+2\left[2\times\frac{\pi}{2}-1-2\times\frac{\pi}{4}\right]=\frac{8}{3}+2\pi-2-\pi=\pi+\frac{2}{3}$ Sq. unit

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Question8

The area of the region enclosed between the circles $x^2+y^2=4$ and $x^2+(y-2)^2=4$ is:

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Options:

A.

$\frac{2}{3}(4\pi-3\sqrt{3})$

B.

$\frac{4}{3}(2\pi - \sqrt{3})$

C.

$\frac{4}{3}(2\pi - 3\sqrt{3})$

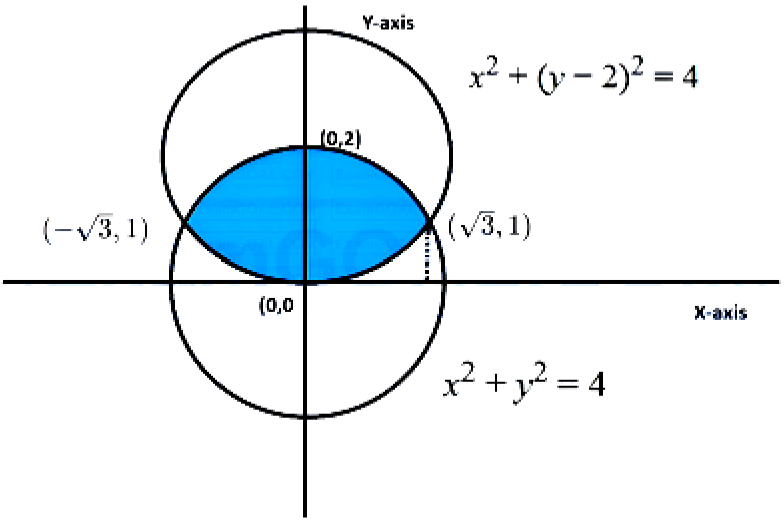
D.

$\frac{2}{3}(2\pi - 3\sqrt{3})$

Answer: A

Solution:

$$x^2 + y^2 = 4 \implies C_1(0,0), r = 2$$
$$x^2 + (y-2)^2 = 4 \implies C_2(0,2), r = 2$$



Intersection points:

$$x^2 + y^2 = x^2 + (y-2)^2$$
$$y^2 = y^2 - 4y + 4 \implies 4y = 4 \implies y = 1$$
$$x^2 + 1 = 4 \implies x^2 = 3 \implies x = \pm\sqrt{3}$$

Points are $(\sqrt{3}, 1)$ and $(-\sqrt{3}, 1)$

For the region between $y = 0$ and $y = 2$:

$$y_{\text{upper}} = \sqrt{4 - x^2} \left(\text{from} x^2 + y^2 = 4 \right)$$
$$y_{\text{lower}} = 2 - \sqrt{4 - x^2} \left(\text{from} x^2 + (y-2)^2 = 4 \right)$$

$$\text{Area} = \int_{-\sqrt{3}}^{\sqrt{3}} \left(y_{\text{upper}} \quad - y_{\text{lower}} \quad\right) dx$$

$$\text{Area} = \int_{-\sqrt{3}}^{\sqrt{3}} \left(\sqrt{4-x^2} - \left(2 - \sqrt{4-x^2}\right)\right) dx$$

$$\text{Area} = \int_{-\sqrt{3}}^{\sqrt{3}} \left(2\sqrt{4-x^2} - 2\right) dx$$

Property: $\int_{-a}^{a} f(x)dx = 2\int_0^a f(x)dx$ (for even functions)

$$\text{Area} = 2\int_0^{\sqrt{3}} \left(2\sqrt{4-x^2} - 2\right) dx$$

$$\text{Area} = 4\int_0^{\sqrt{3}} \left(\sqrt{4-x^2} - 1\right) dx$$

$$\int \sqrt{a^2 - x^2} dx = \frac{x}{2}\sqrt{a^2-x^2} + \frac{a^2}{2}\sin^{-1}\left(\frac{x}{a}\right)$$

$$\text{Area} = 4\left[\frac{x}{2}\sqrt{4-x^2} + 2\sin^{-1}\left(\frac{x}{2}\right) - x\right]_0^{\sqrt{3}}$$

$$\text{Area} = 4\left[\frac{\sqrt{3}}{2}(1) + 2\left(\frac{\pi}{3}\right) - \sqrt{3}\right]$$

$$\text{Area} = 4\left[\frac{2\pi}{3} - \frac{\sqrt{3}}{2}\right]$$

$$\text{Area} = \frac{8\pi}{3} - 2\sqrt{3} = \frac{2}{3}(4\pi - 3\sqrt{3})$$

Question9

Let A_1 be the bounded area enclosed by the curves $y = x^2 + 2, x + y = 8$ and y-axis that lies in the first quadrant. Let A_2 be the bounded area enclosed by the curves $y = x^2 + 2, y^2 = x, x = 2$, and y-axis that lies in the first quadrant. Then $\mathrm{A}_1 - \mathrm{A}_2$ is equal to

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Options:

A.

$\frac{2}{3}(2\sqrt{2}+1)$

B.

$\frac{2}{3}(3\sqrt{2}+1)$

C.

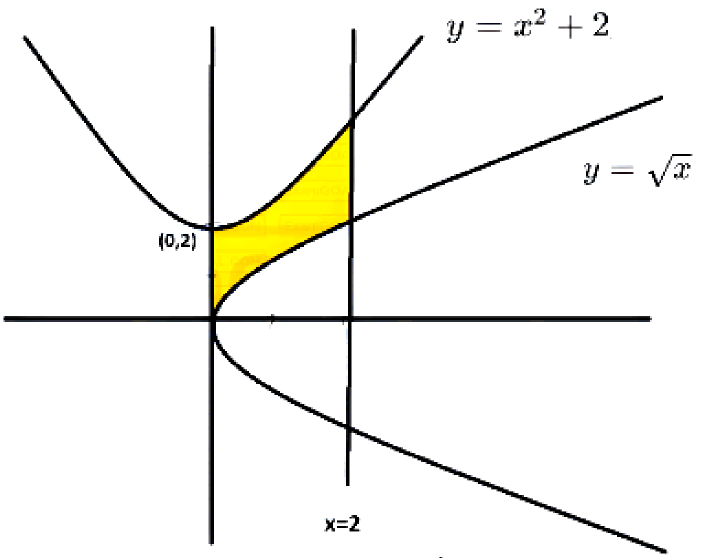
$\frac{2}{3}(\sqrt{2}+1)$

D.

$\frac{2}{3}(4\sqrt{2}+1)$

Answer: A

Solution:



Area A_2

Area A_1 enclosed by curves

$y=x^2+2, x+y=8, y$-axis in first quadrant as shown in graph.

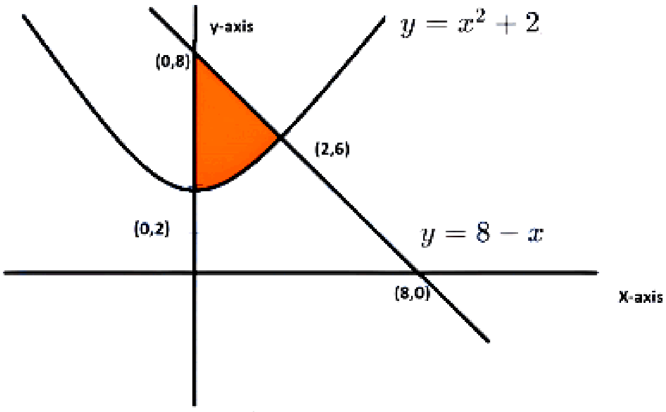
Solve $y=x^2+2$ and $x+y=8$ to find point of intersection.

Substituting $y=8-x$ in $y=x^2+2$

$8-x=x^2+2$
$x^2+x-6=0$
$x^2+3x-2x-6=0$
$x(x+3)-2(x+3)=0$
$(x+3)(x-2)=0$
$x=-3,2$

Since area in first quadrant, point of intersection at $x=2$, $y=8-2=6 \Rightarrow (2,6)$.

$$A_1=\int_0^2\left[(8-x)-\left(x^2+2\right)\right]dx$$
$$=\int_0^2\left(-x^2-x+6\right)dx$$
$$=\left[-\frac{x^3}{3}-\frac{x^2}{2}+6x\right]_0^2$$
$$=\left[-\frac{8}{3}-\frac{4}{2}+12-\{0-0-0\}\right]$$
$$=\frac{-16-12+72}{6}=\frac{72-28}{6}=\frac{44}{6}$$
$$A_1=\frac{22}{3}$$



Area A_1

Now, A_2 is area enclosed by curves

$y=x^2+2, y^2=x, x=2$

Solve $y=x^2+2$ and $y=\sqrt{x}$

$y=y^4+2$

$y^4-y+2=0$

$D=(-1)^2-4\times1\times2=1-8=-7$

discriminant (D) is negative. This means curves do not intersect.

So A_2 is bounded by curves y-~~axis~~, $2, y = x^2 + 2$ and $y = \sqrt{x}$ in first quadrant is:

$$A_2 = \int_0^2 \left(x^2 + 2 - \sqrt{x}\right) dx$$

$$= \left[\frac{x^3}{3} + 2x - \frac{2x^{3/2}}{3}\right]_0^2$$

$$= \left[\frac{8}{3} + 2 \times 2 - \frac{2 \times 2\sqrt{2}}{3} - \{0 + 0 - 0\}\right]$$

$$= \left[\frac{8}{3} + 4 - \frac{4\sqrt{2}}{3}\right] = \left[\frac{20}{3} - \frac{4\sqrt{2}}{3}\right]$$

$$A_1 - A_2 = \frac{22}{3} - \left(\frac{20}{3} - \frac{4\sqrt{2}}{3}\right)$$

$$= \frac{22}{3} - \frac{20}{3} + \frac{4\sqrt{2}}{3} = \frac{2}{3} + \frac{4\sqrt{2}}{3}$$

$$= \frac{2}{3}(2\sqrt{2} + 1)$$

Question10

Let $f(\alpha)$ denote the area of the region in the first quadrant bounded by $x = 0, x = 1, y^2 = x$ and $y = |\alpha x - 5| - |1 - \alpha x| + \alpha x^2$. Then $(f(0) + f(1))$ is equal to

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Options:

A.

12

B.

14

C.

9

D.

Answer: D

Solution:

Given Curves $y^2 = x, y = |\alpha x - 5| - |1 - \alpha x| + \alpha x^2 x = 0$ and $x = 1$

The area of region bounded by these curve is $f(\alpha)$.

To find $f(0) + f(1)$.

for $\alpha = 0$

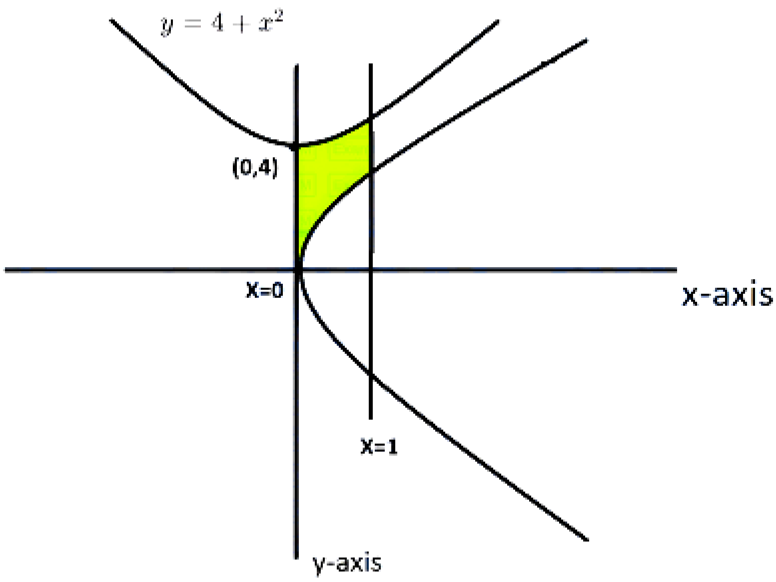
$$y = |0 \times x - 5| - |1 - 0 \times x| + 0 \times x^2$$
$$y = 5 - 1 = 4$$

so the region bounded by $x = 0, x = 1, y = \sqrt{x}$

and $y = 4$ is shown in graph.

$$f(0) = \int_0^1 (4 - \sqrt{x}) dx$$

$$f(0) = \left[4x - \frac{2}{3}x^{3/2}\right]_0^1 = 4 - \frac{2}{3} = \frac{10}{3}$$



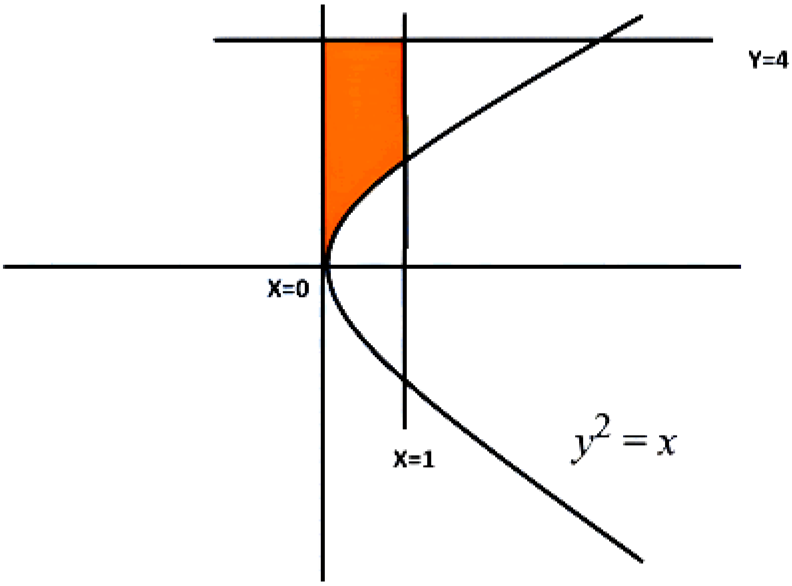
If, When $\alpha = 1$, the second equation becomes:

$$y = |x - 5| - |1 - x| + x^2$$

For the interval $0 \leq x \leq 1$:

$$|x - 5| = -(x - 5) = 5 - x$$
$$|1 - x| = 1 - x$$
$$y = (5 - x) - (1 - x) + x^2 = 4 + x^2$$

The region is bounded by $x=0, x=1, y=\sqrt{x}$, and $y=4+x^2$ as shown in graph is.



$$f(1)=\int_0^1\left(4+x^2-\sqrt{x}\right)dx$$

$$f(1)=\left[4x+\frac{x^3}{3}-\frac{2}{3}x^{3/2}\right]_0^1=4+\frac{1}{3}-\frac{2}{3}=4-\frac{1}{3}=\frac{11}{3}$$

So, $(f(0)+f(1))=\frac{10}{3}+\frac{11}{3}=\frac{21}{3}=7$

Question11

The area of the region $\mathrm{R}=\left\{(x,y): xy\leq 8, 1\leq y\leq x^2, x\geq 0\right\}$ is

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Options:

A.

$\frac{2}{3}(20\log_e(2)+9)$

B.

$\frac{1}{3}(40\log_e(2)+27)$

C.

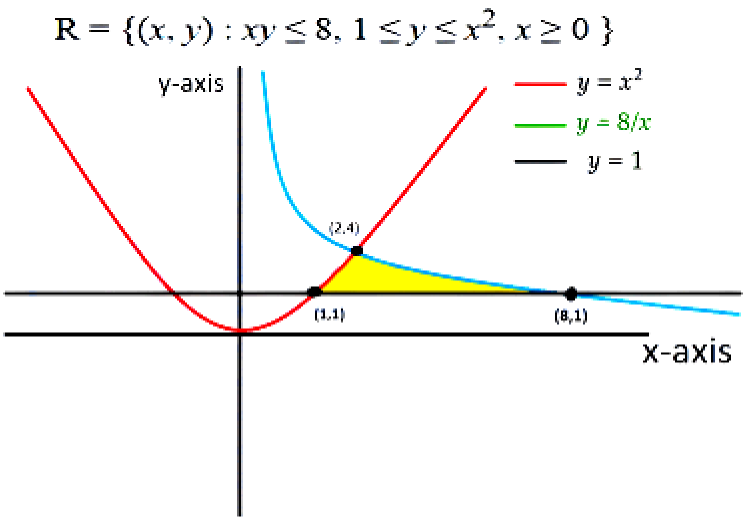
$\frac{1}{3}(49 \log_e(2) - 15)$

D.

$\frac{2}{3}(24 \log_e(2) - 7)$

Answer: D

Solution:



The region is bounded by three curves in the first quadrant ($x \geq 0$) :

- 1. $y = 1$: A horizontal line.
- 2. $y = x^2$: A parabola opening upwards.
- 3. $y = \frac{8}{x}$: A rectangular hyperbola.

Find the Intersection Points

We need to know where these curves meet to set our integration limits:

- Parabola and Line $\left(y = x^2 \text{ and } y = 1\right) : x^2 = 1 \Longrightarrow x = 1$ (since $x \geq 0$). Point: $(1, 1)$.
- Parabola and Hyperbola $\left(y = x^2 \text{ and } y = \frac{8}{x}\right) : x^2 = \frac{8}{x} \Longrightarrow x^3 = 8 \Longrightarrow x = 2$. Point: $(2, 4)$.
- Hyperbola and Line $\left(y = \frac{8}{x} \text{ and } y = 1\right) : \frac{8}{x} = 1 \Longrightarrow x = 8$. Point: $(8, 1)$.

Understanding the Inequalities

- $x \geq 0$: The region is located in the first and fourth quadrants (but limited to the first quadrant by $1 \leq y$).
- $1 \leq y \leq x^2$: This gives us a lower bound of the horizontal line $y = 1$ and an upper bound of the parabola $y = x^2$.
- $xy \leq 8$: This can be rewritten as $y \leq \frac{8}{x}$. This represents the area below the rectangular hyperbola.

Setting up the Integrals

The region starts at $x = 1$ and ends at $x = 8$. Because the "upper" boundary changes at $x = 2$, we must split the area into two parts:

- Part 1 (from $x = 1$ to $x = 2$): Bounded above by the parabola $y = x^2$ and below by $y = 1$.
- Part 2 (from $x = 2$ to $x = 8$): Bounded above by the hyperbola $y = \frac{8}{x}$ and below by $y = 1$.

Area $= \int\limits_1^2 \left(x^2 - 1\right) dx + \int\limits_2^8 \left(\frac{8}{x} - 1\right) dx$

$= \left[\frac{x^3}{3} - x\right]_1^2 + [8 \ln x - x]_2^8$

$= \left[\frac{8}{3} - 2 - \left\{\frac{1}{3} - 1\right\}\right] + [(8 \ln 8 - 8) - \{8 \ln 2 - 2\}]$

$= \frac{7}{3} - 1 + 8 \ln 2^3 - 8 \ln 2 - 6$

$= \frac{7}{3} - 7 + 24 \ln 2 - 8 \ln 2 \quad \{\ln a^m = m \ln a\}$

$= -\frac{14}{3} + 16 \ln 2$

$= \frac{2}{3}[24 \ln 2 - 7]$

Question12

Let $P_1 : y = 4x^2$ and $P_2 : y = x^2 + 27$ be two parabolas. If the area of the bounded region enclosed between P_1 and P_2 is six times the area of the bounded region enclosed between the line $y = \alpha x, \alpha > 0$ and P_1, then α is equal to :

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Options:

A.

12

B.

15

C.

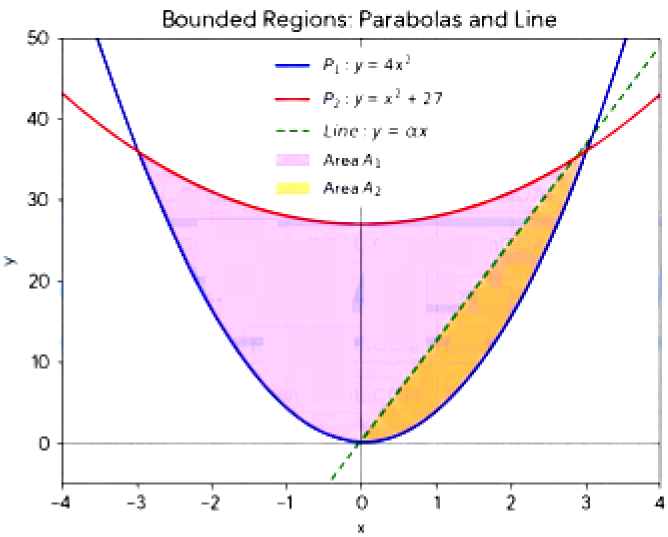
8

D.

6

Answer: A

Solution:



1. Calculate the Area A_1 between P_1 and P_2

The two parabolas are given by:

- $P_1 : y = 4x^2$
- $P_2 : y = x^2 + 27$

Find the intersection points: Setting the equations equal to each other:

$4x^2 = x^2 + 27$

$\Rightarrow 3x^2 = 27$

$\Rightarrow x^2 = 9 \Rightarrow x = \pm 3$

Find the intersection points : Setting the equations equal to each other :

$4x^2 = x^2 + 27$

$3x^2 = 27 \quad , \quad x^2 = 9 \Rightarrow x = \pm 3$

Calculate the area: On the interval $[-3, 3]$, the parabola P_2 lies above P_1. The area A_1 is:

$$A_1 = \int_{-3}^{3} (x^2 + 27 - 4x^2)\,dx = \int_{-3}^{3} (27 - 3x^2)\,dx$$
$$A_1 = \left[27x - x^3\right]_{-3}^{3}$$
$$A_1 = \left(27(3) - 3^3\right) - \left(27(-3) - (-3)^3\right)$$
$$A_1 = (81 - 27) - (-81 + 27) = 54 - (-54) = 108$$

Calculate the Area A_2 between P_1 and the line $y = \alpha x$

Find the intersection points:

$4x^2 = \alpha x \implies x(4x - \alpha) = 0$

The points are $x = 0$ and $x = \frac{\alpha}{4}$.

Calculate the area: On the interval $\left[0, \frac{\alpha}{4}\right]$, the line $y = \alpha x$ lies above $y = 4x^2$. The area A_2 is:

$$A_2 = \int_{0}^{\alpha/4} (\alpha x - 4x^2)\,dx$$
$$A_2 = \left[\frac{\alpha x^2}{2} - \frac{4x^3}{3}\right]_{0}^{\alpha/4}$$
$$A_2 = \frac{\alpha}{2}\left(\frac{\alpha}{4}\right)^2 - \frac{4}{3}\left(\frac{\alpha}{4}\right)^3$$
$$A_2 = \frac{\alpha^3}{32} - \frac{4\alpha^3}{192} = \frac{\alpha^3}{32} - \frac{\alpha^3}{48} = \frac{3\alpha^3 - 2\alpha^3}{96} = \frac{\alpha^3}{96}$$

The problem states that A_1 is six times A_2 :

$$A_1 = 6 \times A_2$$
$$108 = 6 \times \left(\frac{\alpha^3}{96}\right)$$
$$108 = \frac{\alpha^3}{16}$$
$$\alpha^3 = 108 \times 16 = 1728$$
$$\alpha = \sqrt[3]{1728} = 12$$

Question13

The area of the region $\{(x, y) : y \leq \pi - |x|, y \leq |x \sin x|, y \geq 0\}$ is:

JEE Main 2026 (Online) 4th April Morning Shift

Options:

A. $1 + \frac{\pi^2}{8}$

B.

$2+\frac{\pi^2}{4}$

C.

$\frac{\pi^2}{8}-1$

D.

$4+\frac{\pi^2}{2}$

Answer: B

Solution:

We need the area of the region

$\{(x,y): y \le \pi - |x|,\ y \le |x\sin x|,\ y \ge 0\}.$

This means the required area is the area under the curve

$y = \min(\pi - |x|,\ |x\sin x|)$

and above the x-axis.

So the area is

$A = \int \min(\pi - |x|,\ |x\sin x|)\, dx$

over the values of x where the region exists.

Since $y \ge 0$ and $y \le \pi - |x|$, we must have

$\pi - |x| \ge 0 \implies |x| \le \pi.$

Hence the region lies in $[-\pi, \pi]$.

Also, both functions $\pi - |x|$ and $|x\sin x|$ are even functions, so the region is symmetric about the y-axis. Therefore,

$A = 2\int_0^{\pi} \min(\pi - x,\ x\sin x)\, dx.$

Now for $0 \le x \le \pi$, we have $\sin x \ge 0$, so

$|x\sin x| = x\sin x.$

Thus,

$A = 2\int_0^{\pi} \min(\pi - x,\ x\sin x)\, dx.$

Now compare $\pi - x$ and $x\sin x$ on $[0, \pi]$.

At $x = 0$,

$\pi - x = \pi, \qquad x \sin x = 0.$

At $x = \pi$,

$\pi - x = 0, \qquad x \sin x = 0.$

Check where they meet:

$x \sin x = \pi - x.$

It is easy to see that $x = \dfrac{\pi}{2}$ satisfies this, because

$\frac{\pi}{2} \sin \frac{\pi}{2} = \frac{\pi}{2} = \pi - \frac{\pi}{2}.$

Also, on $[0, \pi/2]$, we have $x \sin x \leq \pi - x$, and on $[\pi/2, \pi]$, we have $\pi - x \leq x \sin x$.

So,

$A = 2\left[\int_0^{\pi/2} x \sin x \, dx + \int_{\pi/2}^{\pi} (\pi - x) \, dx\right].$

Now evaluate each integral.

First,

$\int x \sin x \, dx = -x \cos x + \int \cos x \, dx = -x \cos x + \sin x.$

Therefore,

$\int_0^{\pi/2} x \sin x \, dx = [-x \cos x + \sin x]_0^{\pi/2} = 1.$

Next,

$\int_{\pi/2}^{\pi} (\pi - x) \, dx.$

Let $u = \pi - x$. Then as x goes from $\pi/2$ to π, u goes from $\pi/2$ to 0. Or directly,

$\int_{\pi/2}^{\pi} (\pi - x) \, dx = \left[\pi x - \frac{x^2}{2}\right]_{\pi/2}^{\pi}.$

Compute:

At $x = \pi$,

$\pi^2 - \frac{\pi^2}{2} = \frac{\pi^2}{2}.$

At $x = \frac{\pi}{2}$,

$\pi \cdot \frac{\pi}{2} - \frac{1}{2}\left(\frac{\pi}{2}\right)^2 = \frac{\pi^2}{2} - \frac{\pi^2}{8} = \frac{3\pi^2}{8}.$

So,

$\int_{\pi/2}^{\pi} (\pi - x) \, dx = \frac{\pi^2}{2} - \frac{3\pi^2}{8} = \frac{\pi^2}{8}.$

Hence,

$A = 2\left(1 + \frac{\pi^2}{8}\right) = 2 + \frac{\pi^2}{4}.$

Therefore, the correct answer is

$$\boxed{2+\frac{\pi^2}{4}}$$

So, Option B is correct.

Question14

The area of the region bounded by the curves $x+3y^2=0$ and $x+4y^2=1$ is equal to :

JEE Main 2026 (Online) 4th April Evening Shift

Options:

A.

$\frac{1}{3}$

B.

$\frac{2}{3}$

C.

$\frac{4}{3}$

D.

$\frac{5}{3}$

Answer: C

Solution:

Given curves are

$x+3y^2=0 \quad \text{and} \quad x+4y^2=1$

First, write them in the form $x=$:

$x=-3y^2$

and

$x = 1 - 4y^2$

To find the bounded area, we first find their points of intersection.

Set the two values of x equal:

$-3y^2 = 1 - 4y^2$

$y^2 = 1$

$y = \pm 1$

So the curves intersect at $y = -1$ and $y = 1$.

Now, for a fixed value of y, the right curve is

$x = 1 - 4y^2$

and the left curve is

$x = -3y^2$

So the horizontal width of the region is

$(1 - 4y^2) - (-3y^2) = 1 - y^2$

Hence, area of the enclosed region is

$A = \int_{-1}^{1} (1 - y^2)\, dy$

Now evaluate:

$A = \left[y - \frac{y^3}{3}\right]_{-1}^{1}$

$A = \left(1 - \frac{1}{3}\right) - \left(-1 + \frac{1}{3}\right)$

$A = \frac{2}{3} - \left(-\frac{2}{3}\right)$

$A = \frac{4}{3}$

Therefore, the required area is

$\boxed{\frac{4}{3}}$

So, the correct option is:

$\boxed{\text{Option C}}$

Question15

The area of the region $\mathrm{R}=\left\{(x, y): x y \leq 27,1 \leq y \leq x^2\right\}$ is equal to :

JEE Main 2026 (Online) 5th April Morning Shift

Options:

A.

$78 \log_e 3-\frac{52}{3}$

B.

$54 \log_e 3-\frac{52}{3}$

C.

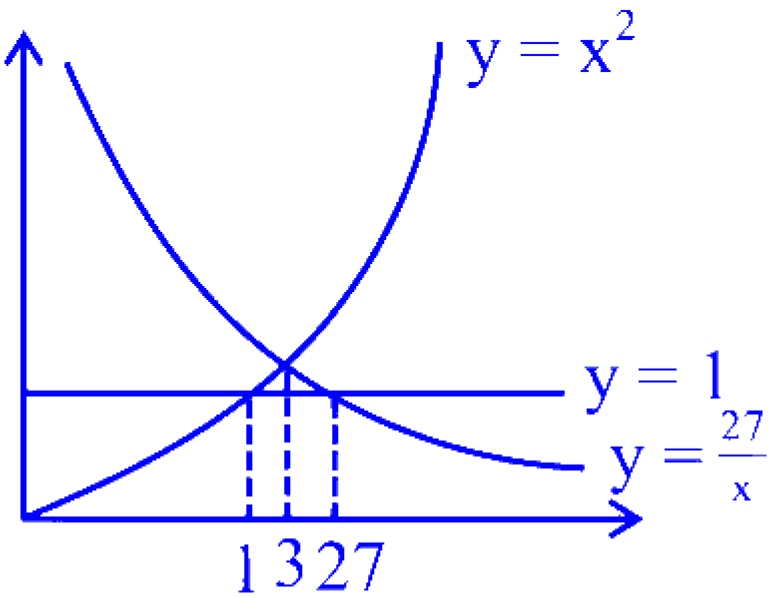
$54 \log_e 3-\frac{26}{3}$

D.

$54 \log_e 3+\frac{26}{3}$

Answer: B

Solution:



$$\text{Area } = \int_1^3 (x^2-1)\,dx + \int_3^{27}\left(\frac{27}{x}-1\right)dx$$
$$= \left[\frac{x^3}{3}-x\right]_1^3 + [27\ell nx - x]_3^{27}$$
$$= \frac{26}{3} - 2 + 27\ln 9 - 24$$
$$= 27\ln 9 - \frac{26\times 2}{3}$$
$$= 54\ln 3 - \frac{52}{3}$$

Question16

Let e be the base of natural logarithm and let $f:\{1,2,3,4\}\rightarrow\left\{1,e,e^2,e^3\right\}$ and g : $\left\{1,e,e^2,e^3\right\}\rightarrow\left\{1,\frac{1}{2},\frac{1}{3},\frac{1}{4}\right\}$ be two bijective functions such that f is strictly decreasing and g is strictly increasing. If $\phi(x)=\left[f^{-1}\left\{g^{-1}\left(\frac{1}{2}\right)\right\}\right]^x$, then the area of the region $\mathrm{R}=\left\{(x,y): x^2\leq y\leq\phi(x), 0\leq x\leq 1\right\}$ is :

JEE Main 2026 (Online) 6th April Morning Shift

Options:

A. $\frac{3-\log_e(2)}{3\log_e(2)}$

B.

$\frac{1}{3\log_e(2)}$

C.

$3+\log_e(2)$

D.

$\frac{3+\log_e(2)}{2+\log_e(3)}$

Answer: A

Solution:

Since $f:\{1,2,3,4\}\to\{1,e,e^2,e^3\}$ is bijective and strictly decreasing, we match the largest element of the domain to the smallest element of the codomain.

So,

$f(1)=e^3,\quad f(2)=e^2,\quad f(3)=e,\quad f(4)=1$

Hence,

$f^{-1}(1)=4,\quad f^{-1}(e)=3,\quad f^{-1}(e^2)=2,\quad f^{-1}(e^3)=1$

Now $g:\{1,e,e^2,e^3\}\to\left\{1,\frac{1}{2},\frac{1}{3},\frac{1}{4}\right\}$ is bijective and strictly increasing.

Since

$$ 1

and

$\frac{1}{4}<\frac{1}{3}<\frac{1}{2}<1$

for g to be strictly increasing, we must assign:

$g(1)=\frac{1}{4},\quad g(e)=\frac{1}{3},\quad g(e^2)=\frac{1}{2},\quad g(e^3)=1$

Therefore,

$g^{-1}\left(\frac{1}{2}\right)=e^2$

Now,

$f^{-1}\left(g^{-1}\left(\frac{1}{2}\right)\right)=f^{-1}(e^2)=2$

So,

$\phi(x)=\left[f^{-1}\left\{g^{-1}\left(\frac{1}{2}\right)\right\}\right]^x=2^x$

Thus the region is

$R=\{(x,y):x^2\le y\le 2^x,\ 0\le x\le 1\}$

Its area is

$\int_0^1(2^x-x^2)\,dx$

Now,

$\int 2^x\,dx=\frac{2^x}{\ln 2}$

So,

$\int_0^1 2^x\,dx=\left[\frac{2^x}{\ln 2}\right]_0^1=\frac{2-1}{\ln 2}=\frac{1}{\ln 2}$

Also,

$\int_0^1 x^2\,dx=\left[\frac{x^3}{3}\right]_0^1=\frac{1}{3}$

Hence area

$= \frac{1}{\ln 2} - \frac{1}{3}$

Taking LCM,

$= \frac{3-\ln 2}{3\ln 2}$

Therefore, the required area is

$\boxed{\frac{3-\log_e 2}{3\log_e 2}}$

So the correct option is $\boxed{\text{A}}$.

Question17

The area of the region $\{(x, y) : 0 \leq y \leq 6 - x, y^2 \geq 4x - 3, x \geq 0\}$ is :

JEE Main 2026 (Online) 6th April Morning Shift

Options:

A.

8

B.

9

C.

12

D.

15

Answer: B

Solution:

We need to find the area of the region

$\{(x,y) : 0 \le y \le 6 - x,\ y^2 \ge 4x - 3,\ x \ge 0\}.$

Let us understand each condition carefully.

1. From $0 \le y \le 6 - x$

This means:

- $y \ge 0$, so the region lies above the x-axis.
- $y \le 6 - x$, or

$x + y \le 6,$

so the region lies below the line

$y = 6 - x.$

2. From $y^2 \ge 4x - 3$

Rewrite it in terms of x:

$4x - 3 \le y^2$

$4x \le y^2 + 3$

$x \le \frac{y^2+3}{4}.$

So the region lies to the left of the parabola

$x = \frac{y^2+3}{4}.$

Also, we are given

$x \ge 0.$

So for a fixed y, x lies between 0 and whichever is smaller:

$x = \frac{y^2+3}{4} \quad \text{and} \quad x = 6 - y.$

So the horizontal width is

$0 \le x \le \min\left(\frac{y^2+3}{4}, 6 - y\right).$

Now we find where these two curves intersect.

3. Point of intersection

Set

$\frac{y^2+3}{4} = 6 - y.$

Multiply by 4:

$y^2 + 3 = 24 - 4y$

$y^2 + 4y - 21 = 0$

$(y+7)(y-3)=0.$

So,

$y=3 \quad \text{or} \quad y=-7.$

Since $y \geq 0$, we take

$y=3.$

Now compare the two right boundaries:

- For $0 \leq y \leq 3$, we have

$\frac{y^2+3}{4} \leq 6-y,$

so width is

$\frac{y^2+3}{4}.$

- For $3 \leq y \leq 6$, we have

$6-y \leq \frac{y^2+3}{4},$

so width is

$6-y.$

Hence area is

$A=\int_0^3 \frac{y^2+3}{4}\,dy+\int_3^6 (6-y)\,dy.$

4. Evaluate the integrals

First integral:

$\int_0^3 \frac{y^2+3}{4}\,dy=\frac{1}{4}\int_0^3 (y^2+3)\,dy$

$=\frac{1}{4}\left[\frac{y^3}{3}+3y\right]_0^3=\frac{1}{4}\left(\frac{27}{3}+9\right)=\frac{1}{4}(9+9)=\frac{18}{4}=\frac{9}{2}.$

Second integral:

$\int_3^6 (6-y)\,dy=\left[6y-\frac{y^2}{2}\right]_3^6$

$=(36-18)-\left(18-\frac{9}{2}\right)=18-\frac{27}{2}=\frac{9}{2}.$

So total area is

$A=\frac{9}{2}+\frac{9}{2}=9.$

Therefore, the correct answer is:

$\boxed{9}$

So, **Option B** is correct.

Question18

The area of the region $\left\{(x, y) : x^2 - 8x \leq y \leq -x\right\}$ is :

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Options:

A.

$\frac{343}{6}$

B.

$\frac{637}{6}$

C.

$\frac{437}{6}$

D.

$\frac{523}{6}$

Answer: A

Solution:

For the region

$\{(x, y) : x^2 - 8x \leq y \leq -x\},$

the lower curve is

$y = x^2 - 8x$

and the upper curve is

$y = -x.$

To find the area between the curves, we first find their points of intersection.

Set

$x^2 - 8x = -x$

$x^2 - 7x = 0$

$x(x-7) = 0$

So, the curves intersect at

$x = 0 \quad \text{and} \quad x = 7.$

Now, on the interval $[0, 7]$, the line $y = -x$ lies above the parabola $y = x^2 - 8x$.

Hence, the required area is

Area $= \int_0^7 \left[(-x) - (x^2 - 8x)\right] dx.$

Simplify the integrand:

$(-x) - (x^2 - 8x) = -x - x^2 + 8x = 7x - x^2.$

Therefore,

Area $= \int_0^7 (7x - x^2)\, dx.$

Now integrate:

$\int (7x - x^2)\, dx = \frac{7x^2}{2} - \frac{x^3}{3}.$

Applying the limits from 0 to 7:

Area $= \left[\frac{7x^2}{2} - \frac{x^3}{3}\right]_0^7$

$= \left(\frac{7\cdot 49}{2} - \frac{343}{3}\right) - 0$

$= \frac{343}{2} - \frac{343}{3}$

Taking LCM 6:

$= \frac{1029-686}{6} = \frac{343}{6}.$

Hence, the correct answer is

$$\boxed{\frac{343}{6}}$$

So, the correct option is **A**.

Question19

Let $f : (1, \infty) \to \mathbf{R}$ be a function defined as $f(x) = \frac{x-1}{x+1}$. Let $f^{i+1}(x) = f\left(f^i(x)\right), i = 1, 2, \ldots, 25$, where $f^1(x) = f(x)$. If $g(x) + f^{26}(x) = 0, x \in (1, \infty)$, then the area of the region bounded by the curves $y = g(x), 2y = 2x - 3, y = 0$ and $x = 4$ is :

JEE Main 2026 (Online) 8th April Evening Shift

Options:

A.

$\frac{1}{8}+\log_e 2$

B.

$\frac{1}{4}+\log_e 2$

C.

$\frac{5}{6}+3\log_e 2$

D.

$\frac{5}{6}+\log_e 2$

Answer: A

Solution:

We first focus on the orbit periodicity of $f^n(\boldsymbol{x})$:

$$f^1(x)=f(x)=\frac{x-1}{x+1}; x\in(1,\infty)\ldots\ldots(i)$$

$$f^2(x)=f\left\{f^1(x)\right\}=f\{f(x)\}=f\left(\frac{x-1}{x+1}\right)=\frac{\left(\frac{x-1}{x+1}\right)-1}{\left(\frac{x-1}{x+1}\right)+1}=-\frac{2}{2x}=-\frac{1}{x}; x\in(1,\infty)\ldots\ldots(ii)$$

$$\boldsymbol{f}^3(\boldsymbol{x})=f\left\{\boldsymbol{f}^2(\boldsymbol{x})\right\}=f\left(-\frac{1}{\boldsymbol{x}}\right)=\frac{\left(-\frac{1}{\boldsymbol{x}}\right)-1}{\left(-\frac{1}{\boldsymbol{x}}\right)+1}=\frac{1+\boldsymbol{x}}{1-\boldsymbol{x}}; \boldsymbol{x}\in(1,\infty)\ldots\ldots(\boldsymbol{iii})$$

$$f^4(x)=f\left\{f^3(x)\right\}=f\left(\frac{1+x}{1-x}\right)=\frac{\left(\frac{1+x}{1-x}\right)-1}{\left(\frac{1+x}{1-x}\right)+1}=\frac{2x}{2}=x; x\in(1,\infty)\ldots\ldots(iv)$$

$$f^5(x)=f\left\{f^4(x)\right\}=f(x)=\frac{x-1}{x+1}; x\in(1,\infty)\ldots\ldots(v)$$

$f(x)$	$f^1(x)$	$f^2(x)$	$f^3(x)$	$f^4(x)$	$f^5(x)$	$f^6(x)$	$f^7(x)$	$f^8(x)$	$f^9(x)$	.....	$f^{13}(x)$	.....	$f^{17}(x)$	.....	$f^{21}(x)$	.....	$f^{25}(x)$	n	Orbit periodicity	Cycle
$\frac{x-1}{x+1}$	$\frac{x-1}{x+1}$	$-\frac{1}{x}$	$\frac{1+x}{1-x}$	x	$\frac{x-1}{x+1}$	$-\frac{1}{x}$	$\frac{1+x}{1-x}$	x	$\frac{x-1}{x+1}$	.....	$\frac{x-1}{x+1}$	.....	$\frac{x-1}{x+1}$	.....	$\frac{x-1}{x+1}$	.....	$\frac{x-1}{x+1}$	25	4	6

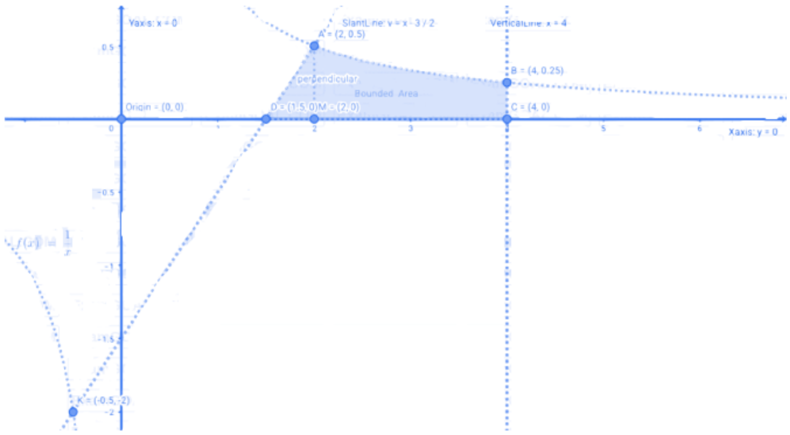
From the nature of table which completes the 25 iterations in 6 cycles with orbit periodicity 4 :

$$f^{25}(x)=f(x)=f^1(x)\Rightarrow f^{26}(x)=f^2(x)=-\frac{1}{x}\ldots\ldots(vi)$$

Equation (vi) exactly identifies the function $\boldsymbol{g}(\boldsymbol{x})$ defined under SOW :

$$\boldsymbol{g}(\boldsymbol{x})+\boldsymbol{f}^{26}(\boldsymbol{x})=0\Rightarrow\boldsymbol{g}(\boldsymbol{x})=-\boldsymbol{f}^{26}(\boldsymbol{x})=\frac{1}{\boldsymbol{x}}\ldots\ldots(vii)$$

Equation (vii) has identified the function $\boldsymbol{g(x)}$ and we can mathematically plot the bounded area between the curves :



$g(x)$	Slant line	Horizontal line	Vertical line
$\frac{1}{x}$	$y = x - \left(\frac{3}{2}\right)$	X-axis	$x = 4$

Pair	$\boldsymbol{g(x)}$	Slant line	Feasible	Intersection Points
1	$\boldsymbol{\frac{1}{x}}$	$\boldsymbol{y = x - \left(\frac{3}{2}\right)}$	Yes	$\boldsymbol{A, K}$

$\Rightarrow \frac{1}{x} = x - \left(\frac{3}{2}\right) \Rightarrow 2x^2 - 3x - 2 = 0 \Rightarrow x = \left(2, -\frac{1}{2}\right) \Rightarrow y = \left(\frac{1}{2}, -2\right)$

Intersections	$\boldsymbol{A}$	$\boldsymbol{K}$	Discarded point	Reason
Coordinates	$\left(2, \frac{1}{2}\right)$	$\left(-\frac{1}{2}, -2\right)$	$\boldsymbol{K}$	Falls outside the scope of bounded area

Pair	$\boldsymbol{g(x)}$	Vertical line	Feasible	Intersection Points
2	$\boldsymbol{\frac{1}{x}}$	$\boldsymbol{x} = 4$	Yes	$\boldsymbol{B}$

$\Rightarrow x = 4 \Rightarrow y = 0.25$

Intersection	$\boldsymbol{B}$
Coordinate	$(4, 0.25)$

Pair	Horizontal line	Vertical line	Feasible	Intersection point
3	$\boldsymbol{X}$-axis	$\boldsymbol{x} = 4$	Yes	$\boldsymbol{C}$

$\Rightarrow x = 4, y = 0$

Intersection	$\boldsymbol{C}$
Coordinate	$(4, 0)$

Pair	Slant line	Horizontal line	Feasible	Intersection point
4	$\boldsymbol{y} = x - \left(\frac{3}{2}\right)$	X-axis	Yes	D

$\Rightarrow \boldsymbol{y} = 0 \Rightarrow \boldsymbol{x} = \frac{3}{2}$

Intersection	$\boldsymbol{D}$
Coordinate	$\left(\frac{3}{2}, 0\right)$

g(x)	X-axis	Feasible	Intersection point	Reason
1/x	y = 0	No	None	X-axis is a horizontal asymptote to g(x)
Slant line	Vertical line	Feasible	Intersection point	Reason
y = x − (3/2)	x = 4	No	(4, 5/2)	Intersects much above B and outside the scope of bounded area

Sides	Left boundary	Equation	Top boundary	Equation	Right boundary	Equation	Bottom boundary	Equation
Curves	Slant line	y = x − (3/2)	Rectangular Hyperbola	y = 1/x	Vertical line	x = 4	X-axis	y = 0
Vertices	Top left	Coordinates	Top right	Coordinates	Bottom right	Coordinates	Bottom left	Coordinates
Points	A	A (2, 1/2)	B	B (4, 1/4)	C	C (4, 0)	D	D (3/2, 0)

We can visualize that both $\boldsymbol{A}$ and $\boldsymbol{D}$ lie on the slant line but $\boldsymbol{A}$ is ahead of $\boldsymbol{D}$ along the $\boldsymbol{X}$-axis. From $\boldsymbol{A}$ we can drop a $\perp$ on the $\boldsymbol{X}$-axis at $\boldsymbol{M}$ to form a right angled $\boldsymbol{\Delta AMD}$. From $\boldsymbol{A}$ and $\boldsymbol{D}$ the remaining area upto the vertical $\boldsymbol{x} = 4$ lies under the rectangular hyperbola $g(x) = \frac{1}{x}$:

$$\begin{aligned}\text{Area } &= \left(\frac{1}{2}\right) \cdot (\boldsymbol{DM}) \cdot (\boldsymbol{AM}) + \int_2^4 \boldsymbol{g}(\boldsymbol{x}) \boldsymbol{dx} = \left(\frac{1}{2}\right) \cdot (2 - 1 \cdot 5) \cdot \left(\frac{1}{2}\right) + \int_2^4 \left(\frac{1}{x}\right) \boldsymbol{dx} \\ &= \frac{1}{8} + [\ln |\boldsymbol{x}|]_2^4 \\ &= \frac{1}{8} + \ln 2\end{aligned}$$

Question20

The area of the region, inside the circle $(x - 2\sqrt{3})^2 + y^2 = 12$ and outside the parabola $y^2 = 2\sqrt{3}x$ is :

JEE Main 2025 (Online) 22nd January Morning Shift

Options:

A. $3\pi - 8$

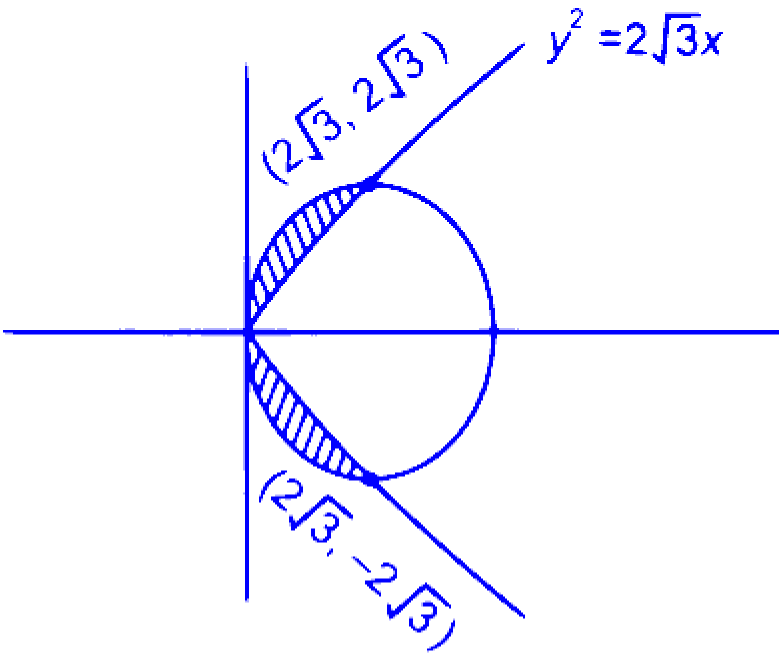
B. $6\pi - 8$

C. $3\pi + 8$

D. $6\pi - 16$

Answer: D

Solution:



Required area $= 2\int_0^{2\sqrt{3}} \left(\sqrt{4\sqrt{3}x - x^2} - \sqrt{2\sqrt{3}x}\right) dx$

$= 2\int_0^{2\sqrt{3}} \left(\sqrt{12 - (x - 2\sqrt{3})^2} - \sqrt{2\sqrt{3}x}\right) dx$

$= 2\left[\frac{x-2\sqrt{3}}{2}\sqrt{12 - (x-2\sqrt{3})^2} + \frac{12}{2}\sin^{-1}\left(\frac{x-2\sqrt{3}}{2\sqrt{3}}\right) - \frac{\sqrt{2\sqrt{3}}x^{\frac{3}{2}}}{\frac{3}{2}}\right]_0^{2\sqrt{3}}$

$= 2\{3\pi - 8\}$

$= 6\pi - 16$ sq. units.

Question21

The area of the region enclosed by the curves $y = x^2 - 4x + 4$ and $y^2 = 16 - 8x$ is :

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Options:

A. $\frac{8}{3}$

B. 5

C. 8

D. $\frac{4}{3}$

Answer: A

Solution:

Consider the curves

$y = x^2 - 4x + 4 = (x-2)^2$

and

$y^2 = 16 - 8x.$

Notice that the second equation can be rewritten in terms of x:

$8x = 16 - y^2 \quad \implies \quad x = 2 - \frac{y^2}{8}.$

Step 1. Find the Intersection Points

To find the points where the curves intersect, substitute

$y = (x-2)^2$

into

$y^2 = 16 - 8x.$

Let

$u = x - 2 \quad \text{so that} \quad y = u^2.$

Then

$y^2 = u^4$

and

$x = u + 2.$

Substitute into the second curve:

$u^4 = 16 - 8(u+2).$

Simplify the right side:

$16 - 8(u+2) = 16 - 8u - 16 = -8u.$

Thus, the equation becomes

$u^4 + 8u = 0 \quad \implies \quad u(u^4/u? **correctingfactor**).$

In fact, factor by taking out a common factor u:

$u(u^3+8)=0.$

Thus, either

$u=0 \quad \text{or} \quad u^3=-8.$

For $u=0$:

$x=u+2=2, \quad y=u^2=0.$

For $u^3=-8$:

$u=-2, \quad \text{so} \quad x=-2+2=0, \quad y=(-2)^2=4.$

The curves intersect at the points $(2,0)$ and $(0,4)$.

Step 2. Express the Curves in Terms of y

It is easier to integrate horizontally by expressing x as a function of y.

From the second curve:

$x=2-\frac{y^2}{8}.$

From the first curve, solving

$y=(x-2)^2$

for x gives

$x-2=\pm\sqrt{y}.$

Since at the intersection $(0,4)$ the x–value is less than 2, we take the negative branch:

$x=2-\sqrt{y}.$

Thus, for a fixed y between 0 and 4, the left boundary is

$x_{\text{left}}=2-\sqrt{y},$

and the right boundary is

$x_{\text{right}}=2-\frac{y^2}{8}.$

Step 3. Set Up the Integral for the Area

The horizontal distance between the curves at a given y is

$\Delta x=x_{\text{right}}-x_{\text{left}}=\left(2-\frac{y^2}{8}\right)-\left(2-\sqrt{y}\right)=\sqrt{y}-\frac{y^2}{8}.$

Integrate with respect to y from $y=0$ to $y=4$:

$A=\int_0^4\left(\sqrt{y}-\frac{y^2}{8}\right)dy.$

Step 4. Evaluate the Integral

Write the integral as

$A = \int_0^4 y^{1/2}\,dy - \frac{1}{8}\int_0^4 y^2\,dy.$

Compute each term:

For the first integral:

$\int y^{1/2}\,dy = \frac{2}{3}y^{3/2}.$

For the second integral:

$\int y^2\,dy = \frac{y^3}{3}.$

Thus,

$A = \left[\frac{2}{3}y^{3/2}\right]_0^4 - \frac{1}{8}\left[\frac{y^3}{3}\right]_0^4.$

Substitute $y = 4$ (note that at $y = 0$ both terms vanish):

Compute $4^{3/2}$:

$4^{3/2} = \left(\sqrt{4}\right)^3 = 2^3 = 8.$

Compute the first term:

$\frac{2}{3} \times 8 = \frac{16}{3}.$

Compute the second term:

$\frac{1}{8} \cdot \frac{64}{3} = \frac{64}{24} = \frac{8}{3}.$

Therefore, the area is

$A = \frac{16}{3} - \frac{8}{3} = \frac{8}{3}.$

Final Answer

The area of the region enclosed by the curves is

$\frac{8}{3}.$

Question22

If the area of the region $\left\{(x, y) : -1 \leq x \leq 1, 0 \leq y \leq \mathrm{a} + \mathrm{e}^{|x|} - \mathrm{e}^{-x}, \mathrm{a} > 0\right\}$ is $\frac{\mathrm{e}^2+8\mathrm{e}+1}{\mathrm{e}}$, then the value of a is :

JEE Main 2025 (Online) 23rd January Evening Shift

Options:

A. 7

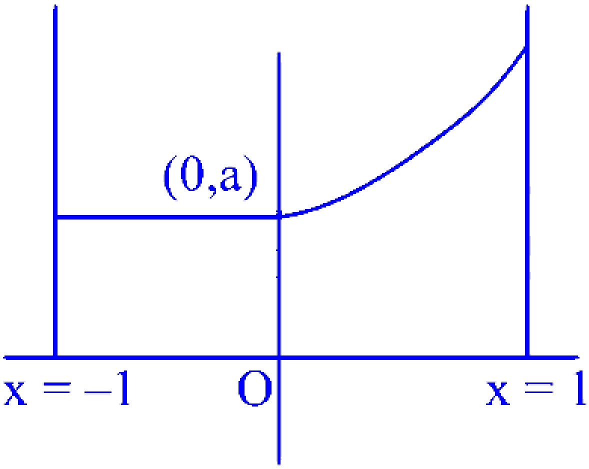
B. 5

C. 6

D. 8

Answer: B

Solution:



required area is $a + \int_0^1 \left(a + e^x - e^{-x}\right) dx$

$a + \left[a + e^x + e^{-x}\right]_0^1$

$2a + e - 1 + e^{-1} - 1 = e + 8 + \frac{1}{e}$

$2a = 10 \Rightarrow a = 5$

Question23

The area of the region $\left\{(x, y) : x^2 + 4x + 2 \leq y \leq |x + 2|\right\}$ is equal to

JEE Main 2025 (Online) 24th January Morning Shift

Options:

A. 7

B. 24/5

C. 20/3

D. 5

Answer: C

Solution:

$x^2 + 4x + 2 \leq y \leq |x+2|$

The area bounded between

$y = x^2 + 4x + 2 = (x+2)^2 - 2$

and $y = |x+2|$ is same as area bounded between $y = x^2 - 2$ and $y = |x|$

For P.O.I $|x|^2 - 2 = |x|$

$\Rightarrow |x| = 2 \Rightarrow x = \pm 2$

$$\therefore \text{ Required area } = -\int_{-2}^{2} (x^2 - 2) dx + \int_{-2}^{2} |x| dx$$

$$= -2 \int_{0}^{2} (x^2 - 2) dx + 2 \int_{0}^{2} x \cdot dx$$

$$= -2 \left[\frac{x^3}{3} - 2x \right]_0^2 + 2 \left[\frac{x^2}{2} \right]_0^2$$

$$= - - 2 \left[\frac{8}{3} - 4 \right] + 2 \left[\frac{4}{2} \right]$$

$$= -2 \times \left(\frac{-4}{3} \right) + 4$$

$$= \frac{20}{3}$$

Question24

The area of the region enclosed by the curves $y = \mathrm{e}^x, y = |\mathrm{e}^x - 1|$ and y-axis is :

JEE Main 2025 (Online) 24th January Evening Shift

Options:

A. $1+\log_e 2$

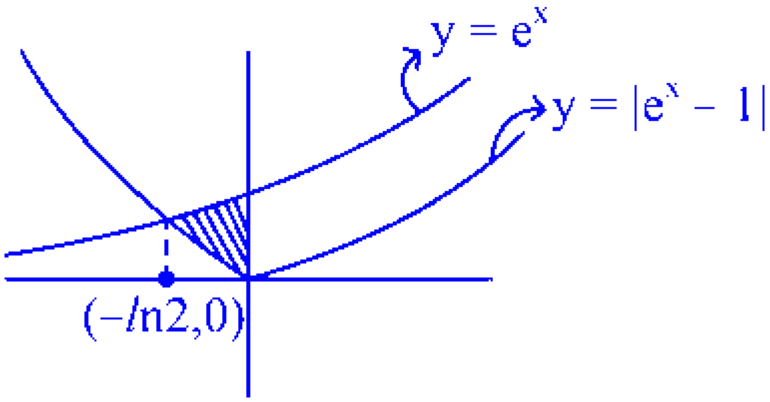
B. $\log_e 2$

C. $1-\log_e 2$

D. $2\log_e 2-1$

Answer: C

Solution:



For Area $\int_{-\ln 2}^{0}[e^x-(1-e^x)]dx$

$\int_{-\mathrm{nn}\,2}^{0}(2\mathrm{e}^{\mathrm{x}}-1)\mathrm{dx}=[2\mathrm{e}^{\mathrm{x}}-\mathrm{x}]_{-\ln 2}^{0}$

$=(2-(1+\ell\mathrm{n}2))$

$=1-\ell\mathrm{n}2$

Question25

The area (in sq. units) of the region $\{(x,\mathrm{y}):0\leq \mathrm{y}\leq 2|x|+1,0\leq \mathrm{y}\leq x^2+1,|x|\leq 3\}$ is

JEE Main 2025 (Online) 28th January Morning Shift

Options:

A. $\frac{32}{3}$

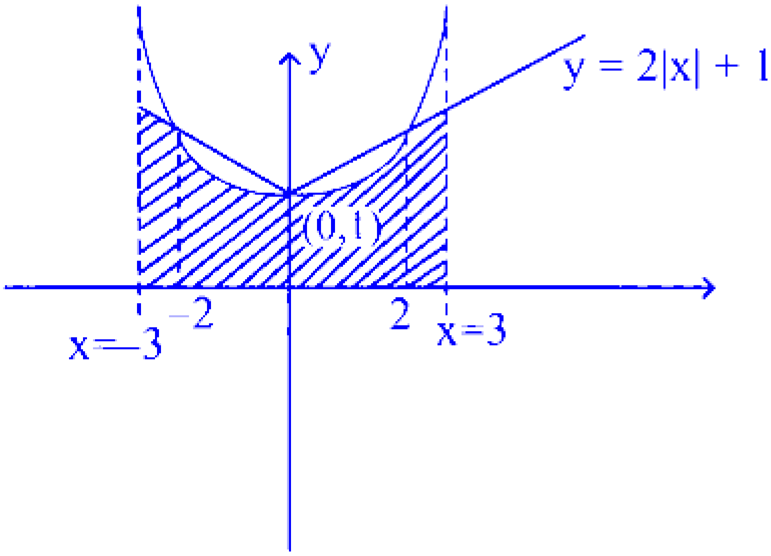
B. $\frac{64}{3}$

C. $\frac{17}{3}$

D. $\frac{80}{3}$

Answer: B

Solution:



$$\text{Area } = 2\left[\int_0^2 (x^2+1)dx + \int_2^3 (2x+1)dx\right]$$

$$\Rightarrow \frac{64}{3} \quad \therefore (2)$$

Question26

The area of the region bounded by the curves $x(1+y^2)=1$ and $y^2=2x$ is:

JEE Main 2025 (Online) 28th January Evening Shift

Options:

A.

$\frac{\pi}{4}-\frac{1}{3}$

B.

$\frac{\pi}{2}-\frac{1}{3}$

C.

$2\left(\frac{\pi}{2}-\frac{1}{3}\right)$

D.

$\frac{1}{2}\left(\frac{\pi}{2}-\frac{1}{3}\right)$

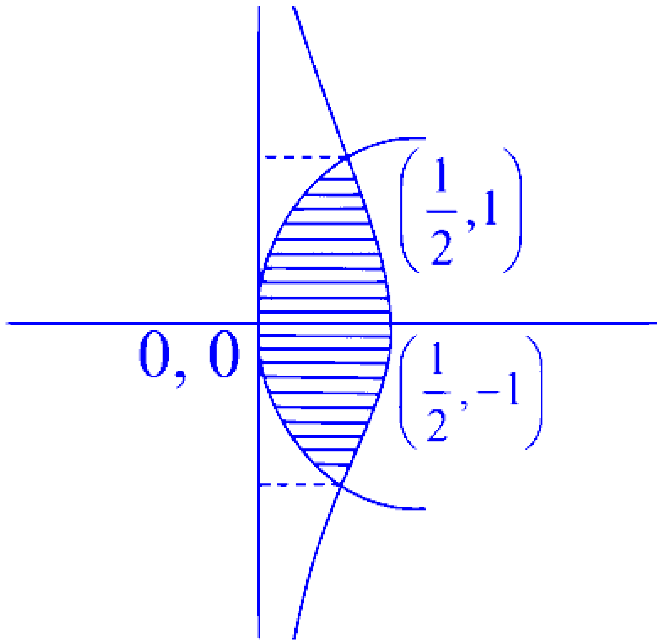
Answer: B

Solution:

$x\left(1+y^2\right)=1 \quad ..... (1)$
$y^2=2x \quad ..... (2)$

From equation (1) & (2)

$$
\begin{aligned}
x(1+2x)=1 &\Rightarrow 2x^2+x-1=0 \\
&\Rightarrow x=\frac{1}{2}, x=-1 \text{ (Reject)} \\
&\Rightarrow y^2=2\left(\frac{1}{2}\right) \\
&\Rightarrow y=\pm 1
\end{aligned}
$$



$$\begin{aligned} \text{Area bounded } &= \int_{-1}^{1}\left(\frac{1}{1+y^2}-\frac{y^2}{2}\right)dy \\ &= \left(\tan^{-1} y-\frac{y^3}{6}\right)\Bigg|_{-1}^{1} \\ &= \frac{\pi}{2}-\frac{1}{3} \end{aligned}$$

Question27

Let the area of the region

$(x, y): 2y \leq x^2+3,\ y+|x| \leq 3,\ y \geq |x-1|$ be A. Then $6A$ is equal to :

JEE Main 2025 (Online) 29th January Morning Shift

Options:

A.

14

B.

18

C.

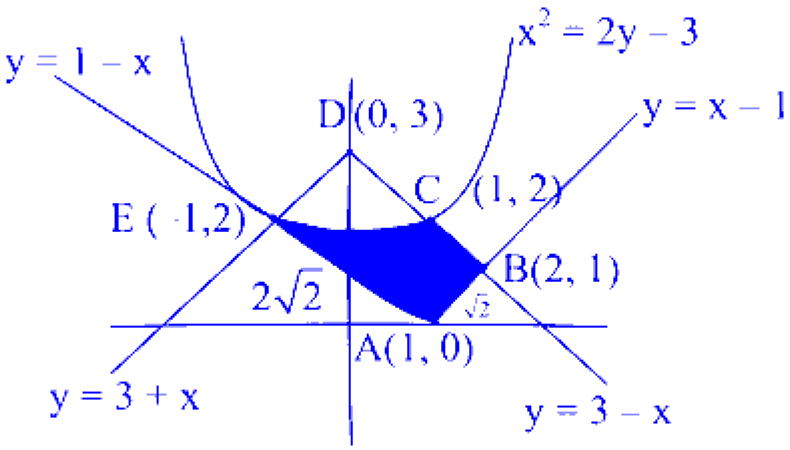
16

D.

12

Answer: A

Solution:



$A \Rightarrow$ Rectangle ABDE $-$ Area of region EDC

$$A \Rightarrow 4-2\int_0^1 (3-x)-\left(\frac{x^2+3}{2}\right)dx$$

$$A \Rightarrow 4-2\left\{3x-\frac{x^2}{2}-\frac{x^3}{6}-\frac{3}{2}x\right\}_0^1$$

$$A \Rightarrow 4-2\left\{3-\frac{1}{2}-\frac{1}{6}-\frac{3}{2}\right\}=\frac{7}{3}$$

So 6 A $=14$

Question28

Let the area enclosed between the curves $|y|=1-x^2$ and $x^2+y^2=1$ be α. If $9\alpha=\beta\pi+\gamma; \beta, \gamma$ are integers, then the value of $|\beta-\gamma|$ equals:

JEE Main 2025 (Online) 29th January Evening Shift

Options:

A.

15

B.

18

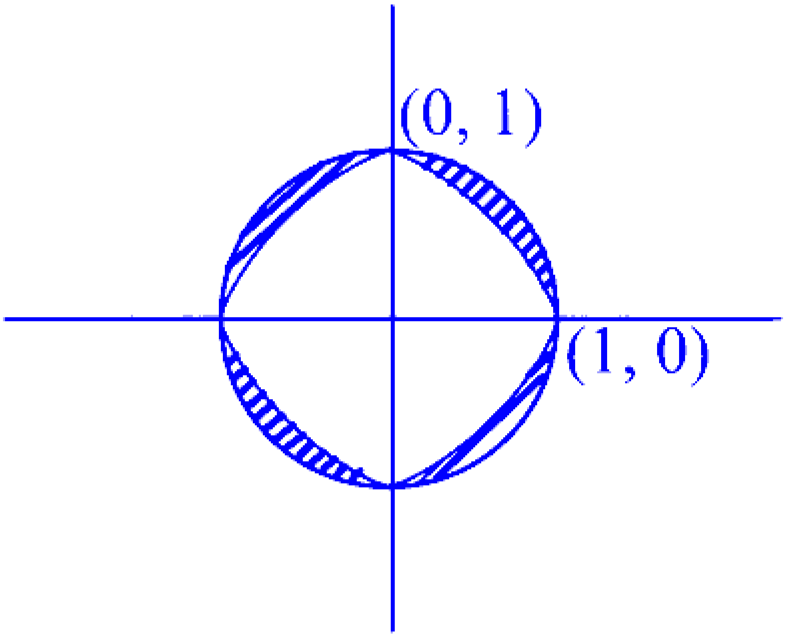
C. 33

D.

Answer: C

Solution:

$C_1 : |y| = 1 - x^2$
$C_2 : x^2 + y^2 = 1$



$\therefore$ Required Area

$= \alpha = 4\left[\text{Area of circle in } 1^{\text{st}} \text{ quad.} - \int_0^1 (1 - x^2)\,dx\right]$

$= 4\left[\frac{\pi}{4} - \left[x - \frac{x^3}{3}\right]_0^1\right]$

$\alpha = \pi - \frac{8}{3}$

$\therefore 3\alpha = 3\pi - 8$

$\therefore 9\alpha = 9\pi - 24$

$\therefore \beta = 9, \gamma = -24$

$\therefore |\beta - \gamma| = 33$

Question29

The area of the region $\{(x, y) : |x - y| \leq y \leq 4\sqrt{x}\}$ is

JEE Main 2025 (Online) 3rd April Evening Shift

Options:

A. $\frac{512}{3}$

B. $\frac{2048}{3}$

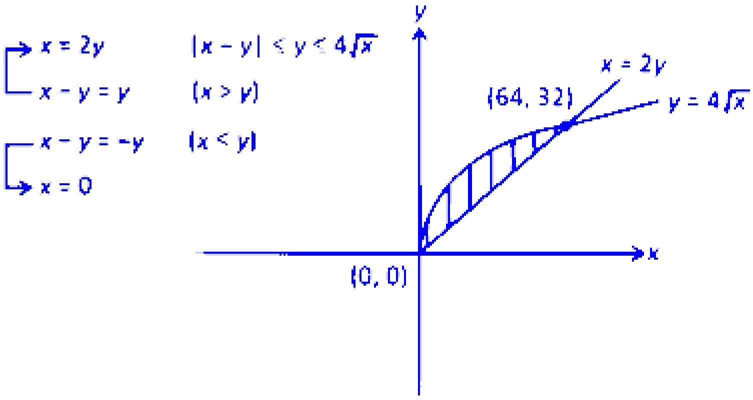
C. 512

D. $\frac{1024}{3}$

Answer: D

Solution:

$|x-y| \leq y \leq 4\sqrt{x}$



$$\text{Area } = \int_0^{64}\left(4\sqrt{x}-\frac{x}{2}\right)dx$$

$$= \left.\frac{4x^{3/2}}{\frac{3}{2}}-\frac{x^2}{4}\right]_0^{64}$$

Question30

Let $f:[0,\infty)\to\mathbb{R}$ be a differentiable function such that

$f(x)=1-2x+\int_0^x e^{x-t}f(t)dt$ for all $x\in[0,\infty)$.

Then the area of the region bounded by $y=f(x)$ and the coordinate axes is

JEE Main 2025 (Online) 4th April Morning Shift

Options:

A. $\sqrt{5}$

B. 2

C. $\sqrt{2}$

D. $\frac{1}{2}$

Answer: D

Solution:

$\because f(x)=1-2x+\int_0^x e^{x-t} f(t) dt$

or, $f(x)=1-2x+e^x \int_0^x e^{-t} f(t) dt$

on differentiating both sides w.r.t. x we get

$$f'(x)=-2+e^x \int_0^x e^{-t} f(t) dt+e^x \cdot e^{-x} f(x)$$
$$f'(x)=-2+f(x)+2x-1+f(x)\{\text{ from eq. (1) }\}$$

$$\begin{aligned}
& \therefore \quad f(x)-2f(x)=2x-3 \\
& \text{I.F. } =e^{\int -2dx}=e^{-2x} \\
& \therefore \quad e^{-2x} \cdot f(x)=\int e^{-2x}(2x-3) dx
\end{aligned}$$

$$\begin{aligned}
& e^{-2x} \cdot f(x)=(2x-3) \cdot \frac{e^{-2x}}{-2}-\int 2 \cdot \frac{e^{-2x}}{-2} dx \\
& e^{-2x} \cdot f(x)=\frac{(2x-3) e^{-2x}}{-2}+\frac{e^{-2x}}{-2}+c \\
& f(x)=-x+1+c' e^{2x} \\
& \because \quad f(x)=1 \text{ from eq. (1)} \\
& \therefore \quad 1=0+1+c' \Rightarrow c'=0 \\
& \therefore \quad f(x)=-x+1
\end{aligned}$$

$\Rightarrow \quad \text{Area}=\frac{1}{2}$

Question31

If the area of the region bounded by the curves $y=4-\frac{x^2}{4}$ and $y=\frac{x-4}{2}$ is equal to α, then 6α. equals

JEE Main 2025 (Online) 7th April Morning Shift

Options:

A. 210

B. 250

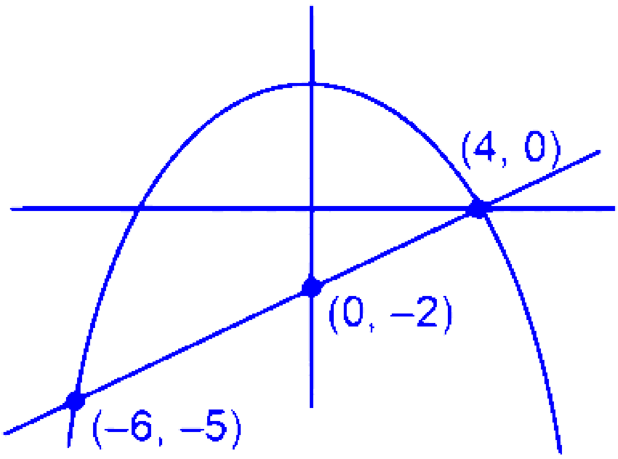
C. 240

D. 220

Answer: B

Solution:

$y = 4 - \frac{x^2}{4}$ and $y = \frac{x-4}{2}$



$$\text{Area } = \int_{-6}^{4} \left(4 - \frac{x^2}{4} - \frac{x}{2} + 2\right) dx$$

$$= \left[6x - \frac{x^3}{12} - \frac{x^2}{4}\right]_{-6}^{4}$$

$$= 6(4+6) - \left(\frac{64}{12} + \frac{216}{12}\right) - \left(\frac{16}{4} - \frac{36}{4}\right)$$

$$= 60 - \frac{70}{3} + 5$$

$$\alpha = \frac{125}{3}$$

$$6\alpha = 6 \times \frac{125}{3} = 250$$

Question32

If the area of the region $\{(x, y) : 1 + x^2 \leq y \leq \min\{x + 7, 11 - 3x\}\}$ is A, then $3A$ is equal to :

JEE Main 2025 (Online) 7th April Evening Shift

Options:

A.

50

B.

46

C.

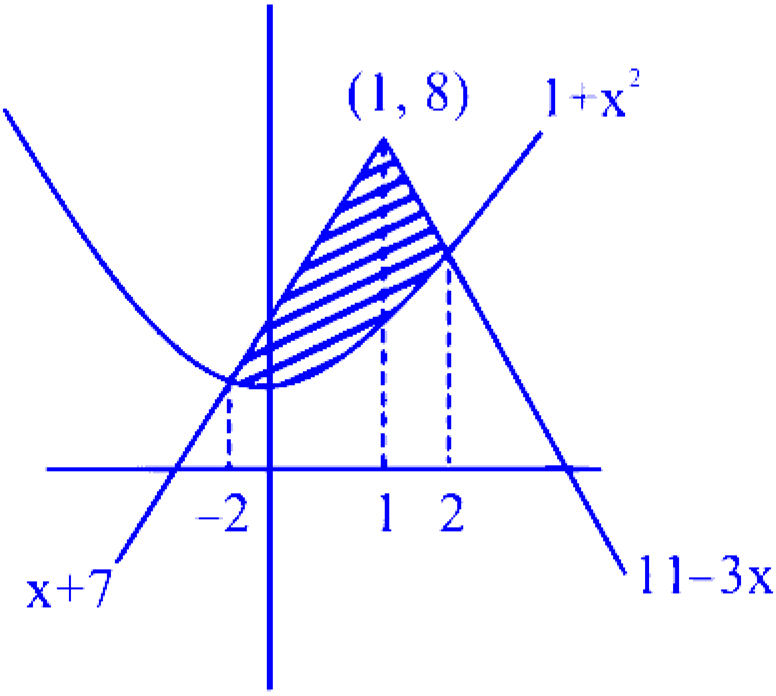
49

D.

47

Answer: A

Solution:



$$A=\int_{-2}^{1}\left(x+7-x^2-1\right)dx+\int_{1}^{2}\left(11+3x-x^2-1\right)dx$$

$$=\left[\frac{x^2}{2}+6x-\frac{x^3}{3}\right]_{-2}^{1}+\left[10x-\frac{3x^2}{2}-\frac{x^3}{3}\right]_{1}^{2}$$

$$=\frac{50}{3}\Rightarrow 3A=50 \quad \text{Option (1)}$$

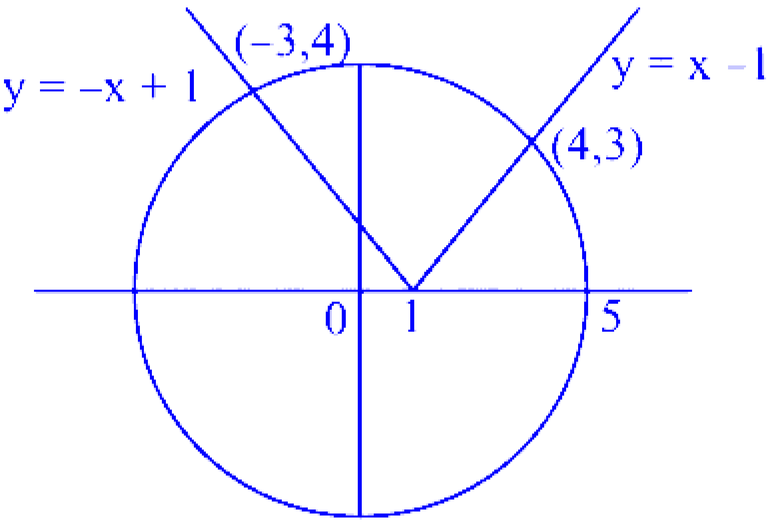
Question33

If the area of the larger portion bounded between the curves $x^2+y^2=25$ and $\mathrm{y}=|\mathrm{x}-1|$ is $\frac{1}{4}(\mathrm{~b}\pi+\mathrm{c}), \mathrm{b}, \mathrm{c} \in N$, then $\mathrm{b}+\mathrm{c}$ is equal to __________

JEE Main 2025 (Online) 23rd January Morning Shift

Answer: 77

Solution:



$x^2 + y^2 = 5$
$x^2 + (x-1)^2 = 25 \Rightarrow x = 4$
$x^2 + (-x+1)^2 = 5 \Rightarrow x = -3$

$$A = 25\pi - \int_{-3}^{4} \sqrt{25 - x^2}dx + \frac{1}{2} \times 4 \times 4 + \frac{1}{2} \times 3 \times 3$$

$$A = 25\pi + \frac{25}{2} - \left[\frac{x}{2}\sqrt{25 - x^2} + \frac{25}{2}\sin^{-1}\frac{x}{5}\right]_{-3}^{4}$$

$$A = 25\pi + \frac{25}{2} - \left[6 + \frac{25}{2}\sin^{-1}\frac{4}{5} + 6 + \frac{25}{2}\sin^{-1}\frac{3}{5}\right]$$

$$A = 25\pi + \frac{1}{2} - \frac{25}{2} \cdot \frac{\pi}{2}$$

$$A = \frac{75\pi}{4} + \frac{1}{2}$$

$$A = \frac{1}{4}(75\pi + 2)$$

$b = 75, c = 2$
$b + c = 75 + 2 = 77$

Question34

If the area of the region $\left\{(x, y) : \ 4 - x^2\right| \leq y \leq x^2, y \leq 4, x \geq 0\}$ is $\left(\frac{80\sqrt{2}}{\alpha} - \beta\right), \alpha, \beta \in \mathbf{N}$, then $\alpha + \beta$ is equal to ________.

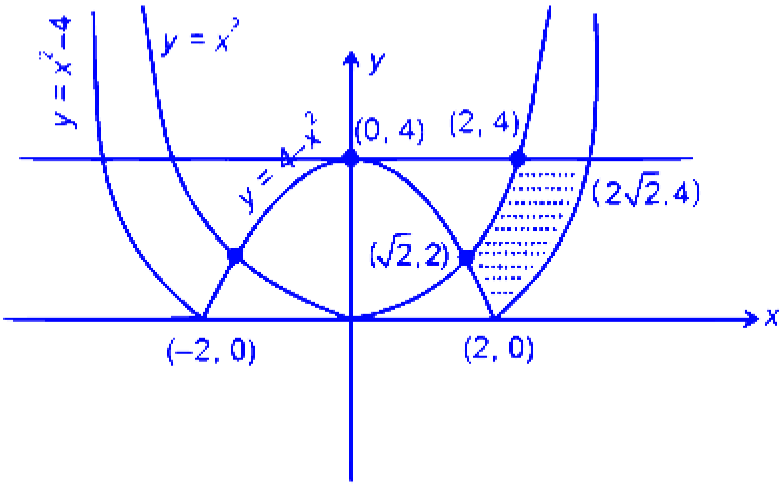
JEE Main 2025 (Online) 2nd April Morning Shift

Answer: 22

Solution:

Area =

$$\int_{\sqrt{2}}^{2}\left(x^2-\left(4-x^2\right)\right)dx+(2\sqrt{2}-2)\times 4-\int_{2}^{2\sqrt{2}}\left(x^2-4\right)dx$$



$$=\left[\frac{2x^3}{3}-4x\right]_{\sqrt{2}}^{2}+8\sqrt{2}-8-\left[\frac{x^3}{3}-4x\right]_{2}^{2\sqrt{2}}$$
$$=\frac{40\sqrt{2}}{3}-16$$
$$\Rightarrow \alpha=6, \beta=16 \Rightarrow \alpha+\beta=22$$

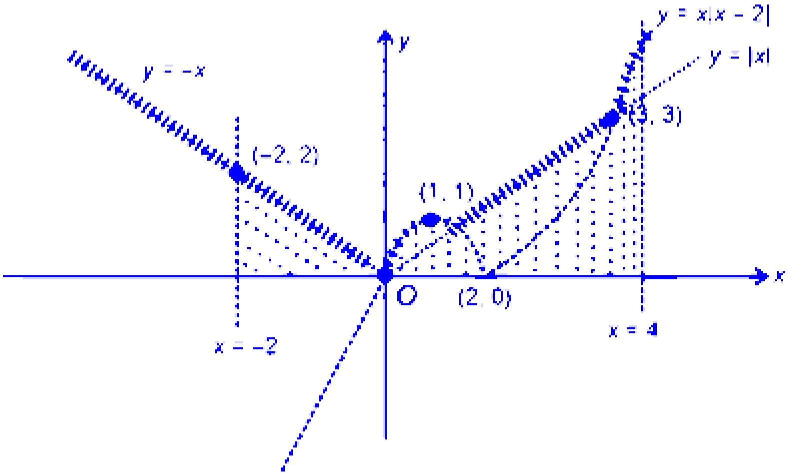
Question35

The area of the region bounded by the curve $y=\max\{|x|, x|x-2|\}$, the x-axis and the lines $x=-2$ and $x=4$ is equal to__________

JEE Main 2025 (Online) 3rd April Morning Shift

Answer: 12

Solution:



Area =

$$\frac{1}{2} \times 2 \times 2 + \int_0^1 (2x - x^2)dx + \int_1^3 xdx + \int_3^4 (x^2 - 2x)dx$$

$$= 2 + \left[\frac{2x^2}{2} - \frac{x^3}{3}\right]_0^1 + \left[\frac{x^2}{2}\right]_1^3 + \left[\frac{x^3}{3} - \frac{2x^2}{2}\right]_3^4$$

= 12 square units

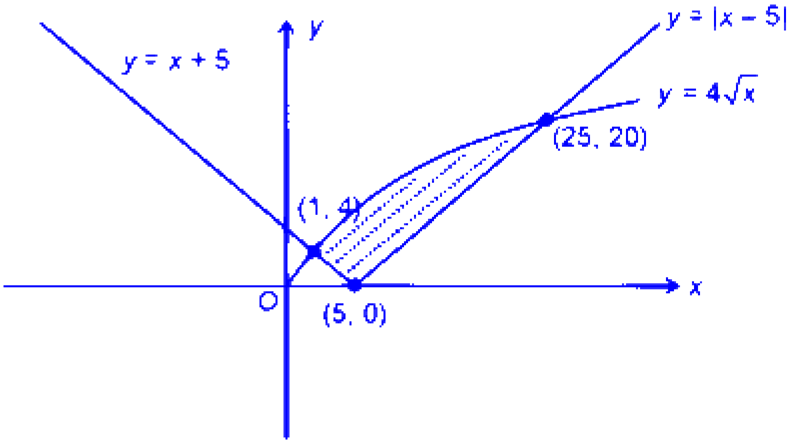
Question36

If the area of the region $\{(x, y) : |x - 5| \le y \le 4\sqrt{x}\}$ is A, then $3A$ is equal to _________.

JEE Main 2025 (Online) 4th April Morning Shift

Answer: 368

Solution:



$$\text{Area } = \int_1^{25} 4\sqrt{x} dx - \frac{1}{2} \times (5-1) \cdot 4 - \frac{1}{2} \times (25-5) \times 20$$

$$= 4\left[\frac{2}{3} x^{\frac{3}{2}}\right]_1^{25} - 8 - 200 = \frac{368}{3}$$

$$\Rightarrow 3A = 368$$

Question37

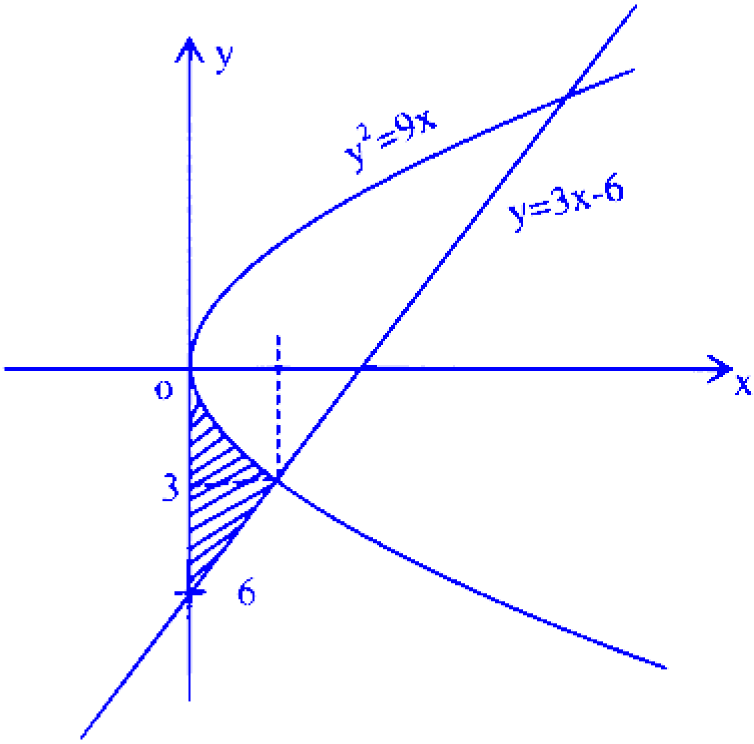
Let the area of the bounded region $\left\{(x, y) : 0 \leq 9x \leq y^2, y \geq 3x - 6\right\}$ be A. Then $6A$ is equal to ________.

JEE Main 2025 (Online) 8th April Evening Shift

Answer: 15

Solution:

$0 \leq 9x \leq y^2$ & $y \geq 3x - 6$



$$A = \text{ Required Area } = \left[\int_0^1 (-3\sqrt{x})dx - \int_0^1 (3x-6)dx\right]$$

$$A = -3\left(\frac{x^{\frac{3}{2}}}{\frac{3}{2}}\right)\Big|_0^1 - \left(\frac{3x^2}{2} - 6x\right)\Big|_0^1$$

$$A = -2[1-0]\left[\frac{3}{2} - 6\right]$$

$$A = -2 - \frac{3}{2} + 6 = \frac{5}{2} \text{ Sq. unit}$$

$$\therefore 6A = 6 \times \frac{5}{2} = 15$$

Question38

Let A_1 be the area of the region bounded by the curves $y = \sin x$, $y = \cos x$ and y-axis in the first quadrant. Also, let A_2 be the area of the region bounded by the curves $y = \sin x$, $y = \cos x$, x-axis and $x = \frac{\pi}{2}$ in the first quadrant. Then, [26 Feb 2021 Shift 2]

Options:

A. $A_1 : A_2 = 1 : \sqrt{2}$ and $A_1 + A_2 = 1$

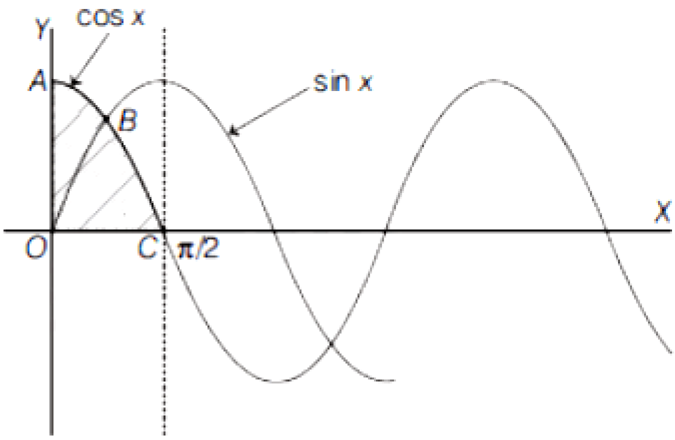
B. $A_1 = A_2$ and $A_1 + A_2 =$ sqrt 2

C. $2A_1 = A_2$ and $A_1 + A_2 = 1 + \sqrt{2}$

D. $A_1 : A_2 = 1 : 2$ and $A_1 + A_2 = 1$

Answer: A

Solution:



A_1 is the area of region OAB.
A_2 is the area of region OBC.

Coordinate of B is $\left(\frac{\pi}{4}, \frac{1}{\sqrt{2}}\right)$

Now, $A_1 = \int_0^{\frac{\pi}{4}} (\cos x - \sin x) d x$

$= [\sin x + \cos x]_0^{\pi/4}$

$= \left(\sin\frac{\pi}{4} + \cos\frac{\pi}{4}\right) - (0 + 1) = \sqrt{2} - 1$

$A_2 = \int_0^{\frac{\pi}{4}} \sin x\, d\, x + \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cos x\, d\, x$

$= [\sin x + \cos x]_0^{\pi/4}$

$= \left(\sin\frac{\pi}{4} + \cos\frac{\pi}{4}\right) - (0 + 1) = \sqrt{2} - 1$

$A_2 = \int_0^{\frac{\pi}{4}} \sin x\, d\, x + \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cos x\, d\, x$

$A_2 = [-\cos x]_0^{\frac{\pi}{4}} + [\sin x]_{\frac{\pi}{4}}^{\frac{\pi}{2}} = \sqrt{2}(\sqrt{2} - 1)$

Now, $A_1 : A_2 = (\sqrt{2} - 1) : \sqrt{2}(\sqrt{2} - 1)$

$A_1 : A_2 = 1 : \sqrt{2}$

and $A_1 + A_2 = (\sqrt{2} - 1) + \sqrt{2}(\sqrt{2} - 1) = (\sqrt{2} - 1)(\sqrt{2} + 1)$

$A_1 + A_2 = 2 - 1 = 1$

Therefore,

and $A_1 : A_2 = 1 : \sqrt{2}$,
$A_1 + A_2 = 1$

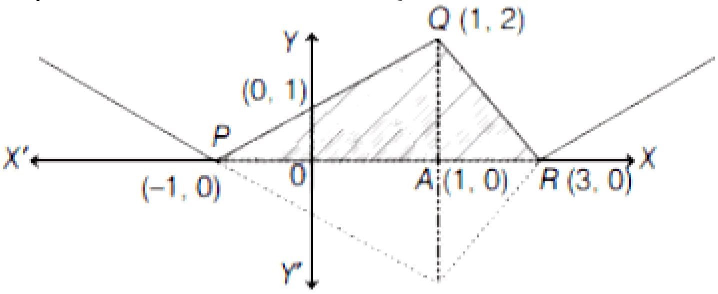
Question39

The area bounded by the lines $y = ||x-1|-2|$ is ....
[26 Feb 2021 Shift 1]

Answer: 4

Solution:

Given, $y = ||x-1|-2|$
Required area is area of $\triangle PQR$.



$$\text{Area} = \frac{1}{2} \times (\text{Base}) \times (\text{Height})$$
$$= \frac{1}{2} \times (PR) \times (AQ)$$
$$= \frac{1}{2} \times 4 \times 2 = 4$$

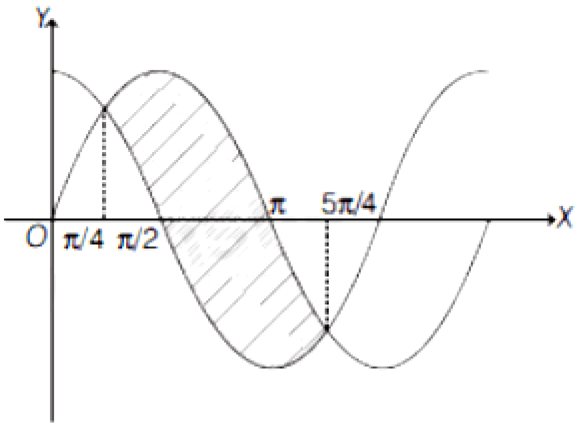
Since, only one curve is given, here assume the area bounded by X-axis. Then, the area will be 4 sq unit.

Question40

The graph of sine and cosine functions, intersect each other at a number of points and between two consecutive points of intersection, the two graphs enclose the same area A. Then A^4 is equal to ....
[25 Feb 2021 Shift 1]

Answer: 64

Solution:



Required area of shaded region

$$A = \int_{\pi/4}^{5\pi/4} \{\sin x - \cos x) dx$$

$$= [-\cos x - \sin x]_{\pi/4}^{5\pi/4}$$

$$= -\left[\left(\cos\frac{5\pi}{4} + \sin\frac{5\pi}{4}\right) - \left(\cos\frac{\pi}{4} + \sin\frac{\pi}{4}\right)\right]$$

$$= -\left[\left(-\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}\right) - \left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}\right)\right]$$

$$\therefore A = \frac{4}{\sqrt{2}} = 2\sqrt{2}$$

$$\Rightarrow A^4 = (2\sqrt{2})^4 = 64$$

Question41

The area of the region $R = \{(x, y) : 5x^2 \le y \le 2x^2 + 9\}$ is [24 Feb 2021 Shift 2]

Options:

A. $11\sqrt{3}$ square units

B. $12\sqrt{3}$ square units

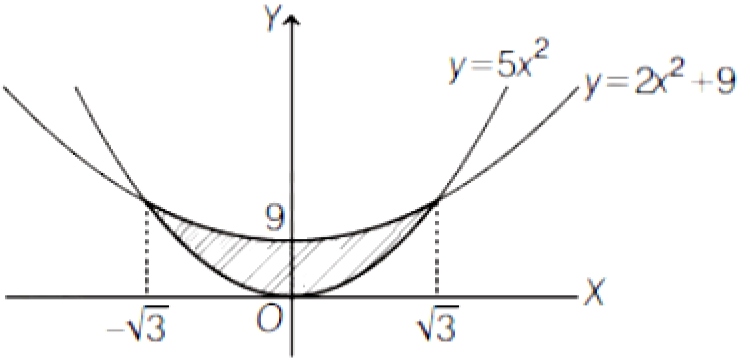
C. $9\sqrt{3}$ square units

D. $6\sqrt{3}$ square units

Answer: B

Solution:

Given, $R=\{(x, y): 5x^2 \le y \le 2x^2+9\}$
Here, we have two curves $y=5x^2$ and $y=2x^2+9$, point of intersection of both curves is find by solving both equations i.e.
$5x^2 = 2x^2+9$
$\Rightarrow x^2 = 3 \Rightarrow x = \pm\sqrt{3}$



$\therefore$ Area $= \int_{-\sqrt{3}}^{\sqrt{3}} (2x^2+9-5x^2)dx$
$= 2\int_0^{\sqrt{3}} (9-3x^2)dx$
$= 2[9x-x^3]_0^{\sqrt{3}}$
$= 2[9\sqrt{3}-3\sqrt{3}]$
$= 12\sqrt{3}$ sq units

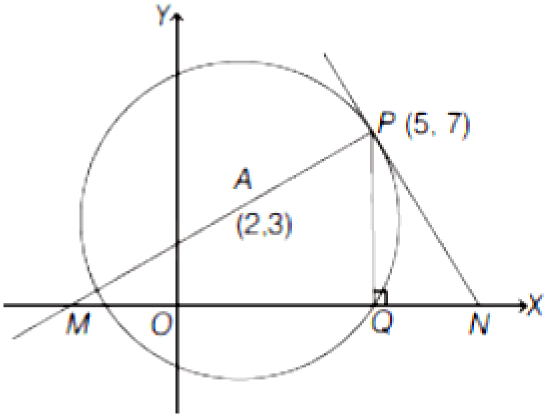
Question42

If the area of the triangle formed by the positive x-axis, the normal and the tangent to the circle $(x-2)^2+(y-3)^2=25$ at the point (5, 7) is A, then 24A is equal to ....
[24 Feb 2021 Shift 2]

Answer: 1225

Solution:

Given, circle $(x-2)^2+(y-3)^2=5^2$
$c=(2,3)$
$r=5$



Equation of normal at P (i.e. PA line)

$\Rightarrow\ (y-7)=\left(\frac{7-3}{5-2}\right)(x-5)$
$\Rightarrow\ 3y-21=4x-20$
$\Rightarrow\ 4x-3y+1=0$

Therefore, $M=\left(\frac{-1}{4},0\right)$ [Put $y=0$ in above equation]

Now, equation of tangent at P.

$y-7=\frac{-3}{4}(x-5)\ \left[\because \text{ slope of PN}=\frac{-1}{\text{Slope of PA}}\right]$
$\Rightarrow 4y-28=-3x+15$
$\Rightarrow 3x+4y=43$

Therefore, $N=\left(\frac{43}{3},0\right)$ [Put $y=0$ in above equation]

$\therefore\ \text{Area (A)}=\frac{1}{2}\times MN\times PQ$

$=\frac{1}{2}\times\left(\frac{43}{3}+\frac{1}{4}\right)\times 7$

$=\frac{1}{2}\times\frac{175}{12}\times 7$

$\therefore\ 24A=24\times\frac{1}{2}\times\frac{175}{12}\times 7=1225$

But this question is wrong as in question. It is mentioned that the triangle is formed with the positive X-axis which contradicts the solution.

Question43

The area (in sq. units) of the part of the circle $x^2+y^2=36$, which is outside the parabola $y^2=9x$, is :
24 Feb 2021 Shift 1

Options:

A. $24\pi+3\sqrt{3}$

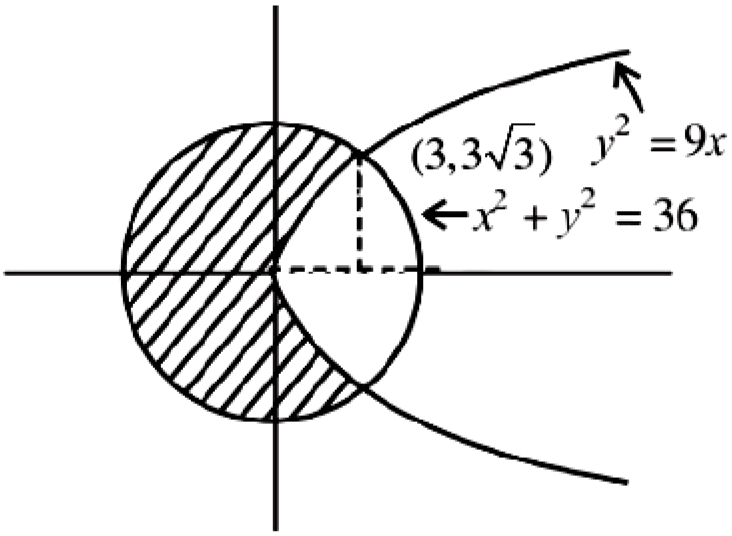
B. $12\pi-3\sqrt{3}$

C. $24\pi-3\sqrt{3}$

D. $12\pi+3\sqrt{3}$

Answer: C

Solution:



Required area

$$=\pi\times(6)^2-2\int_0^3\sqrt{9x}\,dx-2\int_3^6\sqrt{36-x^2}\,dx$$

$$=36\pi-12\sqrt{3}-2\left(\frac{x}{2}\sqrt{36-x^2}+18\sin^{-1}\frac{x}{6}\right)_3^6$$

$$=36\pi-12\sqrt{3}-2\left(9\pi-3\pi-\frac{9\sqrt{3}}{2}\right)$$

$$=24\pi-3\sqrt{3}$$

Question44

Let $g(x)=\int_0^x f(t)\,dt$, where f is continuous function in [0, 3] such that $\frac{1}{3}\le f(t)\le 1$ for all $t\in[0,1]$ and $0\le f(t)\le\frac{1}{2}$ for all $t\in(1,3]$. The largest possible interval in which g(3) lies is
[18 Mar 2021 Shift 2]

Options:

A. $\left[-1, -\frac{1}{2}\right]$

B. $\left[-\frac{3}{2}, -1\right]$

C. $\left[\frac{1}{3}, 2\right]$

D. $[1, 3]$

Answer: C

Solution:

Given, $g(x) = \int_0^x f(t) d t$

$\therefore \; g(3) = \int_0^3 f(t) d t = \int_0^1 f(t) d t + \int_1^3 f(t) d t$

$\Rightarrow \; \int_0^1 \frac{1}{3} d t + \int_1^3 0 \cdot d t \leq g(3) \leq \int_0^1 1 d t + \int_1^3 \frac{1}{2} d t$

$\Rightarrow \; \frac{1}{3} \leq g(3) \leq 1 + 1$

$\Rightarrow \; \frac{1}{3} \leq g(3) \leq 2$

Question45

The area bounded by the curve $4y^2 = x^2(4-x)(x-2)$ is equal to [18 Mar 2021 Shift 2]

Options:

A. $\frac{\pi}{8}$

B. $\frac{3\pi}{8}$

C. $\frac{3\pi}{2}$

D. $\frac{\pi}{16}$

Answer: C

Solution:

Given, equation of curve is $4y^2 = x^2(4-x)(x-2)$

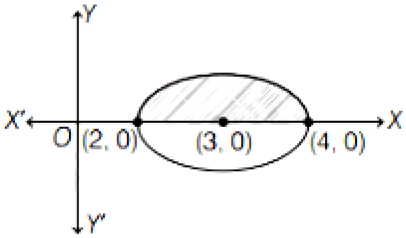
$\Rightarrow \sqrt{4y^2} = \sqrt{x^2(4-x)(x-2)}$

$\Rightarrow |y| = \frac{|x|}{2}\sqrt{(4-x)(x-2)}$

$$y = \begin{cases} \frac{x}{2}\sqrt{(4-x)(x-2)} & x > 0 \\ -\frac{x}{2}\sqrt{(4-x)(x-2)} & x < 0. \end{cases}$$

Let

$y_1 = \frac{x}{2}\sqrt{(4-x)(x-2)} \Rightarrow y_2 = -\frac{x}{2}\sqrt{(4-x)(x-2)}$



and domain $x \text{ in}[2, 4]$

$[\because (4-x)(x-2) \geq 0 \Rightarrow (x-2)(x-4) \leq 0 \Rightarrow 2 \leq x \leq 4]$

$\therefore$ Required area $= \int_2^4 (y_1 - y_2)dx$

$= \int_2^4 x\sqrt{(4-x)(x-2)}dx$...(i)

Using property, $\int_a^b f(x)dx = \int_a^b f(a+b-x)dx$

From Eq. (i),

Area $= \int_2^4 (6-x)\sqrt{(4-x)(x-2)}dx$...(ii)

From Eqs. (i) and (ii), $2A = 6\int_2^4 \sqrt{(4-x)(x-2)}dx$

$\Rightarrow A = 3\int_2^4 \sqrt{1-(x-3)^2}dx$

$A = 3\left[\frac{x-3}{2}\sqrt{1-(x-3)^2} + \frac{1}{2}\sin^{-1}(x-3)\right]_2^4$

$A = \frac{3}{2}\left[\frac{0+\pi}{2} - \frac{0+\pi}{2}\right] \Rightarrow A = \frac{3\pi}{2}$

$\Rightarrow A = 3 \cdot \frac{\pi}{2} \cdot (1)^2 = \frac{3\pi}{2}$

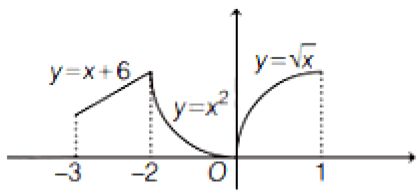
Question46

Let f : [−3, 1] → R be given as $f(x) = \left\{ \begin{array}{ll} \min\{(x+6), x^2\} & -3 \le x \le 0 \\ \max\{\sqrt{x}, x^2\} & 0 \le x \le 1 \end{array} \right\}$

If the area bounded by y = f (x) and x-axis is A, then the value of 6A is equal to
[17 Mar 2021 Shift 2]

Answer: 41

Solution:



$A =$ Area bounded by $y = f(x)$ and X-axis.

$$= \int_{-3}^{-2} (x+6)\,dx + \int_{-2}^{0} x^2\,dx + \int_{0}^{1} \sqrt{x}\,dx$$

$$= \left[\frac{x^2}{2}\right]_{-3}^{-2} + 6[x]_{-3}^{-2} + \left[\frac{x^3}{3}\right]_{-2}^{0} + \left[\frac{2}{3}x^{\frac{3}{2}}\right]_{0}^{1} = \frac{41}{6}$$

$$\therefore 6A = 6 \times \frac{41}{6}$$

$$\Rightarrow 6A = 41$$

Question47

If the area of the bounded region

$R = \left\{ (x, y) : \max\{0, \log_e x\} \le y \le 2^x, \frac{1}{2} \le x \le 2 \right\}$

is, $\alpha(\log_e 2)^{-1} + \beta(\log_e 2) + \gamma$, **then the value of** $(\alpha + \beta - 2\gamma)^2$ **is equal to :**
[27 Jul 2021 Shift 1]

Options:

A. 8

B. 2

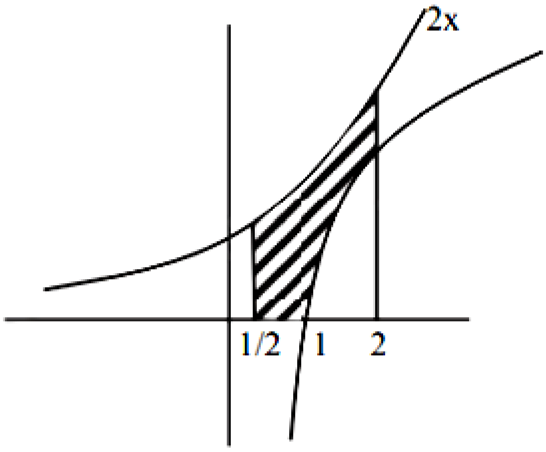
C. 4

D. 1

Answer: B

Solution:

$R = \left\{ (x, y) : \max\{0, \log_e x\} \le y \le 2^x, \frac{1}{2} \le x \le 2 \right\}$



$\int_{\frac{1}{2}}^{2} 2^x dx - \int_{1}^{2} \ln x dx$

$\Rightarrow \left[\frac{2^x}{\ln 2} \right]_{1/2}^{2} - [x \ln x - x]_1^2$

$\Rightarrow \frac{(2^2) - 2^{1/2}}{\log_e 2} - (2 \ln 2 - 1)$

$\Rightarrow \frac{(2^2 - \sqrt{2})}{\log_e 2} - 2 \ln 2 + 1$

$\therefore \alpha = 2^2 - \sqrt{2}, \beta = -2, \gamma = 1$

$\Rightarrow (\alpha + \beta + 2\gamma)^2$

$\Rightarrow (2^2 - \sqrt{2} - 2 - 2)^2$

$\Rightarrow (\sqrt{2})^2 = 2$

Question48

The area of the region bounded by $y - x = 2$ and $x^2 = y$ is equal to :-
[27 Jul 2021 Shift 2]

Options:

A. $\frac{16}{3}$

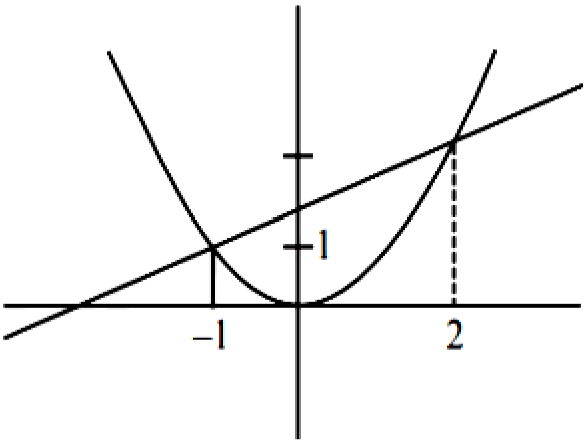
B. $\frac{2}{3}$

C. $\frac{9}{2}$

D. $\frac{4}{3}$

Answer: C

Solution:



$y - x = 2, x^2 = y$

Now, $x^2 = 2 + x$

$\Rightarrow x^2 - x - 2 = 0$

$\Rightarrow (x + 1)(x - 2) = 0$

Area $= \int_{-1}^{2} (2 + x - x^2)$

$= \left|2x + \frac{x^2}{2} - \frac{x^3}{3}\right|_{-1}^{2}$

$= \left(4 + 2 - \frac{8}{3}\right) - \left(-2 + \frac{1}{2} + \frac{1}{3}\right)$

$= 6 - 3 + 2 - \frac{1}{2} = \frac{9}{2}$

Question49

The area (in sq. units) of the region, given by the set $\{(x, y) \in R \times R \mid x \geq 0, 2x^2 \leq y \leq 4 - 2x\}$ is :

[25 Jul 2021 Shift 1]

Options:

A. $\frac{8}{3}$

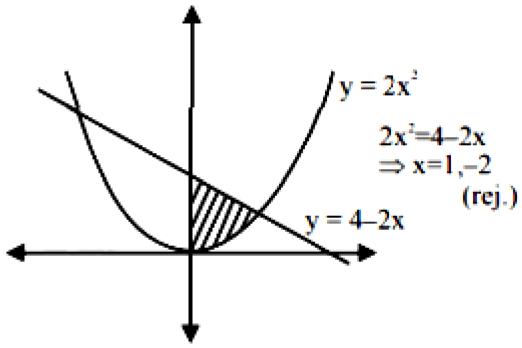
B. $\frac{17}{3}$

C. $\frac{13}{3}$

D. $\frac{7}{3}$

Answer: D

Solution:



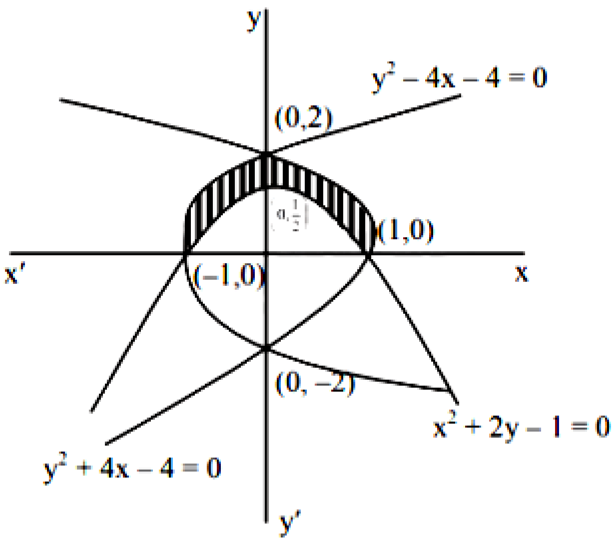
Required area $=\int_0^1 (4-2x-2x^2)dx = 4x-x^2-\frac{2x^3}{3}\Big|_0^1 = 4-1-\frac{2}{3}=\frac{7}{3}$

Question50

The area (in sq. units) of the region bounded by the curves $x^2+2y-1=0$, $y^2+4x-4=0$ and $y^2-4x-4=0$in the upper half plane is _______.
[22 Jul 2021 Shift 2]

Answer: 2

Solution:



Required Area (shaded)

$$= 2\left[\int_0^2 \left(\frac{4-y^2}{4}\right) d\,y - \int_0^1 \left(\frac{1-x^2}{2}\right) d\,x\right]$$

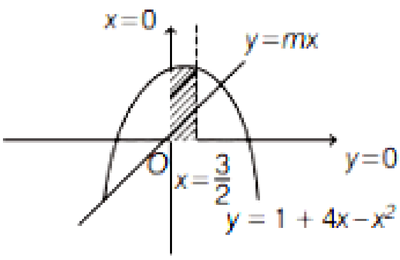
$$= 2\left[\frac{4}{3}-\frac{1}{3}\right] = (2)$$

Question51

If the line $y = mx$ bisects the area enclosed by the lines $x = 0$ and $y = 0$, $x = \frac{3}{2}$ and the curve $y = 1 + 4x - x^2$, then $12m$ is equal to [31 Aug 2021 Shift 2]

Answer: 26

Solution:



According to the question,

$$\frac{1}{2}\int_0^{\frac{3}{2}}(1+4x-x^2)dx=\int_0^{\frac{3}{2}}mxdx$$

$$\Rightarrow\frac{1}{2}\left[\left(x+2x^2-\frac{x^3}{3}\right)\right]_0^{\frac{3}{2}}=\frac{m}{2}[x]_0^{\frac{3}{2}}$$

$$\Rightarrow\frac{3}{2}+\frac{9}{2}-\frac{9}{8}=\frac{9m}{4}$$

$$\Rightarrow m=\frac{39}{18}$$

$$\Rightarrow 12m=26$$

Question52

The area of the region bounded by the parabola $(y-2)^2=(x-1)$, the tangent to it at the point whose ordinate is 3 and the X-axis is
[27 Aug 2021 Shift 2]

Options:

A. 9

B. 10

C. 4

D. 6

Answer: A

Solution:

Given parabola

$(y-2)^2=(x-1)$

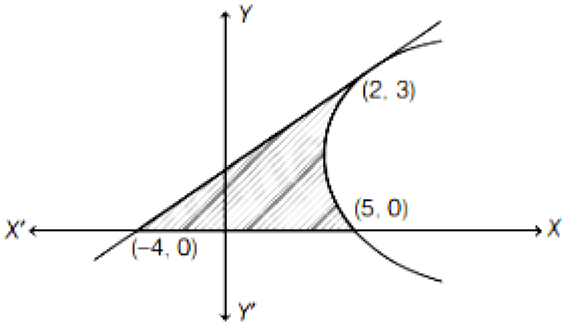
Since, Ordinate = y = 3

Then, x = 2

Point on parabola (2, 3)

Differentiating Eq. (i) w.r.t. x, we get

$2(y-2)\frac{dy}{dx}=1$



$\frac{dy}{dx} = \frac{1}{2(y-2)}$

At (2, 3)

$\frac{dy}{dx} = \frac{1}{2}$

Equation of tangent at (2, 3)

$y - 3 = \frac{1}{2}(x-2)$

or $x - 2y + 4 = 0$

Intersection point of parabola on X-axis is

$y = 0$, $x = 5$ i.e (5, 0)

Intersection point of tangent and X-axis

$y = 0$, $x = -4$ i.e $(-4, 0)$

Area of shaded region $= \int_0^3 [(y-2)^2 + 1 - (2y-4)]dy$

$= \int_0^3 (y^2 - 6y + 9)dy$

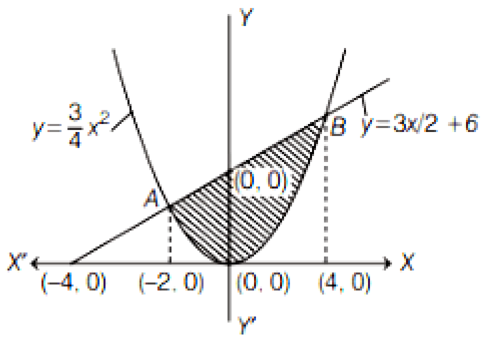
$= \left(\frac{y^3}{3} - 3y^2 + 9y\right)_0^3 = 9$ sq. unit.

Question53

The area of the region $S = \{(x, y) : 3x^2 \leq 4y \leq 6x + 24\}$ is [26 Aug 2021 Shift 1]

Answer: 27

Solution:



We have, $y=\left(\frac{3}{4}\right)x^2$ and $y=\left(\frac{3x}{2}\right)+6$

$\Rightarrow \frac{3x^2}{4}=\frac{3x}{2}+6$

$\Rightarrow 3x^2=6x+24$

$\Rightarrow x^2-2x-8=0$

$\Rightarrow (x-4)(x+2)=0$

$\Rightarrow x=-2, 4$

$\Rightarrow y=3, 12$

$A(-2, 3)$ and $B(4, 12)$

Required area $=\int_{-2}^{4}\left(\frac{3x}{2}+6\right)-\left(\frac{3x^2}{4}\right)dx$

$=\left[\frac{3x^2}{4}+6x-\frac{x^3}{4}\right]_{-2}^{4}$

$=[(12+24-16)-(3-12+2)]$

$=(20+7)=27$ sq units

Question54

Let a and b respectively be the points of local maximum and local minimum of the function $f(x)=2x^3-3x^2-12x$. If A is the total area of the region bounded by y = f(x), the X- axis and the lines x = a and x = b, then 4A is equal to [26 Aug 2021 Shift 2]

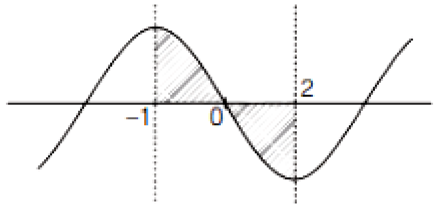
Answer: 114

Solution:

We have, $f(x)=2x^3-3x^2-12x$

$\therefore f'(x)=6x^2-6x-12=6(x^2-x-2)$

$= 6(x+1)(x-2)$
f'(x) = 0
$\Rightarrow x = -1$ and 2
$\therefore x = -1$ and 2 are critical points



$\therefore a = -1$ and $b = 2$

Now, required area, $A_0 = \int_{-1}^{0} f(x)dx + \int_{0}^{2} -f(x)dx$

$\int_{-1}^{0}(2x^3 - 3x^2 - 12x)dx + \int_{0}^{2}(12x + 3x^2 - 2x^3)dx$

$= \left[\frac{x^4}{2} - x^3 - 6x^2\right]_{-1}^{0} + \left[6x^2 + x^3 - \frac{x^4}{2}\right]_{0}^{2} = \frac{114}{4}$

$\therefore 4A = 114$

Question55

The area, enclosed by the curves $y = \sin x + \cos x$ and $y = |\cos x - \sin x|$ and the lines $x = 0$, $x = \frac{\pi}{2}$, is [1 Sep 2021 Shift 2]

Options:

A. $2\sqrt{2}(\sqrt{2} - 1)$

B. $2(\sqrt{2} + 1)$

C. $4(\sqrt{2} - 1)$

D. $2\sqrt{2}(\sqrt{2} + 1)$

Answer: A

Solution:

Area $= \int_0^{\frac{\pi}{2}}((\cos x + \sin x) - |\cos x - \sin x|)dx$

$= \int_0^{\frac{\pi}{4}}((\cos x + \sin x) - (\cos x - \sin x))dx + \int_{\frac{\pi}{4}}^{\frac{\pi}{2}}((\cos x + \sin x) - (\sin x - \cos x))dx$

$$= 2\int_0^{\frac{\pi}{4}} \sin x \, d\,x + 2\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cos x \, d\,x$$

$$= 2\left(\frac{-1}{\sqrt{2}} + 1\right) + 2\left(1 - \frac{1}{\sqrt{2}}\right) \;= 2\sqrt{2}(\sqrt{2} - 1)$$

Question56

The area of the region, enclosed by the circle $x^2 + y^2 = 2$ which is not common to the region bounded by the parabola $y^2 = x$ and the straight line $y = x$, is:
[Jan. 7, 2020 (I)]

Options:

A. $(24\pi - 1)$

B. $(6\pi - 1)$

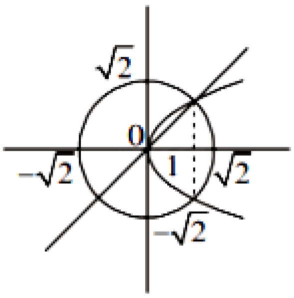
C. $(12\pi - 1)$

D. $(12\pi - 1)/6$

Answer: D

Solution:

Total area – enclosed area between line and parabola



$$= 2\pi - \int_0^1 \sqrt{x} - x d\,x$$

$$= 2\pi - \left(\frac{2x^{3/2}}{3} - \frac{x^2}{2}\right)_0^1$$

$$= 2\pi - \left(\frac{2}{3} - \frac{1}{2}\right) = 2\pi - \left(\frac{1}{6}\right) \;= \frac{12\pi - 1}{6}$$

Question57

The area (in sq. units) of the region $\{(x, y) \in R^2 \mid 4x^2 \leq y \leq 8x + 12\}$ is:
[Jan. 7, 2020 (II)]

Options:

A. $\frac{125}{3}$

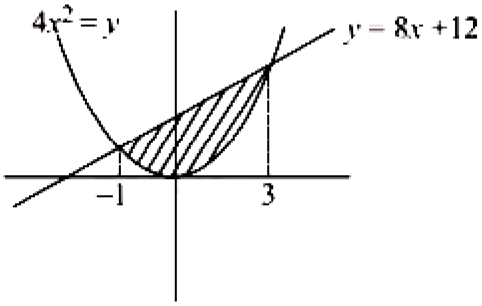
B. $\frac{128}{3}$

C. $\frac{124}{3}$

D. $\frac{127}{3}$

Answer: B

Solution:



Given curves are
$4x^2 = y$ ......(i)
$y = 8x + 12$ .....(ii)
From eqns. (i) and (ii),
$4x^2 = 8x + 12$
$\Rightarrow x^2 - x - 3 = 0$
$\Rightarrow x^2 - 2x - 3 = 0$
$\Rightarrow x^2 - 3x + x - 3 = 0$
$\Rightarrow (x + 1)(x - 3) = 0$
$\Rightarrow x = -1, 3$
Required area bounded by curves is given by
$A = \int_{-1}^{3} (8x + 12 - 4x^2)dx$
$A = \left[\frac{8x^2}{2} + 12x - \frac{4x^3}{3}\right]_{-1}^{3}$

$= (4(9) + 36 - 36) - \left(4 - 12 + \frac{4}{3}\right)$

$= 36 + 8 - \frac{4}{3} = 44 - \frac{4}{3} = \frac{132 - 4}{3} = \frac{128}{3}$

Question58

For a > 0, let the curves $C_1 : y^2 = ax$ and $C_2 : x^2 = ay$ intersect at origin O and a point P. Let the line $x = b(0 < b < a)$ intersect the chord OP and the x -axis at points Q and R, respectively. If the line x = b bisects the area bounded by the curves, C_1 and C_2, and the area of $\Delta OQR = \frac{1}{2}$, then 'a' satisfies the equation: [Jan. 8, 2020 (I)]

Options:

A. $x^6 - 6x^3 + 4 = 0$

B. $x^6 - 12x^3 + 4 = 0$

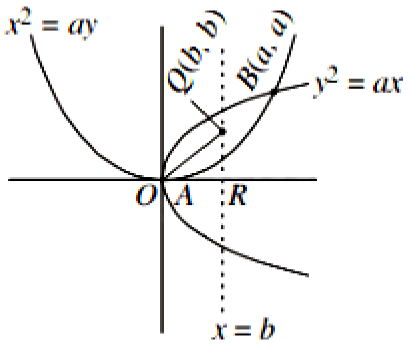
C. $x^6 + 6x^3 - 4 = 0$

D. $x^6 - 12x^3 - 4 = 0$

Answer: B

Solution:

Given eqns. are, $x^2 = ay$ and $y^2 = ax$



After solving, we get $x = a, y = a$
Now, coordinates of B is (a, a) and A is (0,0)
Now, coordinates of Q is (b, b)

$\frac{1}{2}b^2 = \frac{1}{2} \Rightarrow b = 1$

Area bounded by curves and $x = 1$ is

$\int_0^1 \left(\sqrt{a}x^{1/2} - \frac{x^2}{a}\right)dx = \frac{1}{2}\int_0^a \left(\sqrt{a}x^{1/2} - \frac{x^2}{a}\right)dx$

$\Rightarrow \frac{2}{3}\sqrt{a} - \frac{1}{3a} = \frac{a^2}{6}$

$\Rightarrow 4a\sqrt{a} - 2 = a^3$

$\Rightarrow a^6 + 4a^3 + 4 = 16a^3$

$\Rightarrow a^6 - 12a^3 + 4 = 0$

Question59

The area (in sq. units) of the region $\{(x, y) \in R^2 : x^2 \le y \le |3 - 2x|$, is: [Jan. 8, 2020 (II)]

Options:

A. $\frac{32}{3}$

B. $\frac{34}{3}$

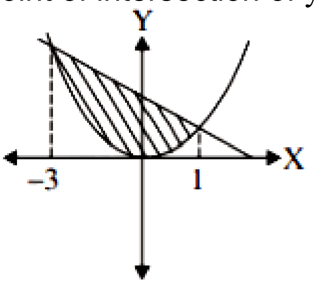
C. $\frac{29}{3}$

D. $\frac{31}{3}$

Answer: A

Solution:

Point of intersection of $y = x^2$ and $y = -2x + 3$ is obtained by $x^2 + 2x - 3 = 0$



$\Rightarrow x = -3, 1$

So, required area $= \int_{-3}^{1} ($ line − parabola $)dz$

$$= \int_{-3}^{1} (3 - 2x - x^2)dx = \left[3x - x^2 - \frac{x^3}{3}\right]_{-3}^{1}$$

$$= (3)4 - 2\left(\frac{1^2 - 3^2}{2}\right) - \left(\frac{1^3 + 3^3}{3}\right) = 12 + 8 - \frac{28}{3} = \frac{32}{3}$$

Question60

Given:

$$\textbf{f (x) =} \begin{cases} x & , \ 0 \le x < \frac{1}{2} \\ \frac{1}{2} & , \ x = \frac{1}{2} \\ 1 - x & , \ \frac{1}{2} < x \le 1 \end{cases}$$

and g(x) = $\left(x - \frac{1}{2}\right)^2$, x ∈ R. Then the area (in sq. units) of the region bounded by the curves, y = f (x) and y = g(x) between the lines, 2x = 1 and 2x = $\sqrt{3}$, is:
[Jan. 9, 2020 (II)]

Options:

A. $\frac{1}{3} + \frac{\sqrt{3}}{4}$

B. $\frac{\sqrt{3}}{4} - \frac{1}{3}$

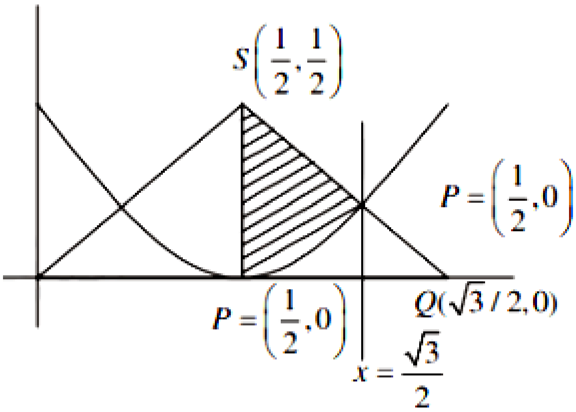
C. $\frac{1}{2} - \frac{\sqrt{3}}{4}$

D. $\frac{1}{2} + \frac{\sqrt{3}}{4}$

Answer: B

Solution:

Coordinates of $P\left(\frac{1}{2}, 0\right)$, $Q\left(\frac{\sqrt{3}}{2}, 0\right)$, $R\left(\frac{\sqrt{3}}{2}, 1-\frac{\sqrt{3}}{2}\right)$ and $S\left(\frac{1}{2}, \frac{1}{2}\right)$



Required area = Area of trapezium PQRS

$$-\int_{1/2}^{\sqrt{3}/2}\left(x-\frac{1}{2}\right)^2 dx$$

$$=\frac{1}{2}\left(\frac{\sqrt{3}-1}{2}\right)\left(\frac{1}{2}+1-\frac{\sqrt{3}}{2}\right)-\frac{1}{3}\left(\left(x-\frac{1}{2}\right)^3\right)_{1/2}^{\sqrt{3}/2}$$

$$=\frac{\sqrt{3}}{4}-\frac{1}{3}$$

Question61

The area (in sq. units) of the region $A=\{(x, y):(x-1)[x] \leq y \leq 2\sqrt{x}, 0 \leq x \leq 2\}$, where [t] denotes the greatest integer function, is: [Sep. 05, 2020 (II)]

Options:

A. $\frac{8}{3}\sqrt{2}-\frac{1}{2}$

B. $\frac{4}{3}\sqrt{2}+1$

C. $\frac{8}{3}\sqrt{2}-1$

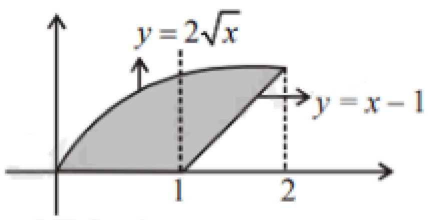
D. $\frac{4}{3}\sqrt{2}-\frac{1}{2}$

Answer: A

Solution:

$[x] = 0$ when $x \in [0, 1)$ and $[x] = 1$ when $x \in [1, 2)$

$$y = \begin{cases} 0 & 0 \le x < 1 \\ x - 1 & 1 \le x < 2 \end{cases}$$



$$\therefore A = \int_0^2 2\sqrt{x}\,dx - \frac{1}{2}(1)(1)$$

$$= \left.\frac{4x^{3/2}}{3}\right|_0^2 - \frac{1}{2} = \frac{8\sqrt{2}}{3} - \frac{1}{2}$$

Question62

The area (in sq. units) of the region $\left\{(x, y) : 0 \le y \le x^2 + 1, 0 \le y \le x + 1, \frac{1}{2} \le x \le 2\right\}$ is [Sep. 03, 2020 (I)]

Options:

A. $\frac{23}{16}$

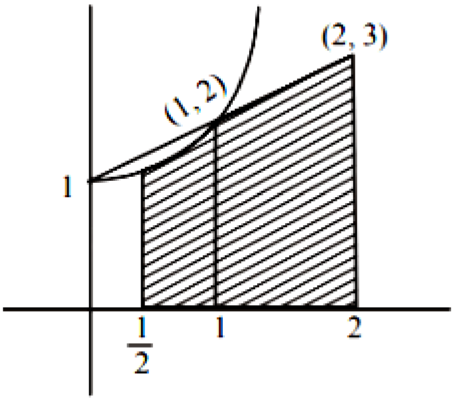
B. $\frac{79}{24}$

C. $\frac{79}{16}$

D. $\frac{23}{6}$

Answer: B

Solution:



Required area $= \int_{\frac{1}{2}}^{1}(x^2+1)dx + \int_{1}^{2}(x+1)dx$

$= \left[\frac{x^3}{3}+x\right]_{\frac{1}{2}}^{1} + \left[\frac{x^2}{2}+x\right]_{1}^{2}$

$= \left[\frac{4}{3}-\frac{13}{24}\right] + \frac{5}{2} = \frac{79}{24}$

Question63

The area (in sq. units) of the region $A = \{(x, y): |x| + |y| \leq 1, 2y^2 \geq |x|\}$ is: [Sep. 06, 2020 (I)]

Options:

A. $\frac{1}{3}$

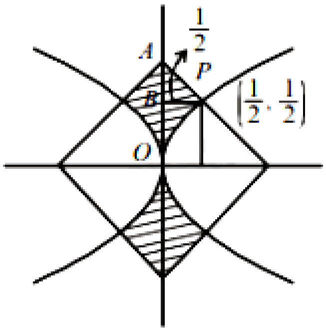
B. $\frac{7}{6}$

C. $\frac{1}{6}$

D. $\frac{5}{6}$

Answer: D

Solution:



Required area $= 4\left[\int_0^{\frac{1}{2}} 2y^2 dy + \frac{1}{2}\text{area}(\Delta PAB)\right]$

$= 4\left[\frac{2}{3}[y^3]_0^{\frac{1}{2}} + \frac{1}{2}\times\frac{1}{2}\times\frac{1}{2}\right] \;= 4\left[\frac{2}{3}\times\frac{1}{8}+\frac{1}{8}\right]$

$= 4\times\frac{5}{24} = \frac{5}{6}$

Question64

The area (in sq. units) of the region enclosed by the curves $y = x^2 - 1$ and $y = 1 - x^2$ is equal to:
[Sep. 06, 2020 (II)]

Options:

A. $\frac{4}{3}$

B. $\frac{8}{3}$

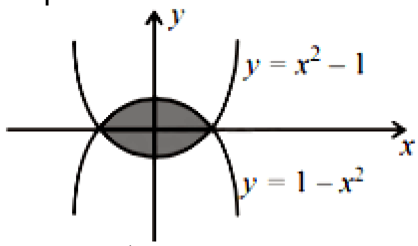
C. $\frac{7}{2}$

D. $\frac{16}{3}$

Answer: B

Solution:

Required area



Area $= 2\int_0^1 ((1-x^2)-(x^2-1))dx$

$= 4\int_0^1 (1-x^2)dx$

$= 4\left(x-\frac{x^3}{3}\right)_0^1 = 4\left(1-\frac{1}{3}\right) \; = 4 \cdot \frac{2}{3} = \frac{8}{3}$ sq. units

----

Question65

Consider a region $R = (x, y) \in \{R^2 : x^2 \le y \le 2x\}$. If a line $y = \alpha$ divides the area of region R into two equal parts, then which of the following is true?
[Sep.02,2020(II)]

Options:

A. $\alpha^3 - 6\alpha^2 + 16 = 0$

B. $3\alpha^2 - 8\alpha^{3/2} + 8 = 0$

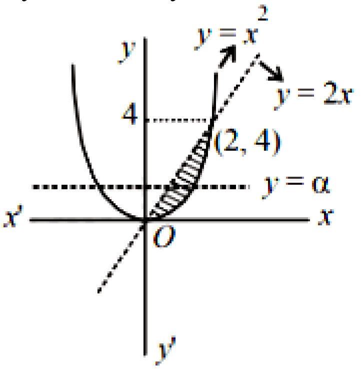
C. $3\alpha^2 - 8\alpha + 8 = 0$

D. $\alpha^3 - 6\alpha^{3/2} - 16 = 0$

Answer: B

Solution:

Let $y = x^2$ and $y = 2x$



According to question

$$\therefore \int_0^{\alpha} \left(\sqrt{y} - \frac{y}{2}\right) dy = \int_{\alpha}^{4} \left(\sqrt{y} - \frac{y}{2}\right) dy$$

$$\Rightarrow \left[\frac{y^{\frac{3}{2}}}{\frac{3}{2}}\right]_0^{\alpha} - \left[\frac{y^2}{4}\right]_0^{\alpha} = \left[\frac{y^{\frac{3}{2}}}{\frac{3}{2}}\right]_{\alpha}^{4} - \left[\frac{y^2}{4}\right]_a^4$$

$$\Rightarrow \frac{2}{3}\alpha^{3/2} - \frac{\alpha^2}{4} = \frac{2}{3}(8 - \alpha^{3/2}) - \frac{1}{4}(16 - \alpha^2)$$

$$\Rightarrow \frac{4}{3}\alpha^{3/2} - \frac{\alpha^2}{2} = \frac{4}{3}$$

$$\Rightarrow 8\alpha^{3/2} - 3\alpha^2 = 8$$

$$\therefore 3\alpha^2 - 8\alpha^{3/2} + 8 = 0$$

Question66

The area (in sq. units) bounded by the parabola $y = x^2 - 1$, the tangent at the point (2,3) to it and the y -axis is:
[Jan. 9,2019 (I)]

Options:

A. $\frac{8}{3}$

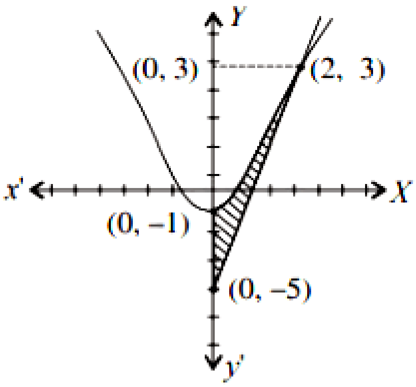
B. $\frac{32}{3}$

C. $\frac{56}{3}$

D. $\frac{14}{3}$

Answer: A

Solution:



$\because$ Curve is given as :

$y = x^2 - 1$

$\Rightarrow \frac{dy}{dx} = 2x$

$\Rightarrow \left(\frac{dy}{dx}\right)_{(2,3)} = 4$

$\therefore$ equation of tangent at (2,3)

$(y-3) = 4(x-2)$

$\Rightarrow y = 4x - 5$

but $x = 0$

$\Rightarrow y = -5$

Here the curve cuts Y-axis

$\therefore$ required area $= \frac{1}{4}\int_{-5}^{3}(y+5)dy - \int_{-1}^{3}\sqrt{y+1}\,dy$

$= \frac{1}{4}\left[\frac{y^2}{2} + 5y\right]_{-5}^{3} \frac{-2}{3}[(y+1)^{3/2}]_{-1}^{3}$

$= \frac{1}{4}\left[\frac{9}{2} + 15 - \frac{25}{2} + 25\right]$

$= -\frac{2}{3}[4^{3/2} - 0]$

$= \frac{32}{4} - \frac{16}{3} = \frac{8}{3}$ sq-units.

Question67

The area (in sq. units) of the region bounded by the parabola, $y = x^2 + 2$ and the lines, $y = x + 1$, $x = 0$ and $x = 3$ is : [Jan. 12, 2019 (I)]

Options:

A. $\frac{15}{4}$

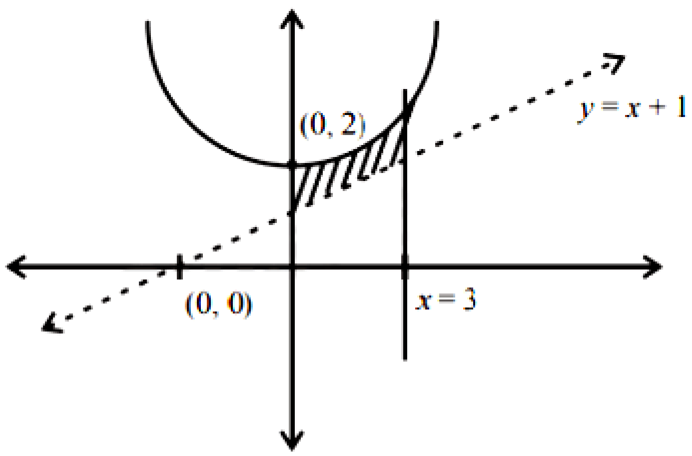
B. $\frac{21}{2}$

C. $\frac{17}{4}$

D. $\frac{15}{2}$

Answer: D

Solution:



Area of the bounded region $\int_0^3 [(x^2 + 2) - (x + 1)]d\,x$

$$= \left[\frac{x^3}{3} - \frac{x^2}{2} + x\right]_0^3$$

$$= 9 - \frac{9}{2} + 3 = \frac{15}{2}$$

Question68

The area (in sq. units) of the region bounded by the curve $x^2 = 4y$ and the straight line $x = 4y - 2$ is:
[Jan. 11, 2019 (I)]

Options:

A. $\frac{5}{4}$

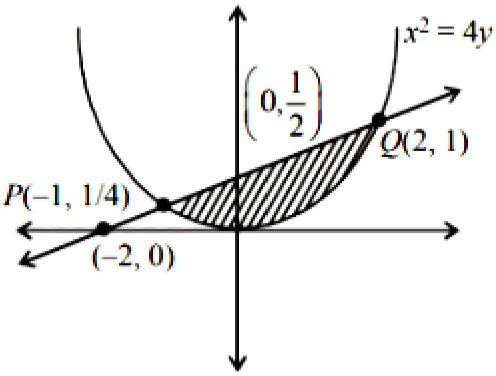
B. $\frac{9}{8}$

C. $\frac{7}{8}$

D. $\frac{3}{4}$

Answer: B

Solution:



Let points of intersection of the curve and the line be P and Q

$x^2 = 4\left(\frac{x+2}{4}\right)$

$x^2 - x - 2 = 0$

$x = 2, -1$

Point are (2,1) and $\left(-1, \frac{1}{4}\right)$

Area $= \int_{-1}^{2}\left[\left(\frac{x+2}{4}\right) - \left(\frac{x^2}{4}\right)\right]dx = \left[\frac{x^2}{8} + \frac{1}{2}x - \frac{x^3}{12}\right]_{-1}^{2} = \left(\frac{1}{2} + 1 - \frac{2}{3}\right) - \left(\frac{1}{8} - \frac{1}{2} + \frac{1}{12}\right) = \frac{9}{8}$

Question69

The area (in sq. units) in the first quadrant bounded by the parabola, $y = x^2 + 1$, the tangent to it at the point (2,5) and the coordinate axes is:
[Jan. 11, 2019 (II)]

Options:

A. $\frac{8}{3}$

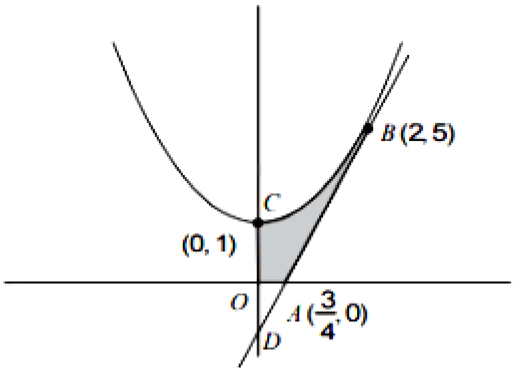
B. $\frac{37}{24}$

C. $\frac{187}{24}$

D. $\frac{14}{3}$

Answer: B

Solution:



The equation of parabola $x^2 = y - 1$
The equation of tangent at (2,5) to parabola is

$y - 5 = \left(\frac{dy}{dx}\right)_{(2,5)}(x - 2)$

$y - 5 = 4(x - 2)$

$4x - y = 3$

Then, the required area

$= \int_0^2 \{(x^2 + 1) - (4x - 3)\} dx -$ Area of ΔAOD

$\overset{2}{\underset{0}{=}} (x^2 - 4x + 4) dx - \frac{1}{2} \times \frac{3}{4} \times 3$

$= \left[\frac{(x-2)^3}{3}\right]_0^2 - \frac{9}{8} = \frac{37}{24}$

Question70

If the area enclosed between the curves $y = kx^2$ and $x = ky^2$, $(k > 0)$, is 1 square unit. Then k is:
[Jan. 10, 2019 (I)]

Options:

A. $\frac{\sqrt{3}}{2}$

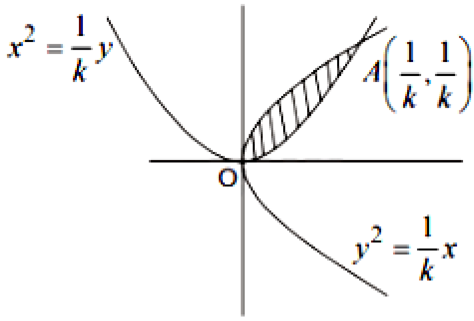
B. $\frac{1}{\sqrt{3}}$

C. $\sqrt{3}$

D. $\frac{2}{\sqrt{3}}$

Answer: B

Solution:



Two curves will intersect in the Ist quadrant at $A\left(\frac{1}{R}, \frac{1}{R}\right)$

$\because$ area of shaded region $= 1$

$$\therefore \int_0^{\frac{1}{k}} \left(\frac{\sqrt{x}}{\sqrt{k}} - kx^2\right) dx = 1$$

$$\Rightarrow \left(\frac{1}{\sqrt{k}} \cdot \frac{x^{\frac{3}{2}}}{\frac{3}{2}}\right)_0^{\frac{1}{k}} - \left(k \cdot \frac{x^3}{3}\right)_0^{\frac{1}{k}} = 1$$

$$\Rightarrow \frac{2}{3\sqrt{k}} \cdot \frac{1}{k^{\frac{3}{2}}} - \frac{k}{3k^3} = 1$$

$$\Rightarrow \frac{2}{3k^2} - \frac{1}{3k^2} = 1$$

$$\Rightarrow 3k^2 = 1$$

$$\Rightarrow k = \pm\frac{1}{\sqrt{3}}$$

$$\therefore k = \frac{1}{\sqrt{3}} (\because k > 0)$$

Question71

The area of the region $A = \{(x, y) : 0 \le y \le x\,|\,x\,| + 1$ and $-1 \le x \le 1\}$ in sq. units is:
[Jan. 09, 2019 (II)]

Options:

A. $\frac{2}{3}$

B. 2

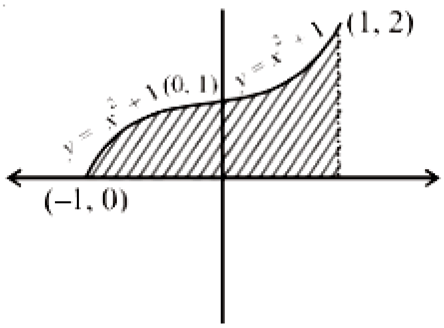
C. $\frac{4}{3}$

D. $\frac{1}{3}$

Answer: B

Solution:

Given $A = \{(x, y) : 0 \leq y \leq xx\,| + 1$ and $-1 \leq x \leq 1\}$



∴ Area of shaded region

$$= \int_{-1}^{0} (-x^2 + 1)dx + \int_{0}^{1} (x^2 + 1)dx$$

$$= \left(-\frac{x^3}{3} + x\right)_{-1}^{0} + \left(\frac{x^3}{3} + x\right)_{0}^{1}$$

$$= 0 - \left(\frac{1}{3} - 1\right) + \left(\frac{1}{3} + 1\right) - (0 + 0)$$

$= \frac{2}{3} + \frac{4}{3} = \frac{6}{3} = 2$ square units

Question72

The area (in sq. units) of the region A = {(x, y) ∈ R × R | 0d "xd "3, 0d "yd "4, yd "x^2 + 3x} is : [April 8, 2019 (I)]

Options:

A. $\frac{53}{6}$

B. 8

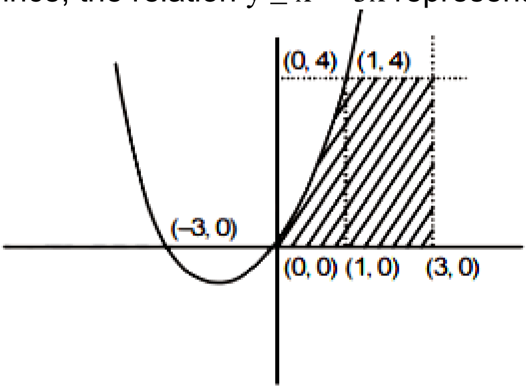
C. $\frac{59}{6}$

D. $\frac{26}{3}$

Answer: C

Solution:

Since, the relation $y \leq x^2 + 3x$ represents the region below the parabola in the 1^{st} quadrant



$\because y = 4$
$\Rightarrow x^2 + 3x = 4 \Rightarrow x = 1, -4$
$\therefore$ the required area = area of shaded region

$$= \int_0^1 (x^2 + 3x)dx + \int_1^3 4 \, . \, dx \; = \left[\frac{x^3}{3} + \frac{3x^2}{2}\right]_0^1 + [4x]_1^3$$

$$= \frac{1}{3} + \frac{3}{2} + 8 = \frac{59}{6}$$

Question73

If the area (in sq. units) of the region $\{(x, y) : y^2 \leq 4x, x + y \leq 1, x \geq 0, y \geq 0\}$ is $a\sqrt{2} + b$, then $a - b$ is equal to
[April 12, 2019 (I)]

Options:

A. $\frac{10}{3}$

B. 6

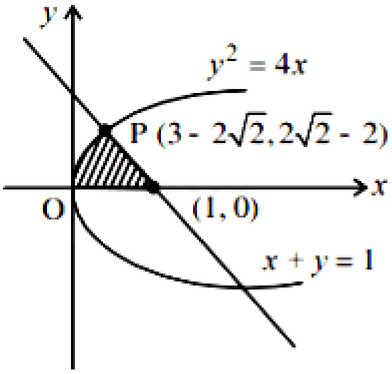
C. $\frac{8}{3}$

D. $-\frac{2}{3}$

Answer: B

Solution:

Consider $y^2 = 4x$ and $x + y = 1$



Substituting $x = 1 - y$ in the equation of parabola,

$y^2 = 4(1 - y) \Rightarrow y^2 + 4y - 4 = 0$

$\Rightarrow (y + 2)^2 = 8 \Rightarrow y + 2 = \pm 2\sqrt{2}$

Hence, required area

$$= \int_0^{3-2\sqrt{2}} 2\sqrt{x}\,dx + \frac{1}{2} \times (2\sqrt{2} - 2) \times (2\sqrt{2} - 2)$$

$$= \left[2 \times \frac{2}{3} x^{3/2}\right]_0^{3-2\sqrt{2}} + \frac{1}{2}(8 + 4 - 8\sqrt{2})$$

$$= \frac{4}{3} \times (3 - 2\sqrt{2})\sqrt{3 - 2\sqrt{2}} + 6 - 4\sqrt{2}$$

$$= \frac{4}{3}(3 - 2\sqrt{2})(\sqrt{2} - 1) + 6 - 4\sqrt{2} \ [\because (\sqrt{2} - 1)^2 = 3 - 2\sqrt{2}]$$

$$= \frac{4}{3}(3\sqrt{2} - 3 - 4 + 2\sqrt{2}) + 6 - 4\sqrt{2}$$

$$= -\frac{10}{3} + \frac{8}{3}\sqrt{2} = a\sqrt{2} + b$$

$\therefore a = 8/3$ and $b = -10/3 \Rightarrow a - b = \frac{10}{3} + \frac{8}{3} = 6$

Question74

If the area (in sq. units) bounded by the parabola $y^2 = 4\lambda x$ and the line $y = \lambda x$, $\lambda > 0$, is $\frac{1}{9}$, then λ is equal to : [April 12, 2019 (II)]

Options:

A. $2\sqrt{6}$

B. 48

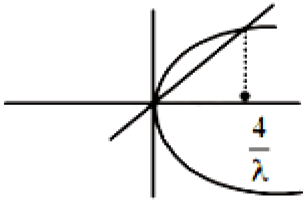
C. 24

D. $4\sqrt{3}$

Answer: C

Solution:

Given parabola $y^2 = 4\lambda x$ and the line $y = \lambda x$



Putting $y = \lambda$ in $y^2 = 4\lambda x$, we get $x = 0, \frac{4}{\lambda}$

$\therefore$ required area $= \int_0^{\frac{4}{\lambda}} (2\sqrt{\lambda x} - \lambda x)d\,x$

$= \frac{2\sqrt{\lambda}\,.\,x^{3/2}}{3/2} - \frac{\lambda x^2}{2_0^{4/\lambda}} = \frac{32}{3\lambda} - \frac{8}{\lambda}$

$= \frac{8}{3\lambda} = \frac{1}{9} \Rightarrow \lambda = 24$

Question75

The region represented by $|x - y| \le 2$ and $|x + y| \le 2$ is bounded by a : [April 10, 2019(I)]

Options:

A. square of side length $2\sqrt{2}$ units

B. rhombus of side length 2 units

C. square of area 16 sq. units

D. rhombus of area $8\sqrt{2}$ sq. units

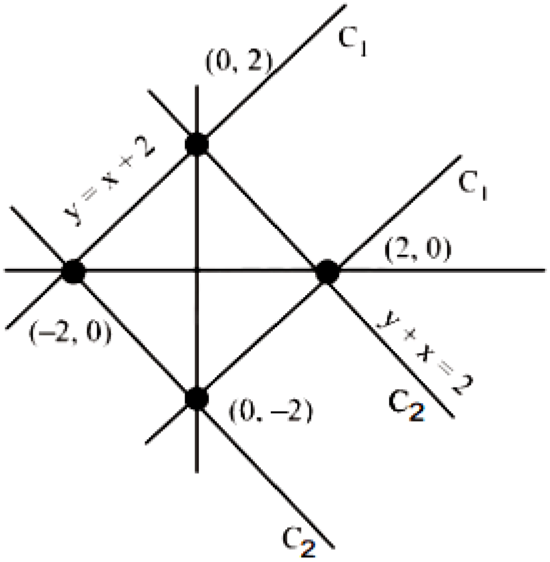
Answer: A

Solution:

Let, $C_1: |y - x| \leq 2$

$C_2: |y + x| \leq 2$

By the diagram, region is square



Now, length of side $= \sqrt{2^2 + 2^2} = 2\sqrt{2}$

Question76

The area (in sq. units) of the region bounded by the curves $y = 2^x$ and $y = |x + 1|$, in the first quadrant is : [April 10, 2019(II)]

Options:

A. $\log_e 2 + \frac{3}{2}$

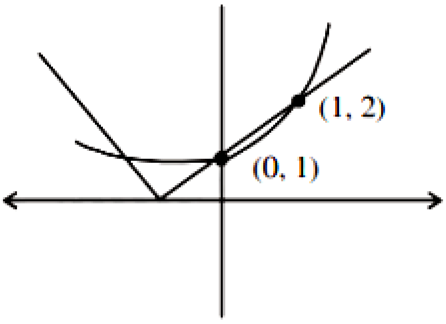
B. $\frac{3}{2}$

C. $\frac{1}{2}$

D. $\frac{3}{2} - \frac{1}{\log_e 2}$

Answer: D

Solution:



Area $=\int_0^1((x+1)-2^x)dx$ ($\because$ Area $=\int y dx$)

$$=\left[\frac{x^2}{2}+x-\frac{2^x}{\ln 2}\right]_0^1=\left(\frac{1}{2}+1-\frac{2}{\ln 2}\right)-\left(\frac{-1}{\ln 2}\right)=\frac{3}{2}-\frac{1}{\ln 2}$$

Question77

The area (in sq. units) of the region $A=\{(x, y): x^2 \leq y \leq x+2\}$ is: [April 9, 2019 (I)]

Options:

A. $\frac{10}{3}$

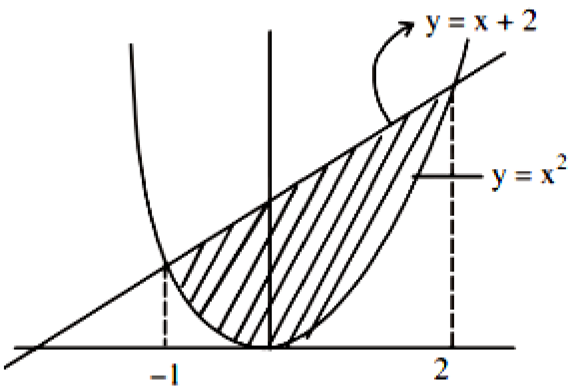
B. $\frac{9}{2}$

C. $\frac{31}{6}$

D. $\frac{13}{6}$

Answer: B

Solution:



Required area is equal to the area under the curves $y \geq x^2$ and $yd''x+2$

$\therefore$ required area $\int_{-1}^{2}((x+2)-x^2)dx$

$$= \left(\frac{x^2}{2}+2x-\frac{x^3}{3}\right)_{-1}^{2} = \left(2+4-\frac{8}{3}\right)-\left(+\frac{1}{2}-2+\frac{1}{3}\right)=\frac{9}{2}$$

Question78

The area (in sq. units) of the region A = $\left\{(x, y) : \frac{y^2}{2} \leq x \leq y+4\right\}$ is:
[April 09, 2019 (II)]

Options:

A. $\frac{53}{3}$

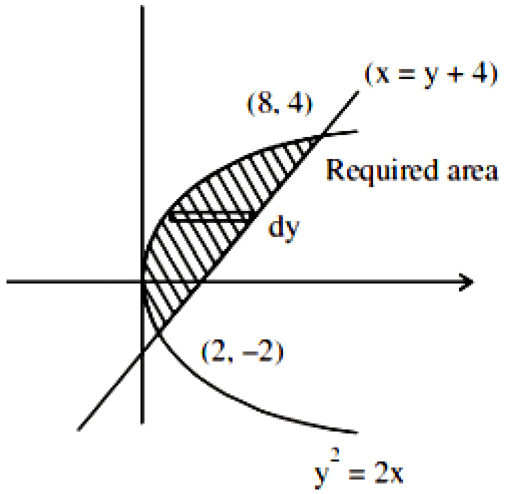
B. 30

C. 16

D. 18

Answer: D

Solution:



Given region, $A = \left\{ (x, y) : \frac{y^2}{2} \le x \le y + 4 \right\}$

Hence, area $= \int_{-2}^{4} x\,dy = \int_{-2}^{4} \left(y + 4 - \frac{y^2}{2} \right) dy$

$= \left[\frac{y^2}{2} + 4y - \frac{y^3}{6} \right]_{-2}^{4} = \left(8 + 16 - \frac{64}{6} \right) - \left(2 - 8 + \frac{8}{6} \right)$

$= \left(24 - \frac{32}{3} \right) - \left(-6 + \frac{4}{3} \right) = \frac{40}{3} + \frac{14}{3} = \frac{54}{3} = 18$

Question79

Let $S(\alpha) = \{(x, y) : y^2 \le x, 0 \le x \le \alpha\}$ and $A(\alpha)$ is area of the region $S(\alpha)$. If for aλ, $0 < \alpha < 4$, $A(\lambda) : A(\alpha) = 2 : 5$, then λequals: [April 08, 2019 (II)]

Options:

A. $2\left(\frac{4}{25}\right)^{\frac{1}{3}}$

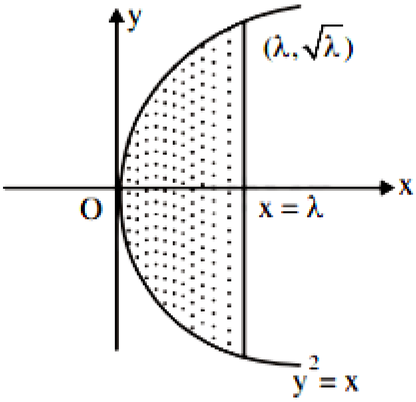
B. $2\left(\frac{2}{5}\right)^{\frac{1}{3}}$

C. $4\left(\frac{2}{5}\right)^{\frac{1}{3}}$

D. $4\left(\frac{4}{25}\right)^{\frac{1}{3}}$

Answer: D

Solution:



Area of the region $= 2 \times \int_0^{\lambda} y\,dx = 2\int_0^{\lambda} \sqrt{x}\,dx$

$= 2 \times \frac{2}{3}\pi^{\frac{3}{2}}$

$A(\lambda) = 2 \times \frac{2}{3}(\lambda \times \sqrt{\lambda}) \;= \frac{4}{3}\lambda^{\frac{3}{2}}$

Given, $\frac{A(\lambda)}{A(4)} = \frac{2}{5} \Rightarrow \frac{\lambda^{\frac{3}{2}}}{8} = \frac{2}{5}$

$\lambda = \left(\frac{16}{5}\right)^{\frac{2}{3}} = 4 \, . \left(\frac{4}{25}\right)^{\frac{1}{3}}$

Question80

Let $g(x) = \cos x^2$, $f(x) = \sqrt{x}$, and alpha, $\beta(\alpha < \beta)$ be the roots of the quadratic equation $18x^2 - 9\pi x + \pi^2 = 0$. Then the area (in sq. units) bounded by the curve $y = (gof)(x)$ and the lines $x = \alpha$, $x = \beta$ and $y = 0$, is :
[2018]

Options:

A. $\frac{1}{2}(\sqrt{3} + 1)$

B. $\frac{1}{2}(\sqrt{3} - \sqrt{2})$

C. $\frac{1}{2}(\sqrt{2} - 1)$

D. $\frac{1}{2}(\sqrt{3}-1)$

Answer: D

Solution:

Here, $18x^2 - 9\pi x + \pi^2 = 0$
$\Rightarrow (3x-\pi)(6x-\pi) = 0$
$\Rightarrow \alpha = \frac{\pi}{6}, \beta = \frac{\pi}{3}$
Also, gof $(x) = \cos x$
$\therefore$ Req. area $= \int_{\pi/6}^{\pi/3} \cos x \, d\,x = \frac{\sqrt{3}-1}{2}$

Question81

If the area of the region bounded by the curves, $y = x^2, y = \frac{1}{x}$ and the lines $y = 0$ and $x = t(t > 1)$ is 1sq. unit, then t is equal to [Online April 16, 2018]

Options:

A. $\frac{4}{3}$

B. $e^{2/3}$

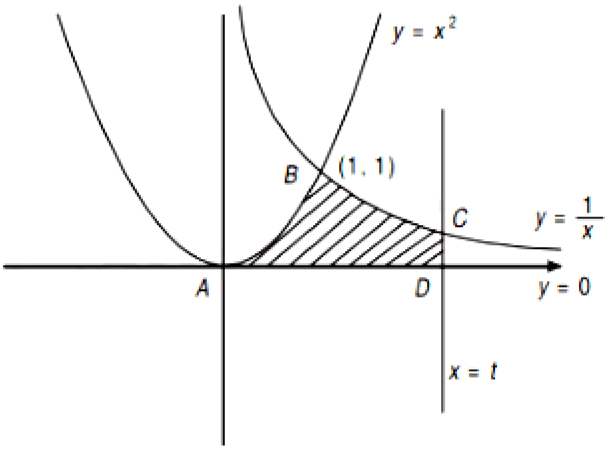
C. $\frac{3}{2}$

D. $e^{3/2}$

Answer: B

Solution:

The intersection point of $y = x^2$ and $y = \frac{1}{x}$ is (1,1)
Area bounded by the curves is the region ABCDA is given as:
Area $= \int_0^1 x^2 d\,x + \int_1^t \frac{1}{x} d\,x$
$= \left[\frac{x^3}{3}\right]_0^1 + [\ln(x)]_1^t = \frac{1}{3} + \ln(t)$



$\because$ area $= 1$

$\Rightarrow \frac{1}{3} + \ln(t) = 1 \Rightarrow \ln(t) = \frac{2}{3} \Rightarrow t = e^{\frac{2}{3}}$

Question82

The area (in sq. units) of the region $\{x \in R : x \geq 0, y \geq 0, y \geq x - 2$ and $y \leq \sqrt{x}\}$, is
[Online April 15, 2018]

Options:

A. $\frac{13}{3}$

B. $\frac{10}{3}$

C. $\frac{5}{3}$

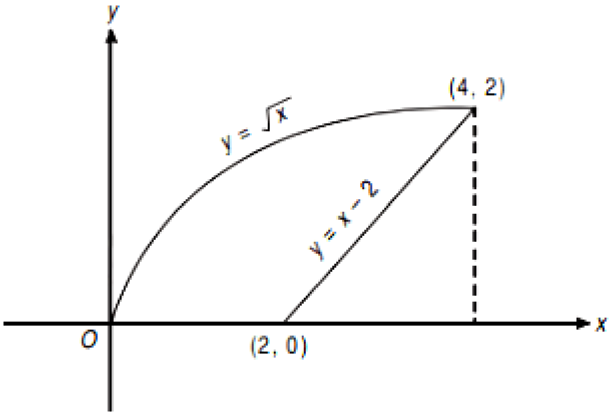
D. $\frac{8}{3}$

Answer: B

Solution:

The intersection point of $y = x - 2$ and $y = \sqrt{x}$ is (4,2) .

The required area $= \int_0^4 \sqrt{x}\, dx - \frac{1}{2} \times 2 \times 2\ \frac{16}{3} - 2 = \frac{10}{3}$



Question83

The area (in sq. units) of the region $\{(x, y) : x \geq 0, x + y \leq 3, x^2 \leq 4y \text{ and } y \leq 1 + \sqrt{x}\}$ is [2017]

Options:

A. $\frac{5}{2}$

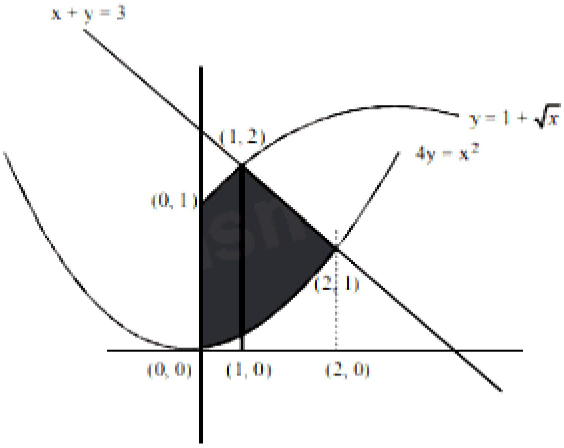
B. $\frac{59}{12}$

C. $\frac{3}{2}$

D. $\frac{7}{3}$

Answer: A

Solution:



Area of shaded region $= \int_0^1 (1+\sqrt{x})dx + \int_1^2 (3-x)dx - \int_0^2 \frac{x^2}{4}dx$

$= [x]_0^1 + \left[\frac{x^{\frac{3}{2}}}{\frac{3}{2}}\right]_0^1 + [3x]_1^2 - \left[\frac{x^2}{2}\right]_1^2 - \left[\frac{x^3}{12}\right]_0^2 = \frac{5}{2}$ sq.units

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Question84

The area (in sq. units) of the smaller portion enclosed between the curves, $x^2 + y^2 = 4$ and $y^2 = 3x$, is:
[Online April 8, 2017]

Options:

A. $\frac{1}{2\sqrt{3}} + \frac{\pi}{3}$

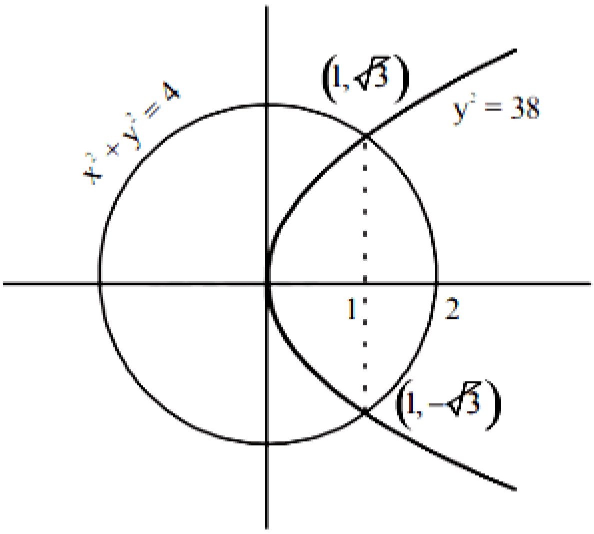
B. $\frac{1}{\sqrt{3}} + \frac{2\pi}{3}$

C. $\frac{1}{2\sqrt{3}} + \frac{2\pi}{3}$

D. $\frac{1}{\sqrt{3}} + \frac{4\pi}{3}$

Answer: D

Solution:



From the equations we get;
$x^2 + 3x - 4 = 0$
$\Rightarrow (x+4)(x-1) = 0 \Rightarrow x = -4, x = 1$
when $x = 1, y = \sqrt{3}$

Area $= \int_0^1 \left(\int_0^1 \sqrt{3} . \sqrt{x} d x + \int_1^2 \sqrt{4-x^2} . d x \right) \times 2$

$= \left(\sqrt{3} \left(\frac{x^{3/2}}{3/2} \right)_0^1 + \left(\frac{x}{2} \sqrt{4-x^2} + 2\sin^{-1}\frac{x}{2} \right)_1^2 \right) \times 2$

$= \left(\sqrt{3} \left(\frac{2}{3} \right) + \left\{ 2 . \frac{\pi}{2} - \left(\frac{\sqrt{3}}{2} + \frac{\pi}{3} \right) \right\} \right) \times 2$

$= \left(\frac{2}{\sqrt{3}} - \frac{\sqrt{3}}{2} + \frac{2\pi}{3} \right) \times 2$

$= \left(\frac{1}{2\sqrt{3}} + \frac{2\pi}{3} \right) \times 2 = \frac{1}{\sqrt{3}} + \frac{4\pi}{3}$

Question85

The area (in sq. units) of the region $\{ (x, y) : y^2 \geq 2x$ and $x^2 + y^2 \leq 4x, x \geq 0, y \geq 0 \}$ is:
[2016]

Options:

A. $\pi - \frac{4\sqrt{2}}{3}$

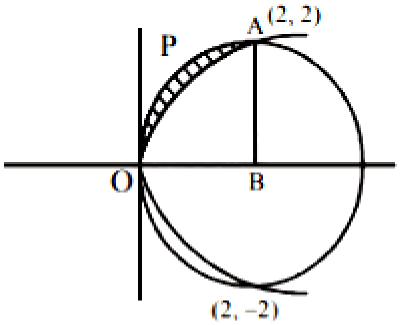
B. $\frac{\pi}{2} - \frac{2\sqrt{2}}{3}$

C. $\pi - \frac{4}{3}$

D. $\pi - \frac{8}{3}$

Answer: D

Solution:



Points of intersection of the two curves are (0,0),(2,2) and (2,-2)
Area = Area (OPAB)− area under parabola (0 to 2)

$= \frac{\pi \times (2)^2}{4} - \int_0^2 \sqrt{2}\sqrt{x}\,dx = \pi - \frac{8}{3}$

Question86

The area (in sq. units) of the region described by $A = \{(x, y) \mid y \geq x^2 - 5x + 4, x + y \geq 1, y \leq 0\}$ is: [Online April 9, 2016]

Options:

A. $\frac{19}{6}$

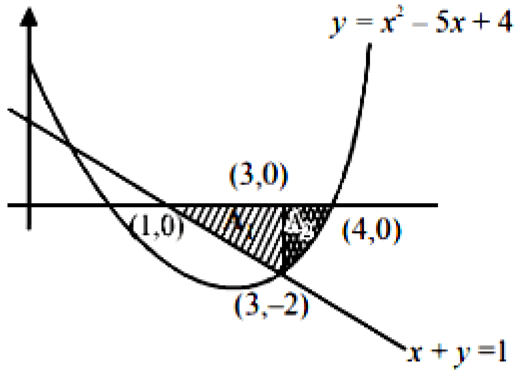
B. $\frac{17}{6}$

C. $\frac{7}{2}$

D. $\frac{13}{6}$

Answer: A

Solution:



Required area $= A_1 + A_2$

$= \frac{1}{2} \times 2 \times 2 + \left| \int_3^4 (x^2 - 5x + 4) dx \right|$

$= 2 + \frac{7}{6} = \frac{19}{6}$ sq. units

Question87

The area (in sq. units) of the region described by $\{(x, y) : y^2 \leq 2x$ and $y \geq 4x - 1\}$ is [2015]

Options:

A. $\frac{15}{64}$

B. $\frac{9}{32}$

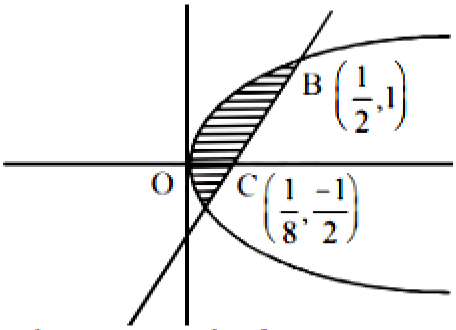
C. $\frac{7}{32}$

D. $\frac{5}{64}$

Answer: B

Solution:

Required area



$$= \int_{-1/2}^{1} \frac{y+1}{4} dy - \int_{-1/2}^{1} \frac{y^2}{2} dy$$

$$= \frac{1}{4}\left[\frac{y^2}{2}+y\right]_{-1/2}^{1} - \frac{1}{2}\left[\frac{y^3}{3}\right]_{-1/2}^{1}$$

$$= \frac{1}{4}\left[\frac{3}{2}+\frac{3}{8}\right] - \frac{9}{48} = \frac{15}{32} - \frac{9}{48} = \frac{27}{96} = \frac{9}{32}$$

Question88

The area (in square units) of the region bounded by the curves $y + 2x^2 = 0$ and $y + 3x^2 = 1$, is equal to [Online April 10, 2015]

Options:

A. $\frac{3}{5}$

B. $\frac{1}{3}$

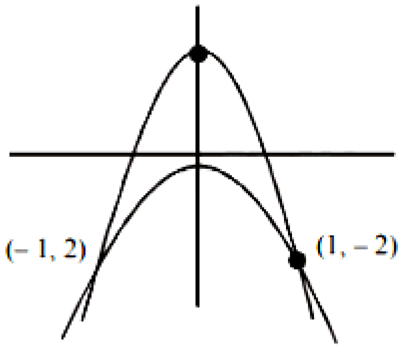
C. $\frac{4}{3}$

D. $\frac{3}{4}$

Answer: C

Solution:

Solving
$y + 2x^2 = 0$
$y + 3x^2 = 1$



Point of intersection (1,-2) and (-1,-2)

Area $= 2\int_0^1 ((1-3x^2)-(-2x^2))dx$

$2\int_0^1 (1-x^2)dx = 2\left(x-\frac{x^3}{3}\right)_0^1 = \frac{4}{3}$

$= 15 - 6 = 9$ sq units

Question89

The area of the region described by $A = \{(x, y) : x^2 + y^2 \leq 1$ and $y^2 \leq 1 - x\}$ is:
[2014]

Options:

A. $\frac{\pi}{2} - \frac{2}{3}$

B. $\frac{\pi}{2} + \frac{2}{3}$

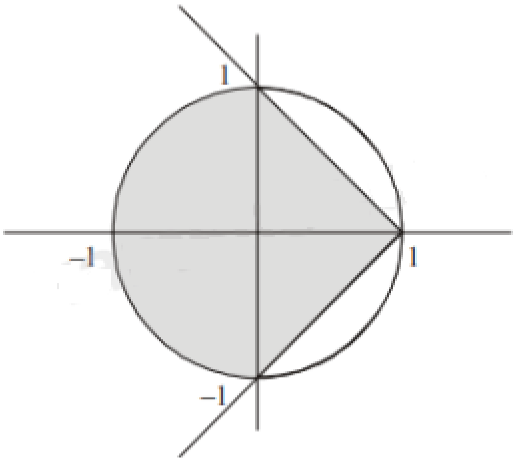
C. $\frac{\pi}{2} + \frac{4}{3}$

D. $\frac{\pi}{2} - \frac{4}{3}$

Answer: C

Solution:

Given curves are $x^2 + y^2 = 1$ and $y^2 = 1 - x$.
Intersecting points are $x = 0, 1$



Area of shaded portion is the required area.
So, Required Area = Area of semi-circle + Area bounded by parabola

$= \frac{\pi r^2}{2} + 2\int_0^1 \sqrt{1-x}\,dx = \frac{\pi}{2} + 2\int_0^1 \sqrt{1-x}\,dx$ ($\because$ radius of circle $= 1$)

$= \frac{\pi}{2} + 2\left[\frac{(1-x)^{3/2}}{-3/2}\right]_0^1 = \frac{\pi}{2} - \frac{4}{3}(-1) = \frac{\pi}{2} + \frac{4}{3}$ sq. unit

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Question90

The area of the region above the x -axis bounded by the curve $y = \tan x, 0 \le x \le \frac{\pi}{2}$ and the tangent to the curve at$x = \frac{\pi}{4}$ is: [Online April 19, 2014]

Options:

A. $\frac{1}{2}\left(\log 2 - \frac{1}{2}\right)$

B. $\frac{1}{2}\left(\log 2 + \frac{1}{2}\right)$

C. $\frac{1}{2}(1 - \log 2)$

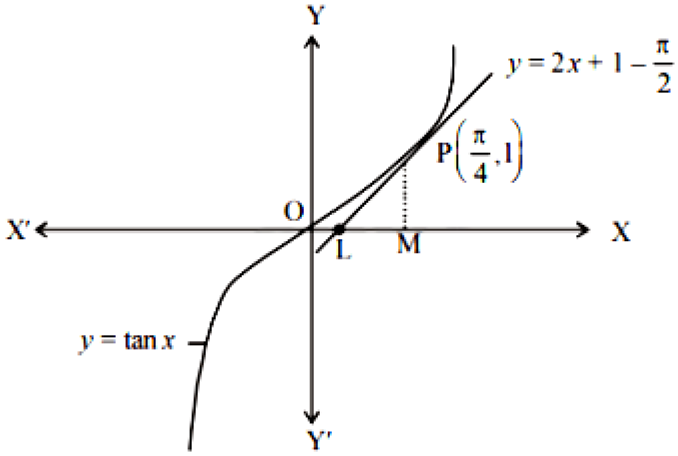
D. $\frac{1}{2}(1 + \log 2)$

Answer: A

Solution:

The given curve is $y = \tan x$ ....(1)
when $x = \frac{\pi}{4}, y = 1$

Equation of tangent at P is $y - 1 = \left(\sec^2\frac{\pi}{4}\right)\left(x - \frac{\pi}{4}\right)$



or $y = 2x + 1 - \frac{\pi}{2}$ ....(2)

Area of shaded region

= area of $OPM\,O - ar(\Delta PLM)$

$$= \int_0^{\frac{\pi}{4}} \tan x \, dx - \frac{1}{2}(OM - OL)PM$$

$$= [\log \sec x]_0^{\frac{\pi}{4}} - \frac{1}{2}\left\{\frac{\pi}{4} - \frac{\pi - 2}{4}\right\} \times 1$$

$$= \frac{1}{2}\left[\log 2 - \frac{1}{2}\right]$$ sq unit

Question91

Let $A = \{(x, y) : y^2 \le 4x, y - 2x \ge -4\}$. The area (in square units) of the region A is:
[Online April 9, 2014]

Options:

A. 8

B. 9

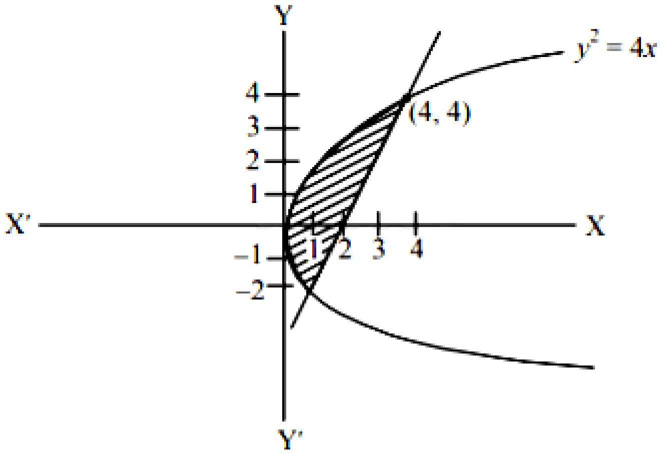
C. 10

D. 11

Answer: B

Solution:

Area of shaded portion



$$=\left|\int_{2}^{4}\left(\frac{y+4}{2}\right)dy\right|-\left|\int_{-2}^{4}\frac{y^2}{4}dy\right|$$

$$=\left|\frac{1}{2}\left[\frac{y^2}{2}+4y\right]_{-2}^{4}\right|-\left|\frac{1}{4}\left[\frac{y^3}{3}\right]_{-2}^{4}\right|$$

$$=\left|\frac{1}{2}[\{8+16\}-\{2-8\}]\right|-\left|\frac{1}{4}\left\{\frac{64}{3}+\frac{8}{3}\right\}\right|=9$$

Question92

Let $f : [-2, 3] \rightarrow [0, \infty)$ be a continuous function such that $f(1-x) = f(x)$ for all $x \in [-2, 3]$.
If R_1 is the numerical value of the area of the region bounded by $y = f(x)$, $x = -2$, $x = 3$ and the axis of x and $R_2 = \int_{-2}^{3} xf(x)dx$, then :
[Online April 25, 2013]

Options:

A. $3R_1 = 2R_2$

B. $2R_1 = 3R_2$

C. $R_1 = R_2$

D. $R_1 = 2R_2$

Answer: D

Solution:

We have

$R_2 = \int_{-2}^{3} xf(x)dx = \int_{-2}^{3} (1-x)f(1-x)dx$ Using $\left[\int_a^b f(x)dx = \int_a^b f(a+b-x)dx\right]$

$\Rightarrow R_2 = \int_{-2}^{3} (1-x)f(x)dx$ $(\because f(x) = f(1-x)$ on $[-2, 3])$

$\therefore R_2 + R_2 = \int_{-2}^{3} xf(x)dx + \int_{-2}^{3} (1-x)f(x)dx = \int_{-2}^{3} f(x)dx = R_1$

$\Rightarrow R_1 = 2R_2$

Question93

The area (in square units) bounded by the curves $y = \sqrt{x}$, $2y - x + 3 = 0$, x -axis, and lying in the first quadrant is: [2013]

Options:

A. 9

B. 36

C. 18

D. $\frac{27}{4}$

Answer: A

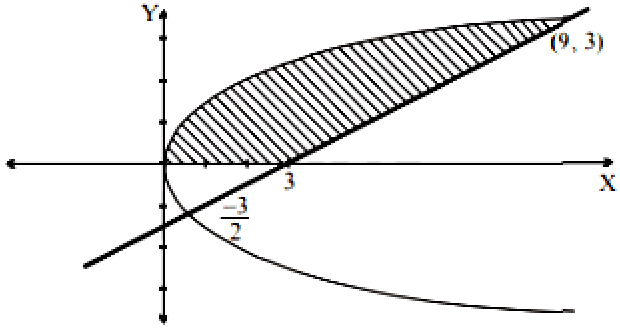
Solution:

Given curves are

$y = \sqrt{x}$ .....(1)

and $2y - x + 3 = 0$ ....(2)

On solving both we get $y = -1, 3$



Required area $= \int_0^3 \{(2y+3) - y^2\} dy$

$= \left[y^2 + 3y - \frac{y^3}{3} \right]_0^3 = 9$

Question94

The area under the curve $y = |\cos x - \sin x|$, $0 \le x \le \frac{\pi}{2}$, and above x -axis is :
[Online April 23, 2013]

Options:

A. $2\sqrt{2}$

B. $2\sqrt{2} - 2$

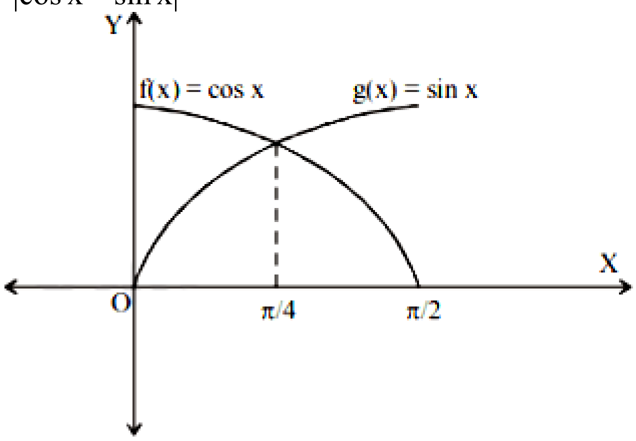
C. $2\sqrt{2} + 2$

D. 0

Answer: B

Solution:

$y = |\cos x - \sin x|$



Required area $= 2\int_0^{\pi/4} (\cos x - \sin x) dx$

$= 2[\sin x + \cos x]_0^{\pi/4}$

$= 2\left[\frac{2}{\sqrt{2}} - 1\right] = (2\sqrt{2} - 2)$ sq. units

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Question95

The area of the region (in sq. units), in the first quadrant bounded by the parabola $y = 9x^2$ and the lines x = 0, y = 1 and y = 4, is:
[Online April 22, 2013]

Options:

A. 7/9

B. 14/3

C. 7/3

D. 14/9

Answer: D

Solution:

Required area $= \int_{y=1}^{4} \sqrt{\frac{y}{9}}\, dy$

$= \frac{1}{3}\int_{y=1}^{4} y^{1/2} dy = \frac{1}{3} \times \frac{2}{3}(y^{3/2})|^4$

$= \frac{2}{9}[(4^{1/2})^3 - (1^{1/2})^3] = \frac{2}{9}[8-1]$

$= \frac{2}{9} \times 7 = \frac{14}{9}$ sq. units.

Question96

The area bounded by the curve y = ln(x) and the lines y = 0, y = ln(c) and x = 0 is equal to :
[Online April 9, 2013]

Options:

A. 3

B. 3 ln(c) − 2

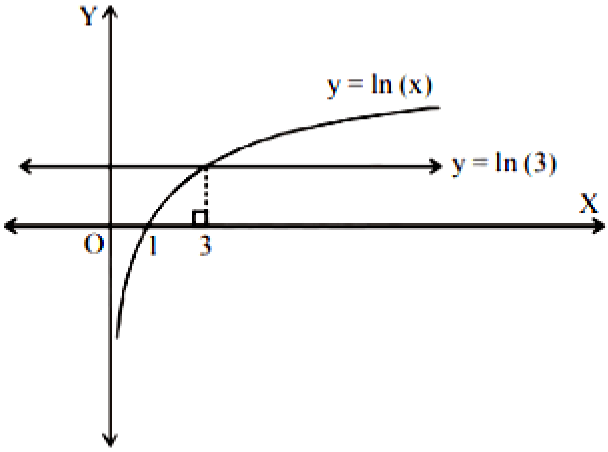
C. $3\ln(c) + 2$

D. 2

Answer: D

Solution:

To find the point of intersection of curves $y = \ln(x)$ and $y = \ln(3)$, put $\ln(x) = \ln(3)$
$\Rightarrow \ln(x) - \ln(3) = 0$
$\Rightarrow \ln(x) - \ln(3) = \ln(1)$
$\Rightarrow \frac{x}{3} = 1, \Rightarrow x = 3$



Required area $= \int_0^3 \ln(3)\,d\,x - \int_1^3 \ln(x)\,d\,x$
$= [x\ln(3)]_0^3 - [x\ln(x) - x]_1^3 = 2$

Question97

The area between the parabolas $x^2 = \frac{y}{4}$ and $x^2 = 9y$ and the straight line $y = 2$ is:
[2012]

Options:

A. $20\sqrt{2}$

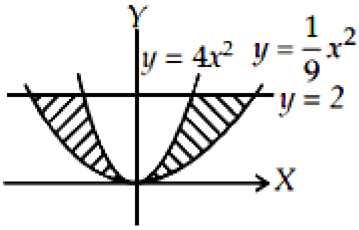
B. $\frac{10\sqrt{2}}{3}$

C. $\frac{20\sqrt{2}}{3}$

D. $10\sqrt{2}$

Answer: C

Solution:



Required area $= 2\int_0^2 \left(\sqrt{9y} - \sqrt{\frac{y}{4}}\right) dy$

$$= 2\int_0^2 \left(3\sqrt{y} - \frac{\sqrt{y}}{2}\right) dy = 2\left[\frac{2}{3} \times 3 \,.\, y^{\frac{3}{2}} - \frac{1}{2} \times \frac{2}{3} \,.\, y^{\frac{3}{2}}\right]_0^2$$

$$= 2\left[2y^{\frac{3}{2}} - \frac{1}{3}y^{\frac{3}{2}}\right]_0^2 = 2 \times \left[\frac{5}{3}y^{\frac{3}{2}}\right]_0^2$$

$$= 2 \,.\, \frac{5}{3} 2\sqrt{2} = \frac{20\sqrt{2}}{3}$$

Question98

The area bounded by the parabola $y^2 = 4x$ and the line $2x - 3y + 4 = 0$, in square unit, is [Online May 26, 2012]

Options:

A. $\frac{2}{5}$

B. $\frac{1}{3}$

C. 1

D. $\frac{1}{2}$

Answer: B

Solution:

Intersecting points are $x = 1, 4$

$\therefore$ Required area $= \int_1^4 \left[2\sqrt{x} - \left(\frac{2x+4}{3} \right) \right] dx$

$= \frac{2x^{3/2}}{3/2}\Big|_1^4 - \frac{2x^2}{3 \times 2}\Big|_1^4 - \frac{4}{3}x\Big|_1^4$

$= \frac{4}{3}(4^{3/2} - 1^{3/2}) - \frac{1}{3}(16-1) - \left[\frac{4}{3}(4) - \frac{4}{3} \right]$

$= \frac{4}{3}(7) - 5 - 4 = \frac{28}{3} - 9 = \frac{28-27}{3} = \frac{1}{3}$

Question99

The area of the region bounded by the curve $y = x^3$, and the lines, $y = 8$, and $x = 0$, is [Online May 19, 2012]

Options:

A. 8

B. 12

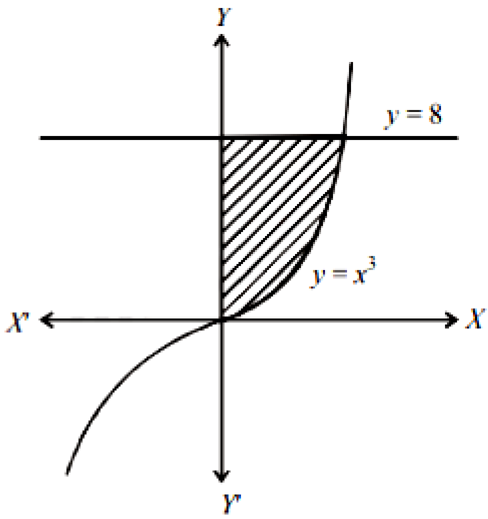
C. 10

D. 16

Answer: B

Solution:

Required Area $= \int_{y=0}^{8} y^{1/3} dy$



$$= \left.\frac{y^{1/3+1}}{\frac{1}{3}+1}\right|_0^8 = \frac{3}{4}(y^{4/3})\Big|_0^8$$

$$= \frac{3}{4}\left[(8)^{\frac{4}{3}} - 0\right] = \frac{3}{4}[2^4] = \frac{3}{4} \times 16 \ = 12\text{sq. unit}$$

Question100

If a straight line $y - x = 2$ divides the region $x^2 + y^2 \leq 4$ into two parts, then the ratio of the area of the smaller part to the area of the greater part is
[Online May 12, 2012]

Options:

A. $3\pi - 8 : \pi + 8$

B. $\pi - 3 : 3\pi + 3$

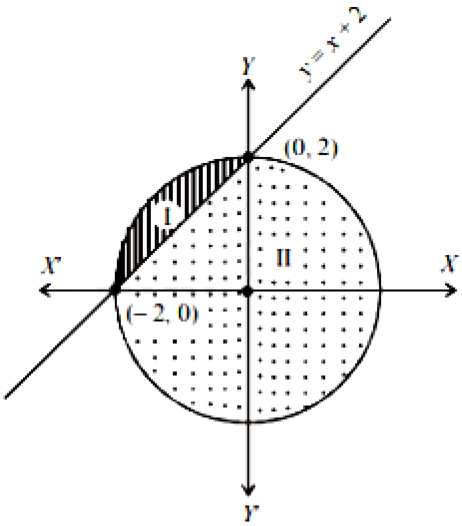
C. $3\pi - 4 : \pi + 4$

D. $\pi - 2 : 3\pi + 2$

Answer: D

Solution:

Let I be the smaller portion and II be the greater portion of the given figure then,



Area of $\mathrm{I} = \int_{-2}^{0}\left[\sqrt{4-x^2}-(x+2)\right]dx$

$= \left[\frac{x}{2}\sqrt{4-x^2}+\frac{4}{2}\sin^{-1}\left(\frac{x}{2}\right)\right]_{-2}^{0}-\left[\frac{x^2}{2}+2x\right]_{-2}^{0}$

$= [2\sin^{-1}(-1)]-\left[-\frac{4}{2}+4\right] = 2\times\frac{\pi}{2}-2=\pi-2$

Now, area of II = Area of circle-area of I. $= 4\pi-(\pi-2)=3\pi+2$

Hence, required ratio $= \frac{\text{area of I}}{\text{area of II}} = \frac{\pi-2}{3\pi+2}$

Question101

The area enclosed by the curves $y = x^2$, $y = x^3$, $x = 0$ and $x = p$, where $p > 1$, is $1/6$. The p equals
[Online May 12, 2012]

Options:

A. 8/3

B. 16/3

C. 2

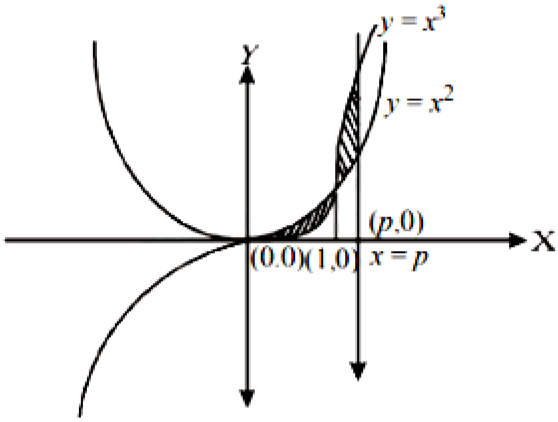
D. 4/3

Answer: D

Solution:

(d) Given curves are $y = x^2$ and $y = x^3$
Also, $x = 0$ and $x = p, p > 1$

Now, intersecting point is (1,1)



Required Area $= \int_0^1 (x^2 - x^3)dx + \int_1^p (x^3 - x^2)dx$

$\frac{1}{6} = \frac{x^3}{3} - \frac{x^4}{4}\Big|_0^1 + \frac{x^4}{4} - \frac{x^3}{3}\Big|_1^p$

$\Rightarrow \frac{1}{6} = \left(\frac{1}{3} - \frac{1}{4}\right) + \left(\frac{p^4}{4} - \frac{p^3}{3} - \frac{1}{4} + \frac{1}{3}\right)$

$\Rightarrow \frac{1}{6} - \frac{1}{3} + \frac{1}{4} + \frac{1}{4} - \frac{1}{3} = \frac{3p^4 - 4p^3}{12}$

$\Rightarrow \frac{p^3(3p-4)}{12} = 0 \Rightarrow p^3(3p-4) = 0$

$\Rightarrow p = 0$ or $\frac{4}{3}$

Since, it is given that $p > 1$

$\therefore p$ can not be zero.

Hence, $p = \frac{4}{3}$

Question102

The parabola $y^2 = x$ divides the circle $x^2 + y^2 = 2$ into two parts whose areas are in the ratio [Online May 7, 2012]

Options:

A. $9\pi + 2 : 3\pi - 2$

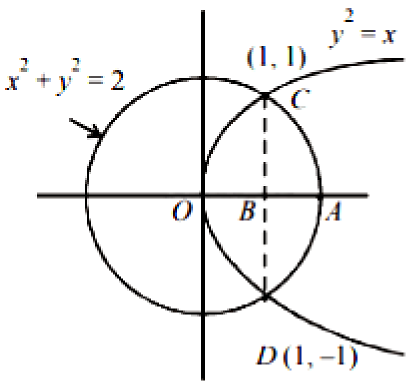
B. $9\pi - 2 : 3\pi + 2$

C. $7\pi i - 2 : 2\pi - 3$

D. $7\pi + 2 : 3\pi + 2$

Answer: B

Solution:



Area of circle $=\pi(\sqrt{2})^2=2\pi$

Area of OCADO = 2{Area of OCAO}

= 2 {area of OCB + area of BCA}

$=2\int_0^1 y_p\,dx+2\int_1^{\sqrt{2}} y_c\,dx$

where $y_p=\sqrt{x}$ and $y_c=\sqrt{2-x^2}$

∴ Required Area $=2\int_0^1 \sqrt{x}\,dx+2\int_1^{\sqrt{2}} \sqrt{2-x^2}\,dx$

$=2\left[\frac{2}{3}.1-0\right]+2\left[\frac{x\sqrt{2-x^2}}{2}+\sin^{-1}\frac{x}{\sqrt{2}}\right]_1^{\sqrt{2}}$

$=\frac{4}{3}+2\left\{\frac{\pi}{2}-\frac{\pi}{4}-\frac{1}{2}\right\}=\frac{4}{3}+2\left\{\frac{\pi}{4}-\frac{1}{2}\right\}=\frac{3\pi+2}{6}$

Bigger area $=2\pi-\frac{3\pi+2}{6}=\frac{9\pi-2}{6}$

∴ Required Ratio $=\frac{9\pi-2}{3\pi+2}$ i.e., $9\pi-2:3\pi+2$

Question103

The area bounded by the curves $y^2 = 4x$ and $x^2 = 4y$ is: [2011 RS]

Options:

A. $\frac{32}{3}$sq units

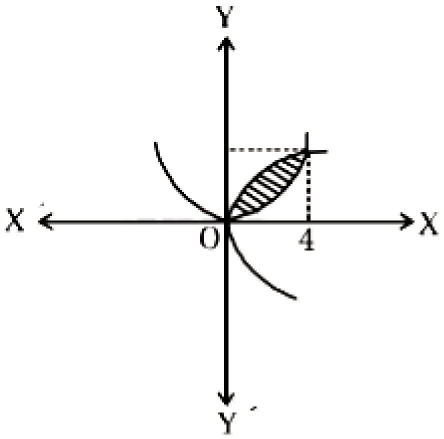
B. $\frac{16}{3}$sq units

C. $\frac{8}{3}$sq. units

D. 0sq. units

Answer: B

Solution:



Required area $= \int_0^4 \left(2\sqrt{x} - \frac{x^2}{4}\right) dx$

$= \left[2\left(\frac{x^{3/2}}{3/2}\right) - \frac{x^3}{12}\right]_0^4$

$= \frac{4}{3} \times 8 - \frac{64}{12} = \frac{32}{3} - 16 \; = \frac{16}{3}$ sq. units

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Question104

The area of the region enclosed by the curves y = x, x = e, y = $\frac{1}{x}$ and the positive x -axis is
[2011]

Options:

A. 1 square unit

B. $\frac{3}{2}$ square units

C. $\frac{5}{2}$ square units

D. $\frac{1}{2}$ square unit

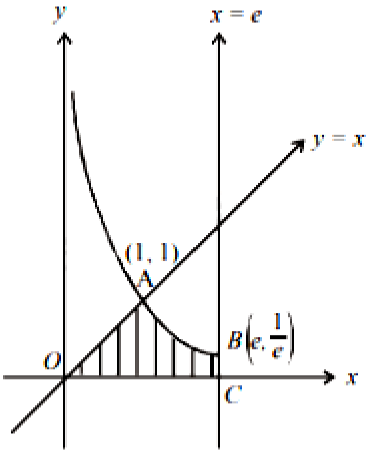
Answer: B

Solution:

Area of required region AOCBO

$= \int_0^1 x\,dx + \int_1^e \frac{1}{x}\,dx = \left[\frac{x^2}{2}\right]_0^1 + [\log x]_1^e$

$= \frac{1}{2} + 1 = \frac{3}{2}$ sq. units



Question105

The area bounded by the curves y = cos x and y = sin x between the ordinates x = 0 and x = $\frac{3\pi}{2}$ is [2010]

Options:

A. $4\sqrt{2} + 2$

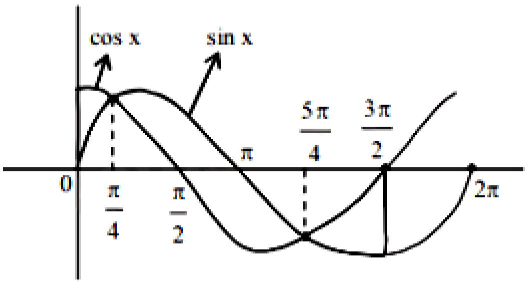
B. $4\sqrt{2} - 1$

C. $4\sqrt{2} + 1$

D. $4\sqrt{2} - 2$

Answer: D

Solution:



Area above x -axis = Area below x -axis

$\therefore$ Required area $= 2\left[\int_0^{\pi/4}(\cos x - \sin x)dx + \int_{\pi/4}^{\pi}\sin x\, dx - \int_{\pi/4}^{\pi/2}\cos x\, dx\right]$

$= 2[(\sin x + \cos x)_0^{\pi/4} + (-\cos x)_{\pi/4}^{\pi} - (\sin x)_{\frac{\pi}{4}}^{\frac{\pi}{2}}]$

$= 2\left[\left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}\right) - (0+1) + \left(1 + \frac{1}{\sqrt{2}}\right) - \left(1 - \frac{1}{\sqrt{2}}\right)\right]$

$= 2\left[\sqrt{2} - 1 + 1 + \frac{1}{\sqrt{2}} - 1 + \frac{1}{\sqrt{2}}\right]$

$= 2[\sqrt{2} + \sqrt{2} - 1] = 4\sqrt{2} - 2$

Question106

The area of the region bounded by the parabola $(y-2)^2 = x - 1$, the tangent of the parabola at the point (2,3) and the x -axis is: [2009]

Options:

A. 6

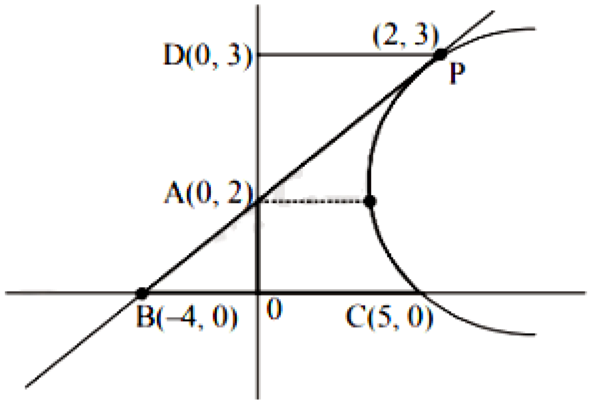
B. 9

C. 12

D. 3

Answer: B

Solution:



For slope of tangents at (2,3)

$(y-2)^2 = x-1$

$2(y-2)\frac{dy}{dx} = 1 \Rightarrow \frac{dy}{dx} = \frac{1}{2(y-2)} m = \left(\frac{dy}{dx}\right)_{(2,3)} = \frac{1}{2(3-2)} = \frac{1}{2}$

Equation of tangent

$y-3 = \frac{1}{2}(x-2)$

$\Rightarrow x-2y+4=0$ .....(i)

The given parabola is $(y-2)^2 = x-1$ ....(ii)

vertex (1,2) and it meets x -axis at (5,0)

Then required area $= Ar\Delta BOA + Ar(OCPD) - Ar(\Delta APD)$

$= \frac{1}{2} \times 4 \times 2 + \int_0^3 x\,dy - \frac{1}{2} \times 2 \times 1$

$= 3 + \int_0^3 (y-2)^2 + 1\,dy = 3 + \left[\frac{(y-2)^3}{3} + y\right]_0^3$

$= 3 + \left[\frac{1}{3} + 3 + \frac{8}{3}\right] = 3 + 6 = 9$sq. units

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Question107

The area of the plane region bounded by the curves $x + 2y^2 = 0$ and $x + 3y^2 =$ lis equal to [2008]

Options:

A. $\frac{5}{3}$

B. $\frac{1}{3}$

C. $\frac{2}{3}$

D. 43

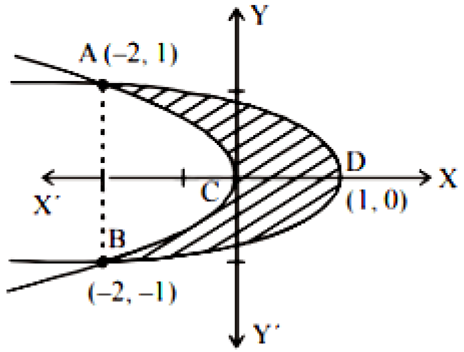
Answer: D

Solution:

Given $x+2y^2=0 \Rightarrow y^2=-\frac{x}{2}$

and $x+3y^2=1 \Rightarrow y^2=-\frac{1}{3}(x-1)$

On solving these two equations we get the points of intersection as (-2,1),(-2,-1)



The required area is ACBDA, given by

$$A=2\left\{\int_{-2}^{1}\frac{1}{\sqrt{3}}\sqrt{1-x}\,dx-\frac{1}{\sqrt{2}}\int_{-2}^{0}\sqrt{-x}\,dx\right\}$$

$$\Rightarrow 2\left\{\frac{1}{\sqrt{3}}\left[\frac{2}{3}(1-x)^{3/2}\right]_{-2}^{1}-\frac{1}{\sqrt{2}}\left[\frac{2}{3}(-x)^{3/2}\right]_{-2}^{0}\right\}$$

$$\Rightarrow 2\left\{\left[-\frac{1}{\sqrt{3}}\times\frac{2}{3}(0-3^{3/2})\right]-\left[\frac{-1}{\sqrt{2}}\times\frac{2}{3}(0-2^{3/2})\right]\right\}\Big\}$$

$$\Rightarrow 2\left\{\frac{2}{3\sqrt{3}}\times 3\sqrt{3}-\frac{1}{\sqrt{2}}\times\frac{2}{3}.2\sqrt{2}\right\}$$

$\Rightarrow 2\left\{2-\frac{4}{3}\right\}=2\left\{\frac{6-4}{3}\right\}=\frac{4}{3}$ sq. units

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Question108

The area enclosed between the curves $y^2 = x$ and $y = |x|$ is [2007]

Options:

A. 1/6

B. 1/3

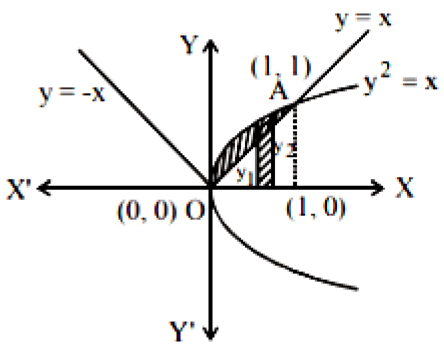
C. 2/3

D. 1

Answer: A

Solution:

It is clear from the figure, area lies between $y^2 = x$ and $y = x$
Intersection point $y = x$ and $y^2 = x$ is (1,1)



$\therefore$ Required area $= \int_0^1 (y_2 - y_1)dx$

$= \int_0^1 (\sqrt{x} - x)dx = \left[\frac{x^{3/2}}{3/2} - \frac{x^2}{2}\right]_0^1$

$= \frac{2}{3}[x^{3/2}]_0^1 - \frac{1}{2}[x^2]_0^1 = \frac{2}{3} - \frac{1}{2} = \frac{1}{6}$

Question109

Let f (x) be a non − negative continuous function such that the area bounded by the curve y = f (x), x -axis and the ordinates $x = \frac{\pi}{4}$ and $x = \beta > \frac{\pi}{4}$ is

$\left(\beta \sin \beta + \frac{\pi}{4} \cos \beta + \sqrt{2}\beta\right)$. Then $f\left(\frac{\pi}{2}\right)$ is
[2005]

Options:

A. $\left(\frac{\pi}{4} + \sqrt{2} - 1\right)$

B. $\left(\frac{\pi}{4} - \sqrt{2} + 1\right)$

C. $\left(1 - \frac{\pi}{4} - \sqrt{2}\right)$

D. $\left(1-\frac{\pi}{4}+\sqrt{2}\right)$

Answer: D

Solution:

From given condition

$\int_{\pi/4}^{\beta} f(x)dx = \beta\sin\beta + \frac{\pi}{4}\cos\beta + \sqrt{2}\beta$

Differentiating w. r . tβ, we get

$f(\beta) = \beta\cos\beta + \sin\beta - \frac{\pi}{4}\sin\beta + \sqrt{2}$

$f\left(\frac{\pi}{2}\right) = \beta . 0 + \left(1-\frac{\pi}{4}\right)\sin\frac{\pi}{2} + \sqrt{2} = 1 - \frac{\pi}{4} + \sqrt{2}$

Question110

The area enclosed between the curve $y = \log_e(x+e)$ and the coordinate axes is
[2005]

Options:

A. 1

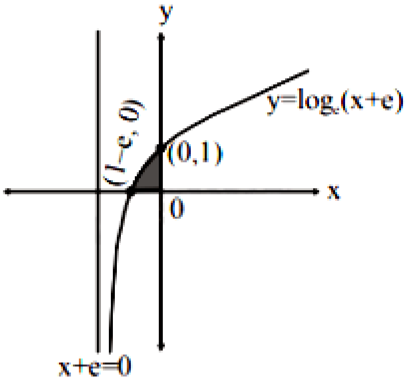
B. 2

C. 3

D. 4

Answer: A

Solution:



Required area $A = \int_{1-e}^{0} y\,dx = \int_{1-e}^{0} \log_e(x+e)\,dx$

put $x + e = t \Rightarrow dx = dt$ also when $x = 1 - e, t = 1$ and when $x = 0, t = e$

$\therefore A = \int_{1}^{e} \log_e t\,dt = [t\log_e t - t]_1^e$

$e - e - 0 + 1 = 1$

Hence the required area is 1 square unit.

Question111

The parabolas $y^2 = 4x$ and $x^2 = 4y$ divide the square region bounded by the lines $x = 4$, $y = 4$ and the coordinate axes. If S_1, S_2, S_3 are respectively the areas of these parts numbered from top to bottom; then $S_1 : S_2 : S_3$ is
[2005]

Options:

A. 1 : 2 : 1

B. 1 : 2 : 3

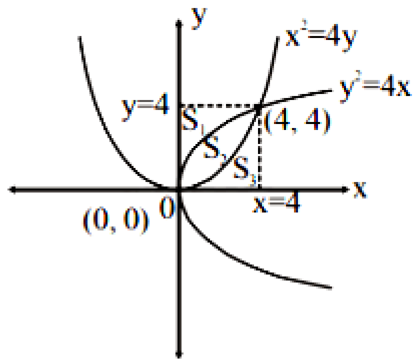
C. 2 : 1 : 2

D. 1 : 1 : 1

Answer: D

Solution:

On solving, we get intersection points of $x^2 = 4y$ and $y^2 = 4x$ are (0,0) and (4,4)



By symmetry, we observe

$S_1 = S_3 = \int_0^4 y\,dx$

$= \int_0^4 \frac{x^2}{4} dx = \left[\frac{x^3}{12}\right]_0^4 = \frac{16}{3}$ sq. units

Also $S_2 = \int_0^4 \left(2\sqrt{x} - \frac{x^2}{4}\right) dx = \left[\frac{2x^{\frac{3}{2}}}{\frac{3}{2}} - \frac{x^3}{12}\right]_0^4$

$= \frac{4}{3} \times 8 - \frac{16}{3} = \frac{16}{3}$sq. units

$\therefore S_1 : S_2 : S_3 = 1 : 1 : 1$

Question112

The area of the region bounded by the curves $y = |x - 2|, x = 1, x = 3$ and the x -axis is [2004]

Options:

A. 4

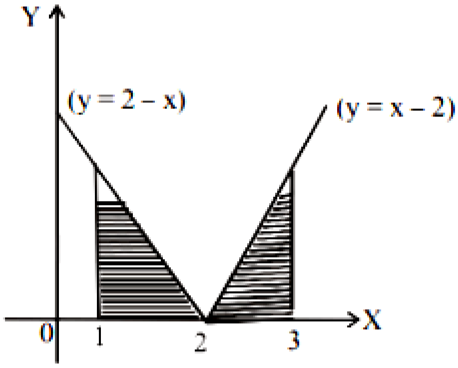
B. 2

C. 3

D. 1

Answer: D

Solution:



Required Area

$$A = 2\int_2^3 (x-2)dx = 2\left[\frac{x^2}{2} - 2x\right]_2^3 = 1$$

Question113

The area of the region bounded by the curves $y = |x-1|$ and $y = 3-|x|$ is [2003]

Options:

A. 6 sq. units

B. 2 sq. units

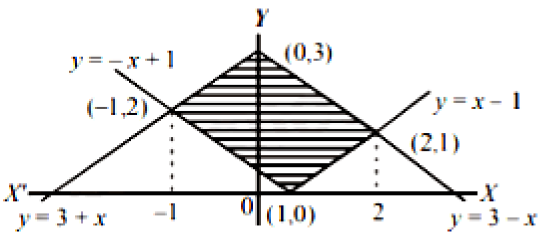
C. 3 sq. units

D. 4 sq. units.

Answer: D

Solution:

Intersection point of $y = x - 1$ and $y = 3 - x$ is (2,1) and eqns. $y = -x + 1$ and $y = 3 + x$ is (-1,2)



$$A = \int_{-1}^{0} \{(3+x)-(-x+1)\}dx +$$

$$\int_0^1 \{(3-x)-(-x+1)\}dx + \int_1^2 \{(3-x)-(x-1)\}dx$$

$= \int_{-1}^{0} (2+2x)dx + \int_{0}^{1} 2dx + \int_{1}^{2} (4-2x)dx$
$= [2x+x^2]_{-1}^{0} + [2x]_0^1 + [4x-x^2]_1^2$
$= 0-(-2+1)+(2-0)+(8-4)-(4-1)$
$= 1+2+4-3 = 4$ sq. units

Question114

If $y = f(x)$ makes +ve intercept of 2 and 0 unit on x and y axes and encloses an area of $3/4$ square unit with the axes then $\int_0^2 xf'(x)dx$ is
[2002]

Options:

A. 3/2

B. 1

C. 5/4

D. –3/4

Answer: D

Solution:

Given that $\int_0^2 f(x)dx = \frac{3}{4}$; Now,

$\int_0^2 xf'(x)dx = x\int_0^2 f'(x)dx - \int_0^2 f(x)dx$
$= [xf(x)]_0^2 - \frac{3}{4} = 2f(2) - \frac{3}{4}$
$= 0 - \frac{3}{4} (\because f(2) = 0) = -\frac{3}{4}$

Question115

The area bounded by the curves $y = \ln x$, $y = \ln|x|$, $y = |\ln x|$ and $y = |\ln|x||$ is
[2002]

Options:

A. 4sq. units

B. 6 sq. units

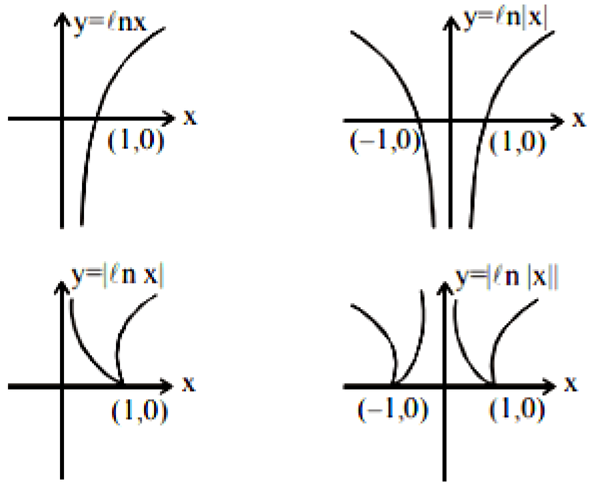
C. 10 sq. units

D. none of these

Answer: A

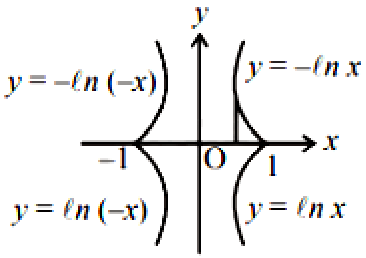
Solution:

Separate graph of each curve



[Note: Graph of $y = |f(x)|$ can be obtained from the graph of the curve $y = f(x)$ by drawing the mirror image of the portion of the graph below x -axis, with respect to x -axis.

Hence the bounded area is as shown by combined all figure.



Required area $= 4\int_0^1 (-\ln x)\,dx$

$= -4[x\ln x - x]_0^1 = 4$sq. units
